

Speedrun: Your E2E AIP Workflow

Palantir Foundry & AIP 端到端工作流速通课程 · 按左侧导航顺序完整收录（51 个 lesson，含图片）

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课程总览

课程名称: Speedrun: Your E2E AIP Workflow

来源链接: <https://learn.palantir.com/speedrun-your-e2e-aip-workflow>

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Course Introduction

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This course is continuously updated. If you have feedback, please email training-feedback@palantir.com.

In this course, you will build a fully functional, end-to-end workflow using AIP.

You will start off with a set of PDF documents provided in the curriculum. (If you are feeling up for a challenge, do bring your own samples.) You will then build a data transformation to extract their contents and split them into short chunks of text. Going forward, we'll assume that each of these chunks contains a single point of information, which can be combined to answer related questions. In this first section, you will also use an LLM to batch process these chunks and extract the most important entities they mention. Your output datasets will then be added to the Ontology, Foundry's semantic layer which underpins the rapid development of a host of interactive applications.

With your Ontology Objects ready - Chunks and Entities - you'll get your first view of your knowledge graph. This graph allows you to understand the most significant entities covered by your documents, and investigate their relationships by reviewing mutually related text chunks. In fact, you will also configure your first AIP Agent to leverage this graph and respond to your questions reliably, without hallucinations.

Finally, you will build an interactive app to allow users to ask such questions and see the segment of your knowledge graph used to formulate the answers.

By the end of the Speedrun you will have experienced a large portion of the full-stack Foundry toolkit. You will be ready to take on the Deep Dive training courses or even dive into our documentation and get building your use-cases.

If you have any feedback on this course, please leave a note in the optional feedback section of this course, or reach out to us directly at training-feedback@palantir.com so we can continue improving this course.

Our partners at Ontologize have developed [a step-by-step YouTube video](#) of this course. Feel free to use the video to follow along as you complete the course material.

Permissions & Help

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As you work through this course, you may encounter certain issues. For example, there may be specific features, limitations, or security controls in place on your Foundry instance that prevent you from following these generic instructions.

Depending on the issue that you encounter, here is how to get help!

Training Application

Your organization's Foundry Platform Administrator may have posted local documentation within Foundry's **Training** application. This will typically cover any user restrictions that exist in your specific Foundry enrollment regarding creating projects, uploading data, creating or editing objects, or deploying Marketplace bundles. It may also include organization-specific training content.

You can open the Training app from the bottom of Foundry’s left navigation bar by selecting **Support > Training**.

Permissions

To complete this specific learning course, you will need access to the following platform components:

- Pipeline Builder
- Ontology Manager
- Vertex
- Logic
- Workshop
- AIP features

You will need the following permissions in order to run all steps of this learning course:

Permissions Required	Workaround if permissions are not available
Create Learning Project	Platform Admin team will need to create Learning Project for each user.
Upload Files	Platform Admin will need to Upload Files and provide instructions to copy them to the Learning Course Folder.
Create Ontology assets (Object Types, Links and Actions)	None. Permission needs to be granted.
Edit Learning Project	None. Permission needs to be granted.

Create Foundry Resources (Builder files, Vertex graphs, Workshop apps)	None. Permission needs to be granted.
Use AIP Developer Capabilities (LLM block in Pipeline Builder, AIP Logic)	None. Permission needs to be granted.

Documentation

You can find out more about any of the Foundry applications or concepts presented in this course by searching in the Foundry documentation: <https://www.palantir.com/docs/foundry>

Screenshots and Instructions

Our platform is evolving quickly. The screenshots in this course may differ slightly from how Foundry appears on your screen. In most cases, these differences should be minor and cosmetic, and it should still be possible to continue with the course. If you notice such differences or are unable to continue because of them, please report them to training-feedback@palantir.com.

Feedback

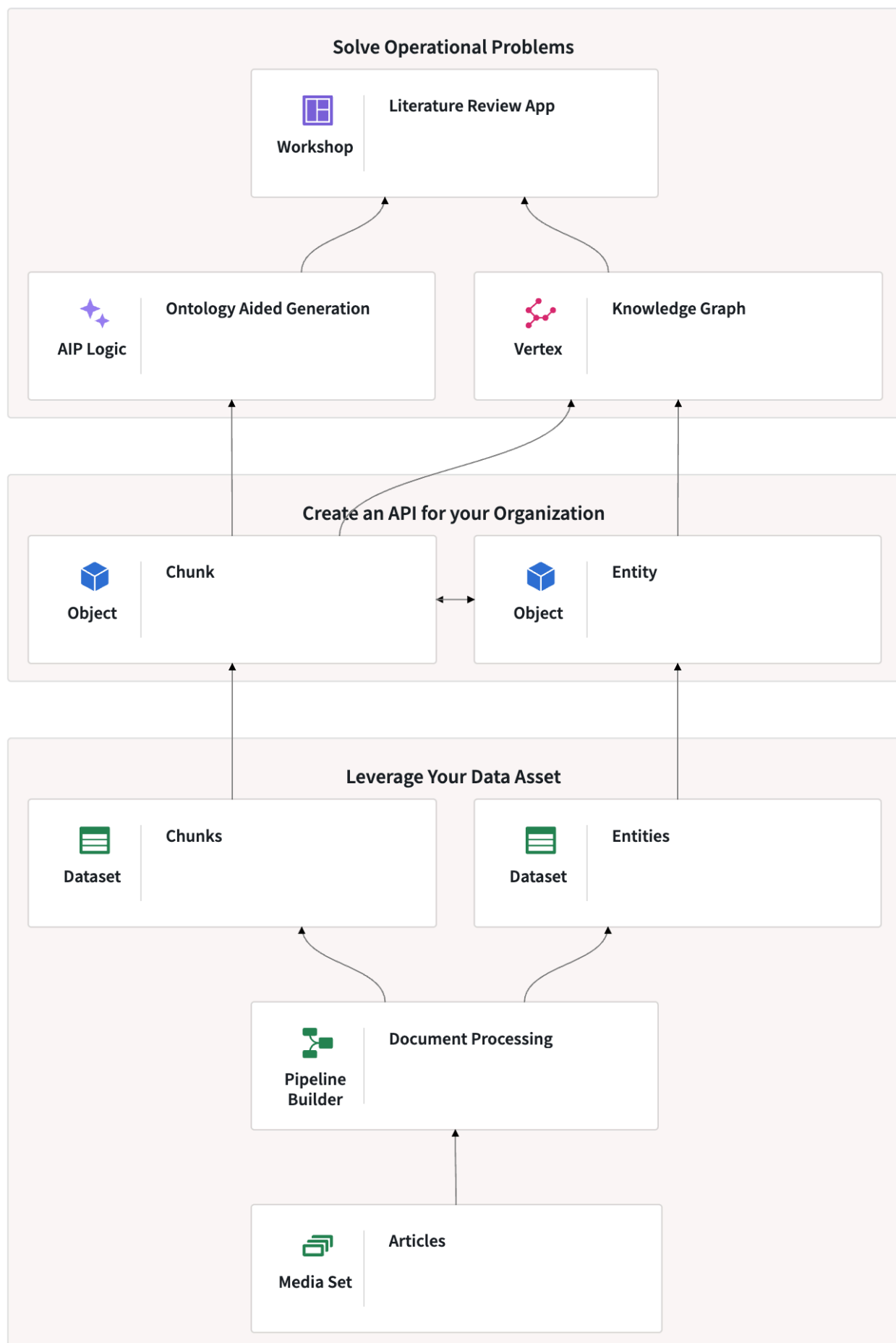
This training is being actively developed and we'd love to hear your feedback! If you have comments or suggestions, or if you encounter issues you cannot resolve by any of the means listed above, please reach out to training-feedback@palantir.com.

The Use Case

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For the purpose of this course, you work for a charity focusing on expanding access to medication in underdeveloped regions. You just joined a new team focusing on Tuberculosis, one of history's most devastating diseases. Onboarding to the team went smoothly and you are excited for the work. Unfortunately, there's a daunting library of studies detailing the latest advances in fighting this disease.

It would take months to read through this material, so you decide to leverage your team's access to AIP and build a Foundry workflow to get answers to your most important questions based on the knowledge library at hand. In preparation for the work you drew the following architecture graph.



Solve Operational Problems

- Interactive Workshop App - *Allow users to ask questions and inspect the data underpinning the answers.*
- Knowledge Graph in Vertex - *Discover the relationship between the most important concepts via connections in your source truth.*
- AIP Agent in Logic - *Outsource literature search to an AI agent for improved efficiency.*

Create the API of Your Organization

- Semantic Layer in the Ontology - *Abstract away and hide the data engineering from non-technical operators.*

Leverage Your Data Asset

- Data Transformation in Pipeline Builder - *Supercharge your data engineering speed by combining the hundreds of optimized, common data operations.*

Introduction

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All of the resources you create in Foundry need to live inside a Project. For production use cases, these resources likely need to be shared across your organization and it is recommended you create a Project for each stage of the workflow, as noted in our [online documentation](#).

During this learning course, you will be creating resources that will be specific to your learning experience. Depending on how your environment is set up, you will want to use a naming convention to easily distinguish your resources from those of other users. We will use the convention `<username>` throughout this course to represent your user login id. Wherever you see `<username>` please replace with your own user login id.

We advise using a single Foundry Project to store all your training materials for all courses on this site. You should then create a dedicated Folder within your training Project for each learning course. If your organization prefers a different Project set-up convention, you may need to adapt some instructions. Contact your platform support team for more details.

Create a Foundry Learning Project

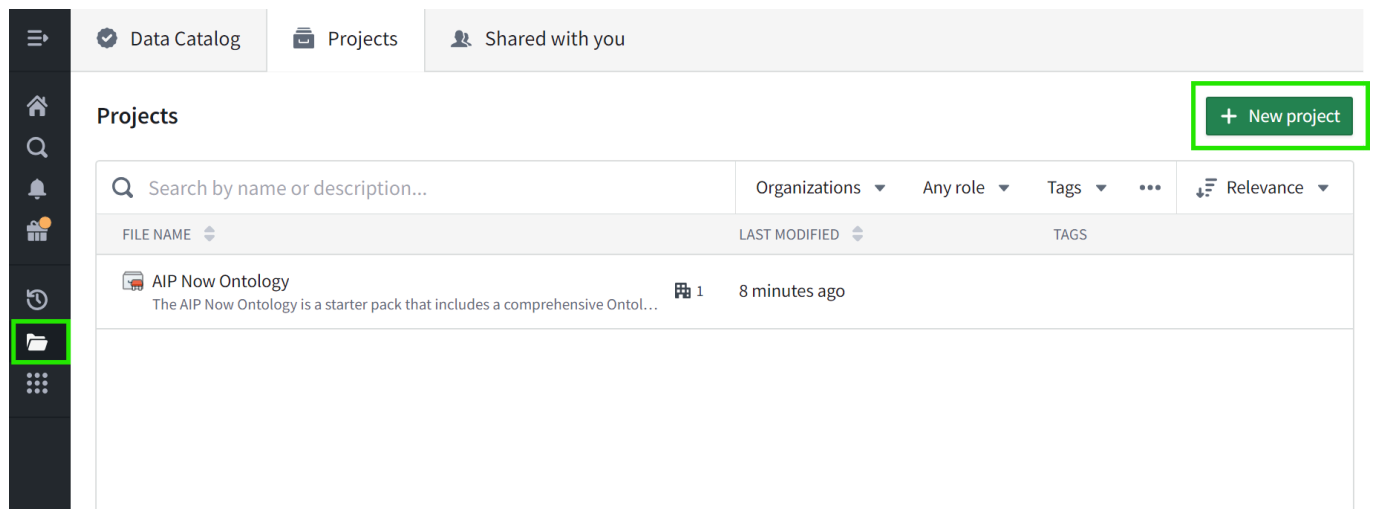
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Note if you already have your own Learning Project on your Foundry instance you can skip this step and move on to the next step to create your course-specific folder.

Step 1: Create a new project

1. Click on the Files icon on the left sidebar
2. Click on *New project* in the top right

Note: If you do not have permissions to create a project, please select a project you feel comfortable using and continue with the next instructions to create a folder. Do not use "Your files" nor your home folder as certain tools cannot be accessed from your home folder.



Step 2: Set your project's details

1. Give your project a name, such as "*Learning (<username>)*". In this case *jsmith* is being used as a notional example.
2. Give your project a description that would help another user understand what the project contains and is used for
3. Select your organization's Namespace from the dropdown
 - If you don't see this dropdown option then you don't need to do it
4. Set the Default role to *Discover*
5. Click on *Create*

Create new project



Name

learning (jsmith)

Description (optional)

This project contains folders for each learning course for the user jsmith.

Default role

Discoverer ▼



Everyone from  [redacted] can see the existence of this project and is granted the Discoverer role.

Cancel

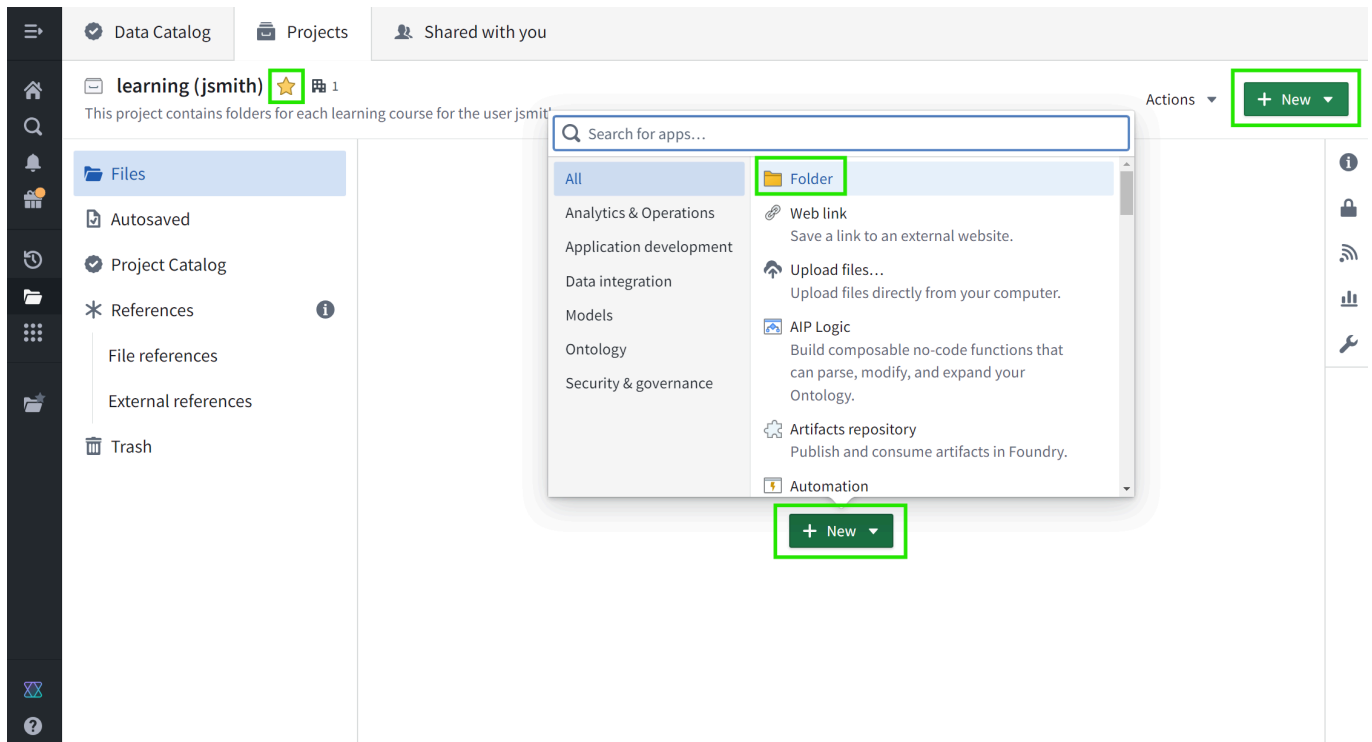
Create

(Tip: Right-click on the image and click "Open Image in New Tab" to see the full-size version)

Create a Course-Specific Training Folder

Lesson 6 / 51 · [原文链接](#)

1. In the top left, click on the star to Favorite your project. This will allow you to find it quickly later on.
 - Favourited projects, apps, and files can be found in the left-side bar of Foundry and in your home page
2. Click on *New*
3. Click on *Folder*



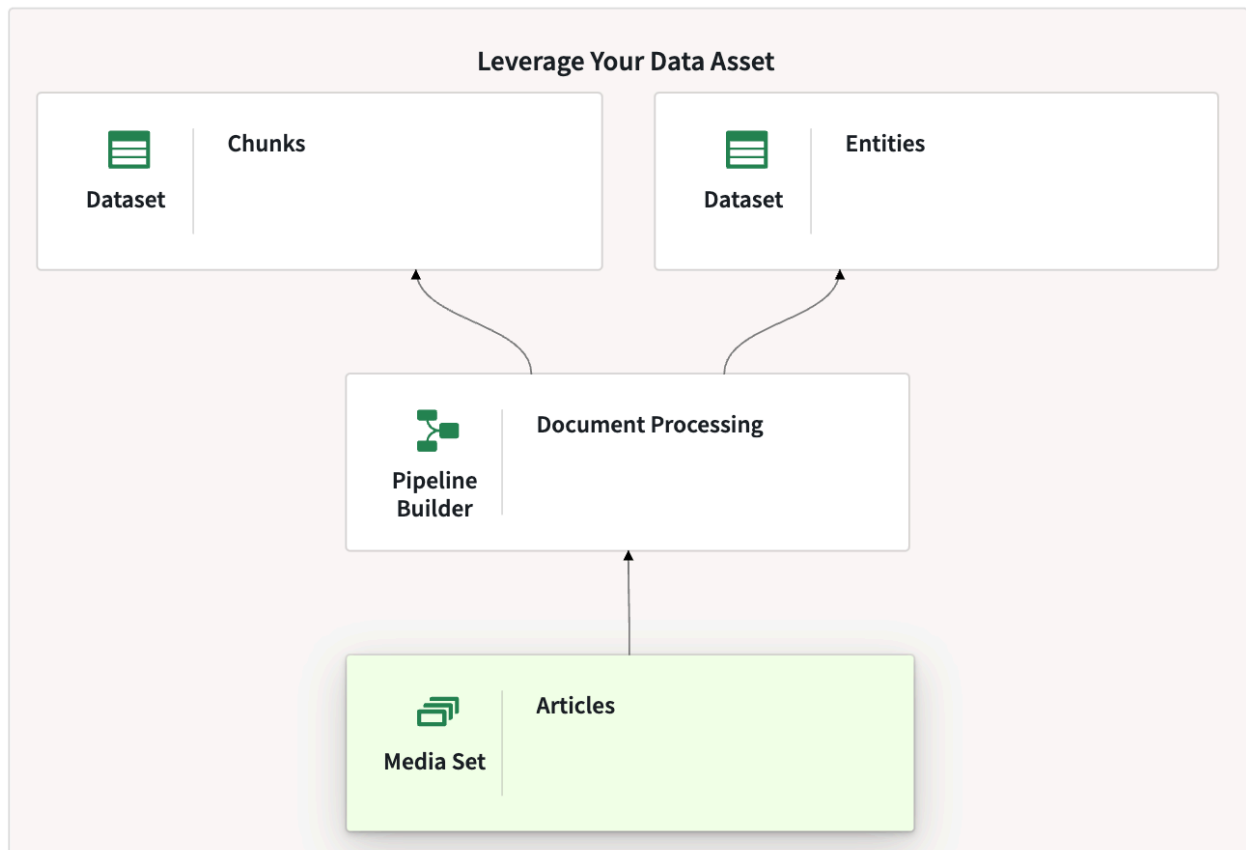
(Tip: Right-click on the image and click "Open Image in New Tab" to see the full-size version)

4. Name the new folder after the current learning course
 - For example: "*Speedrun: Your First E2E AIP Workflow*"
 - You can rename folders and files any time by right-clicking them and selecting *Rename*

Introduction

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In this section you will set up a Project, where you'll develop your workstream. You will also populate it with your unstructured input data, stored in a Media Set.



For this exercise, we will be using the below articles from the National Library of Medicine. These have been released under the CC0 - No Rights Reserved license.

- [Programmatic Management of Drug-Resistant Tuberculosis](#) (CC Zero License)
- [Standardized Treatment of Active Tuberculosis](#) (CC Zero License)
- [Comparisons and Lessons Learned from Neglected Tropical Diseases and Tuberculosis](#) (CC Zero License)
- [Impact of Alcohol Abuse on the Adaptive Immune System](#) (CC Zero License)
- [Tuberculosis in the United States in 2023](#) (CC Zero License)

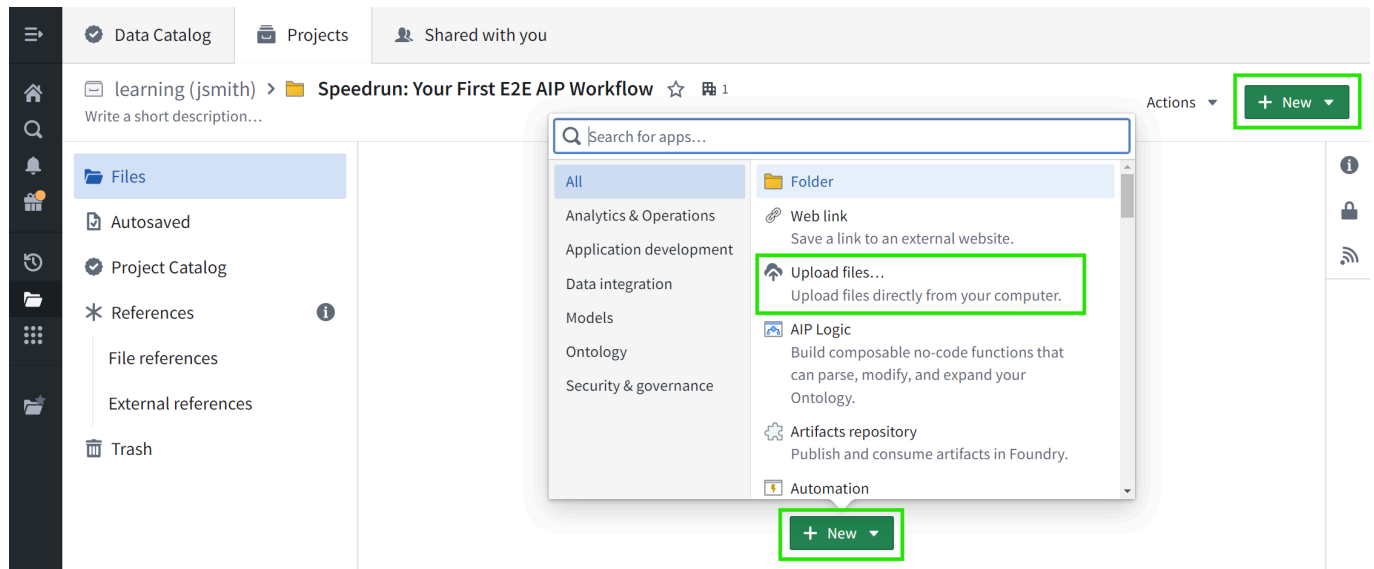
Please click on each article below to download the corresponding PDF file.

If you are unable to download datasets from this site or are unable to manually upload datasets to your Foundry instance due to lack of permissions, you may be able to deploy the datasets to Foundry via the Marketplace application. Move on to the [\[Optional\] Deploy Datasets via Marketplace section](#) for instructions.

Upload Data

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1. Click on *New* in your course folder, inside your personal Learning Project
2. Click on *Upload files...*



You can also import files by dragging them directly into your folder.

1. Click on *choose from your computer*
2. Select the files downloaded earlier
3. Make sure that *Upload to a new media set* is selected
 - Media sets allow us to interact with the file in media format and, in this use-case, is preferable to rows in a dataset or raw data
4. Click on *Next...*

Upload files

Drop files here or [choose from your computer](#)

Comparison_and_Lessons_Learned_from_Neglected_Trop... 373.16 KB

Impact_of_Alcohol_Abuse_on_the_Adaptive_Immune_Sys... 629.86 KB

Programmatic_Management_of_Drug-Resistant_Tuberculosi... 4.32 MB

Standardized_Treatment_of_Active_Tuberculosis.pdf 268.66 KB

Tuberculosis_in_the_United_States_in_2023.pdf 342.35 KB

☒ Upload to a new media set (recommended)

Media sets enable media-specific capabilities for media files (e.g. audio, imagery, video, and documents). [Documentation](#)

☐ Upload to a new unstructured dataset

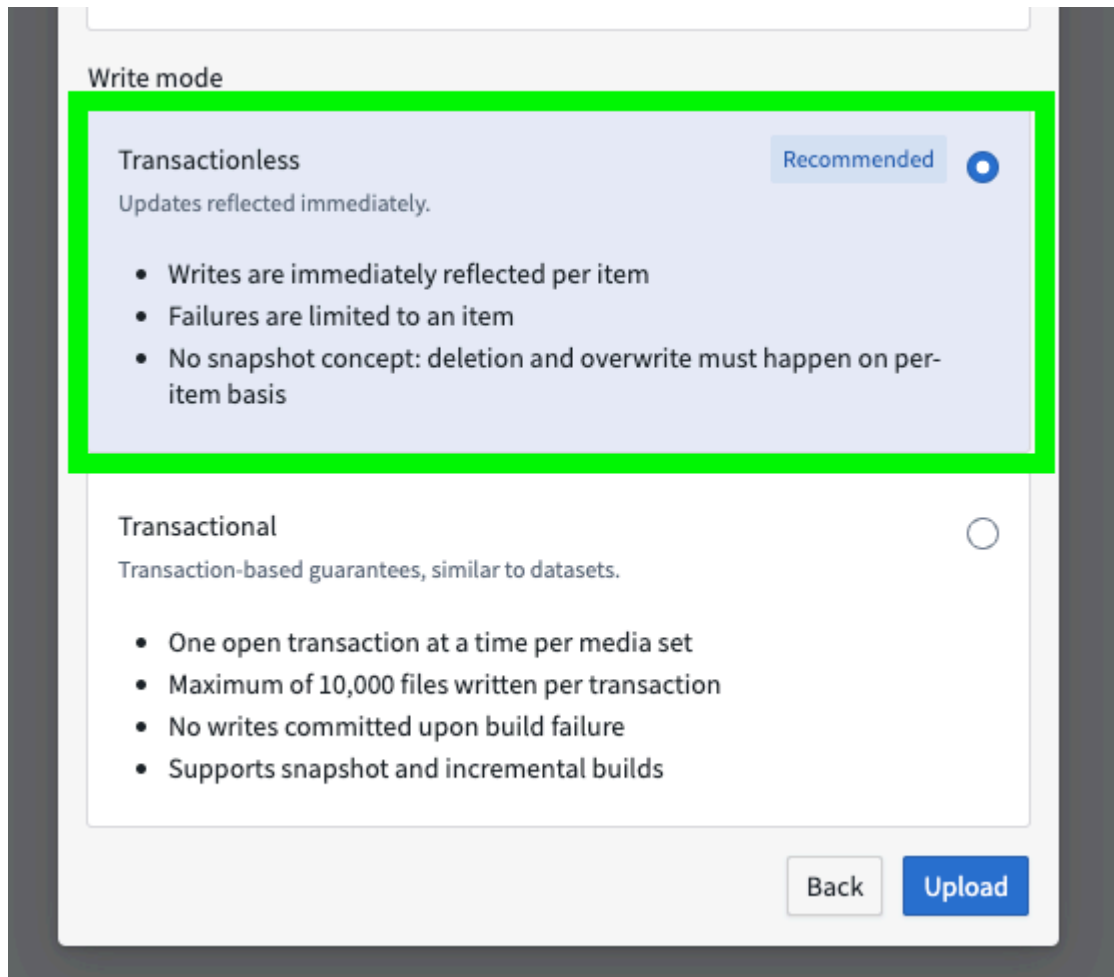
Unstructured datasets can store arbitrary files for processing and analysis. Structured data can be extracted from unstructured datasets using Pipeline Builder or Transforms. [Documentation](#)

☐ Upload as individual raw files

Raw files cannot be used in data pipelines, analyses, or models.

Next...

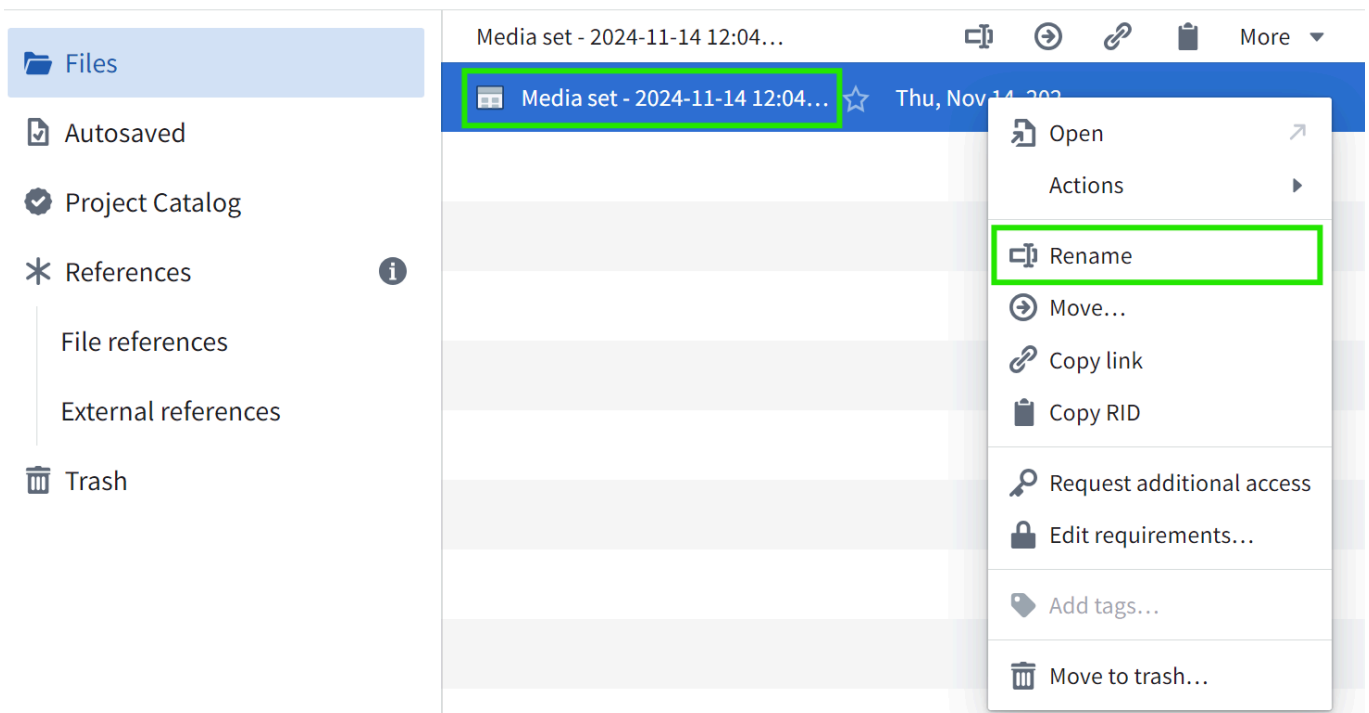
1. Select *Transactionless* as the write mode
2. Click *Upload*



1. Right-click on your Media Set
2. Rename it to "Articles"

learning (jsmith) > Speedrun: Your First E2E AIP Workflow ☆ 1

Write a short description...



1. Double click your Media Set to preview the documents you uploaded

Preview Details

Articles

Report issue

Enter description...

Updated 10 minutes ago

Created 10 minutes ago by [REDACTED]

Location [REDACTED] (jsmith)/...

Type Media set

Schema Document • PDF (.pdf)

Size 5 items • 5.9MB

Updated via File imports • [Import](#)

[Show more](#)



Programmatic_Management_of_Drug-Resistant_Tuberculosis.pdf



Comparison_and_Lessons_Learned_from_Neglected_T...osis.pdf

2. Click on one of the Media Items to review the full document

Programmatic_Management_of_Drug-Resistant_Tuberculosis.pdf →

1 of 16 + - 🔍

RESEARCH ARTICLE

Programmatic Management of Drug-Resistant Tuberculosis: An Updated Research Agenda

Carole D. Mitnick^{1,2*}, Carly A. Rodriguez³, Marita L. Hatton³, Grania Brigden⁴, Frank Cobelens⁵, Martin P. Grobusch^{6,7}, Robert Horsburgh⁸, Christoph Lange^{9,10}, Christian Lienhardt¹¹, Eyal Oren¹², Laura J. Podewils¹³, Barbara Seaworth¹⁴, Susan van den Hof¹⁵, Charles L. Daley¹⁶, Agnes C. Gebhard¹⁷, Fraser Wares¹¹, RESIST-TB (Research Excellence to Stop TB Resistance) and GDI (Global Drug Resistant TB Initiative)¹⁸

1 Department of Global Health and Social Medicine, Harvard Medical School, Boston, MA, United States of America, **2** Partners In Health, Boston, MA, United States of America, **3** Boston University School of Public Health, Boston, MA, United States of America, **4** Access Campaign, Modocins Sans Frontières, Geneva, Switzerland, **5** KNCV Tuberculosis Foundation, The Hague, Netherlands, **6** Amsterdam Institute for Global Health and Development, Academic Medical Center, University of Amsterdam, Amsterdam, Netherlands, **7** Center of Tropical Medicine and Travel Medicine, Department of Infectious Diseases, Division of Internal Medicine, Academic Medical Center, University of Amsterdam, Amsterdam, Netherlands, **8** Division of Clinical Infectious Diseases and German Center for Infection Research Tuberculosis Unit, Research Center Borstel, Borstel, Germany, **9** Department of Medicine, University of Namibia School of Medicine, Windhoek, Namibia, **10** Department of Medicine, Karolinska Institute, Stockholm, Sweden, **11** Global Tuberculosis Program, World Health Organization, Geneva, Switzerland, **12** Department of Epidemiology & Biostatistics, Mel and Enid Zuckerman College of Public Health, Tucson, AZ, United States of America, **13** Division of Global HIV/AIDS and Tuberculosis, US Centers for Disease Control and Prevention, Atlanta, GA, United States of America, **14** Heartland National TB Center, Texas Center for Infectious Diseases, San Antonio, TX, United States of America, **15** Central Office, KNCV Tuberculosis Foundation, The Hague, Netherlands, **16** Department of Medicine, National Jewish Health and the University of Colorado Denver, Denver, CO, United States of America, **17** Technical Division, KNCV Tuberculosis Foundation, The Hague, The Netherlands

*** Membership of the RESIST-TB and GDI is listed in the Acknowledgments.**
*** Carole_mitnick@hms.harvard.edu**

Citation: Mitnick CD, Rodriguez CA, Hatton ML, Brigden G, Cobelens F, Grobusch MP, et al. (2016) Programmatic Management of Drug-Resistant Tuberculosis: An Updated Research Agenda. PLOS ONE 11(5): e0155968. doi:10.1371/journal.pone.0155968

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Abstract

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Author

Carole D. Mitnick, Carly A. Rodriguez, Marita L. Hatton, Grania Brigden, Frank Cobelens, Martin P. Grobusch, Robert Horsburgh, Christoph Lange, Christian Lienhardt, Eyal Oren, Laura J. Podewils, Barbara Seaworth, Susan van den Hof, Charles L. Daley, Agnes C. Gebhard, Fraser Wares, RESIST-TB (Research Excellence to Stop TB Resistance) and GDI (Global Drug Resistant TB Initiative)

Title

Programmatic Management of Drug-Resistant Tuberculosis: An Updated Research Agenda

RID

ri.mio.main.media-item.4f68e...

Similar preview functionality exists for other audio and visual formats as well.

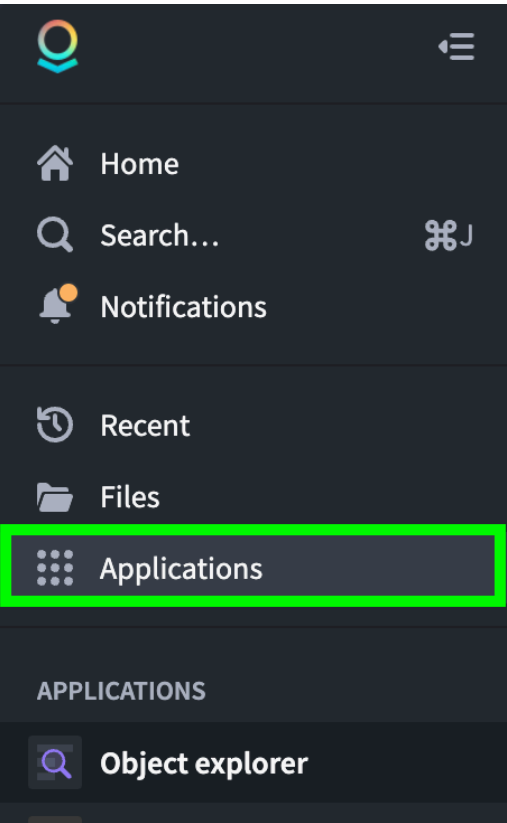
[Optional] Deploy Datasets via Marketplace

Lesson 9 / 51 · [原文链接](#)

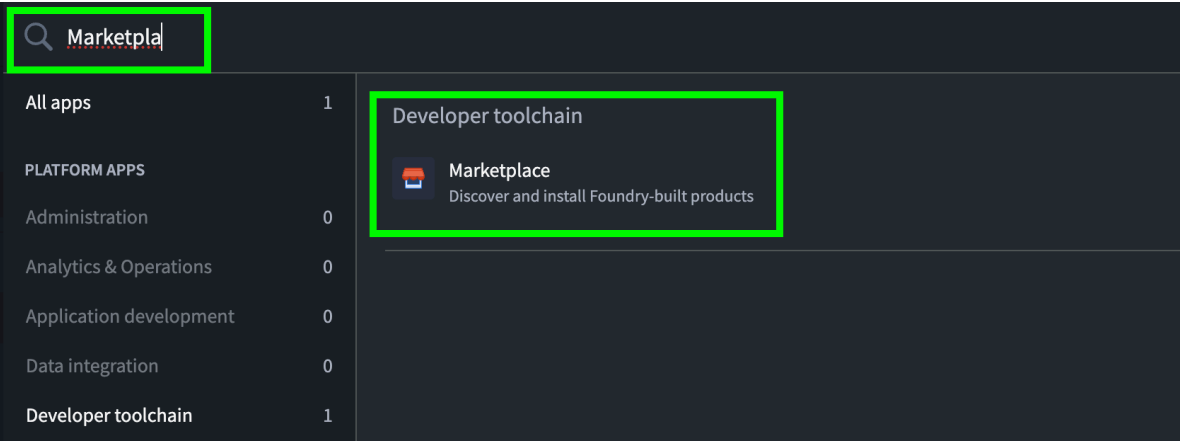
Only complete these steps if you are unable to download files from this site or are unable to upload files to your Foundry instance due to lack of permissions.

Step 1: Open Marketplace

- 1. Click on **Applications** on the left side of the screen

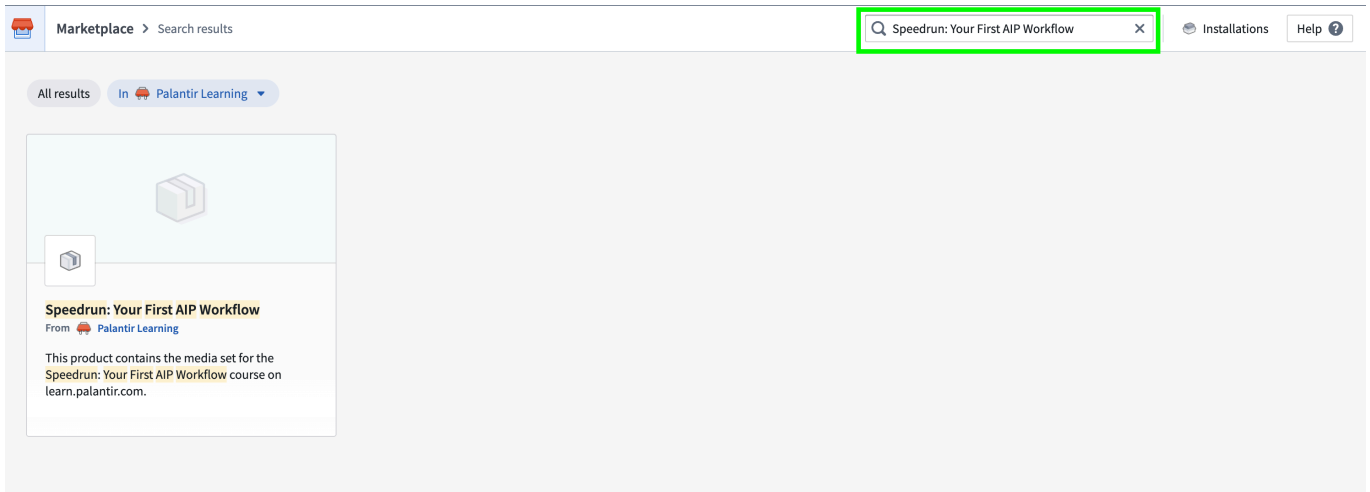


- 2. Search for `Marketplace` and open it.



Step 2: Open the Marketplace product

1. Search for the Speedrun: Your First AIP Workflow product in the top search bar labeled **Search products...** (not the lower search bar labeled **Search stores...**)
2. Click into the corresponding product
3. Click through the prompts to install the datasets via Marketplace and save them in your folder for this course. Detailed instructions for installing a similar Marketplace package are listed in the [Deep Dive: Building Your First Application course](#).



Summary

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You now have a Project set-up to develop your workflow. You've also successfully uploaded your initial data asset to Foundry. In the following sections, you will process this data and make it available to both human and AIP agents to use.

Although we added data manually this time, Foundry includes a fully fledged Data Connection toolkit, to manage recurring batch and streaming ingestions. It comes with 200+ bespoke connectors for the most common systems and a flexible plugin architecture to cover the rest. Both tabular and unstructured data assets are natively handled by the platform. These can be operated on using the broad Foundry ecosystem, as you'll see later in this course.

To learn more about data ingestion, you might follow this course up with the [Data Connection Deep Dive](#), or by taking a look at the [documentation](#).

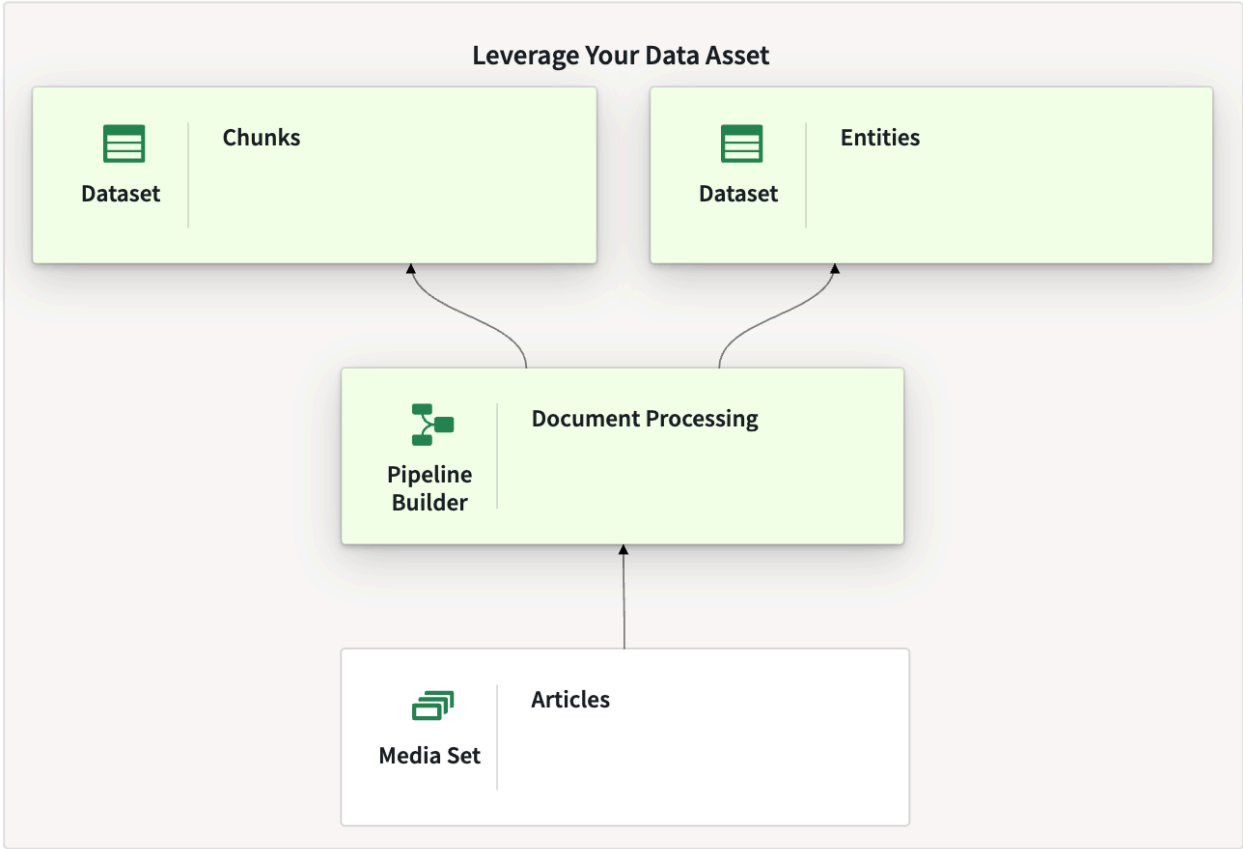
Foundry has data security measures baked into every aspect of the platform. Projects, like the one you created earlier, are the atomic units of "discretionary" access controls. People with the Owner role on the Project can share the contents with other users, or groups of users, by setting them as Editors or Viewers.

To learn more about data security, please review our [documentation](#).

Introduction

Lesson 11 / 51 · [原文链接](#)

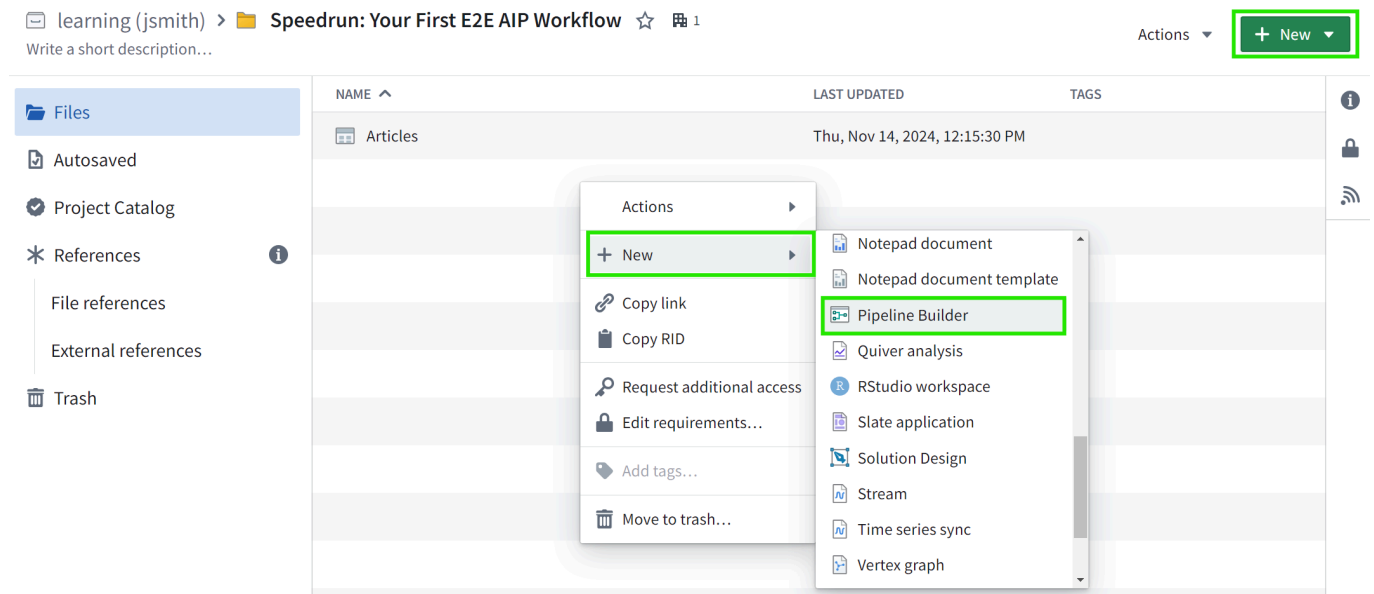
In this section, you will process the previously uploaded PDFs to break them down into short chunks of text. You will also leverage an LLM to summarize these chunks and extract the most important entities they mention.



Create a New Pipeline

Lesson 12 / 51 · [原文链接](#)

1. Navigate to your course folder, inside your training project
2. Right-click inside the folder and pick *New*
 - Alternatively, you can click *New* in the top right corner
3. Select *Pipeline Builder* from the dropdown



1. Name your new pipeline "*Document Processing Transform*"
2. Select the pipeline style as *Batch pipeline*
3. Select the batch compute as *Standard*
4. Click on *Create pipeline*

Name and location



Document Processing Form



Edit location

Configuration



Batch pipeline

Builds and transforms entire datasets on each deploy. For data that is ingested periodically.



Streaming pipeline

Transforms data continuously as new data is made available. For high frequency ingestion.

Select batch compute

[Compare types](#)



Standard

Default

Build your pipelines using advanced expressions. Backed by Spark.



Faster

New

Speed up your pipelines and save compute. Backed by DataFusion.
[Learn More](#)



External

Beta

Build your pipelines using external compute platforms. [Learn More](#)

[← Back](#)

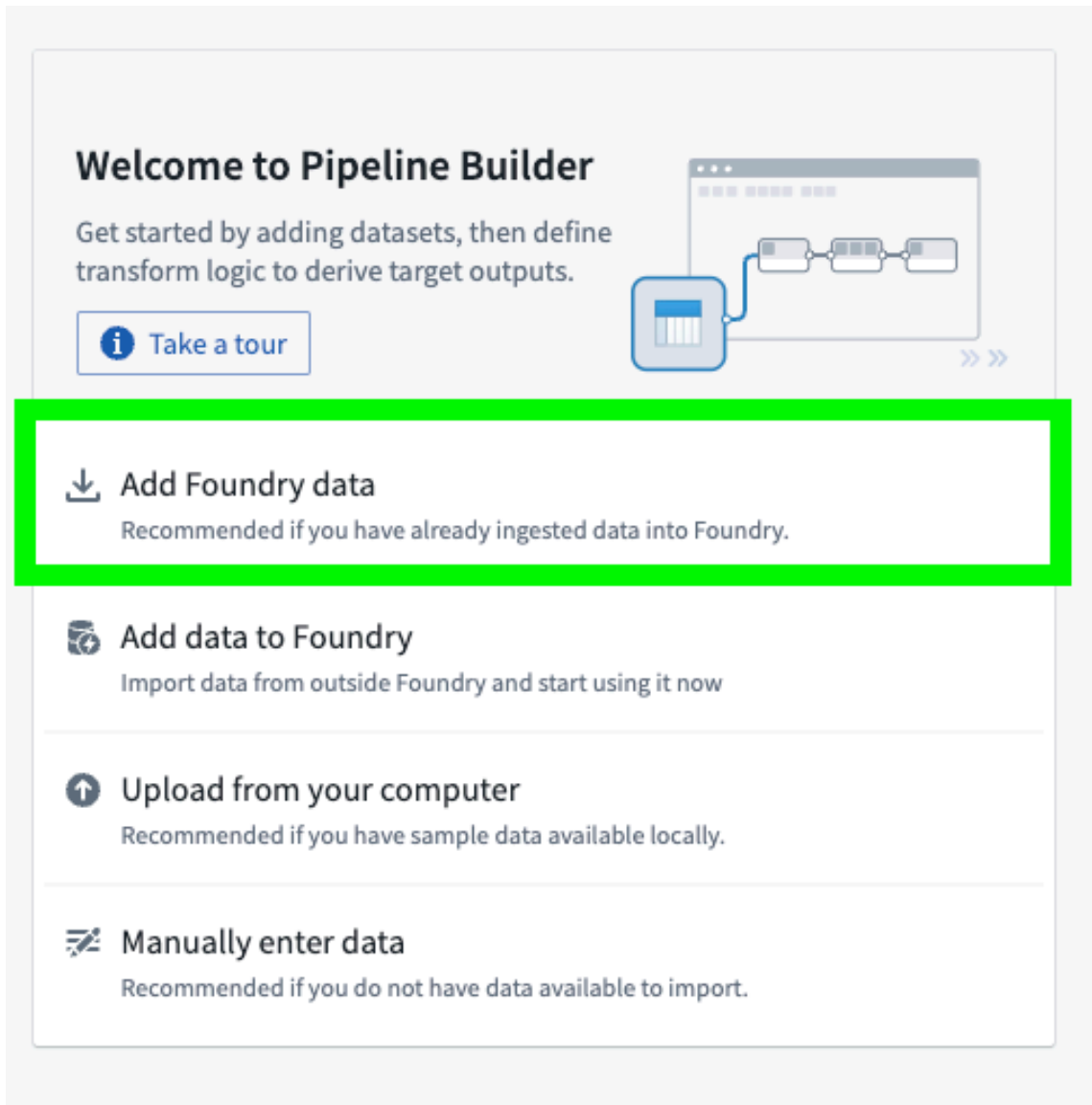
Create pipeline

Add Data

Lesson 13 / 51 · [原文链接](#)

1. Click on *Add Foundry data*

- You can find this *Add Data* dialog in the top left of the Pipeline Builder screen if you want to add more data later. It might be particularly useful for setting up mock datasets (or configuration datasets) using the *Manually Enter Data* option.

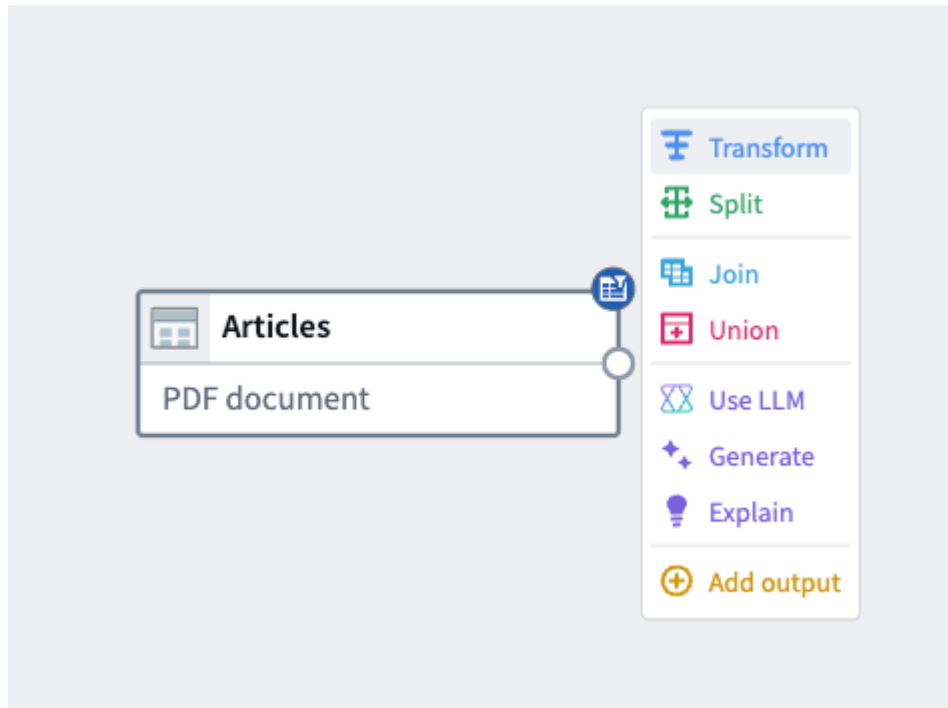


1. Add the *Articles* media set to your selection using the *+* button
2. Click *Add data* in the bottom right to confirm selection

Process PDFs

Lesson 14 / 51 · [原文链接](#)

1. Back on the main Pipeline Builder graph, select the new *Articles* transform node
2. Click the *Transform* button
 - This will create a transform *Node*, where transformation steps can be chained and a resulting dataset can be visualized

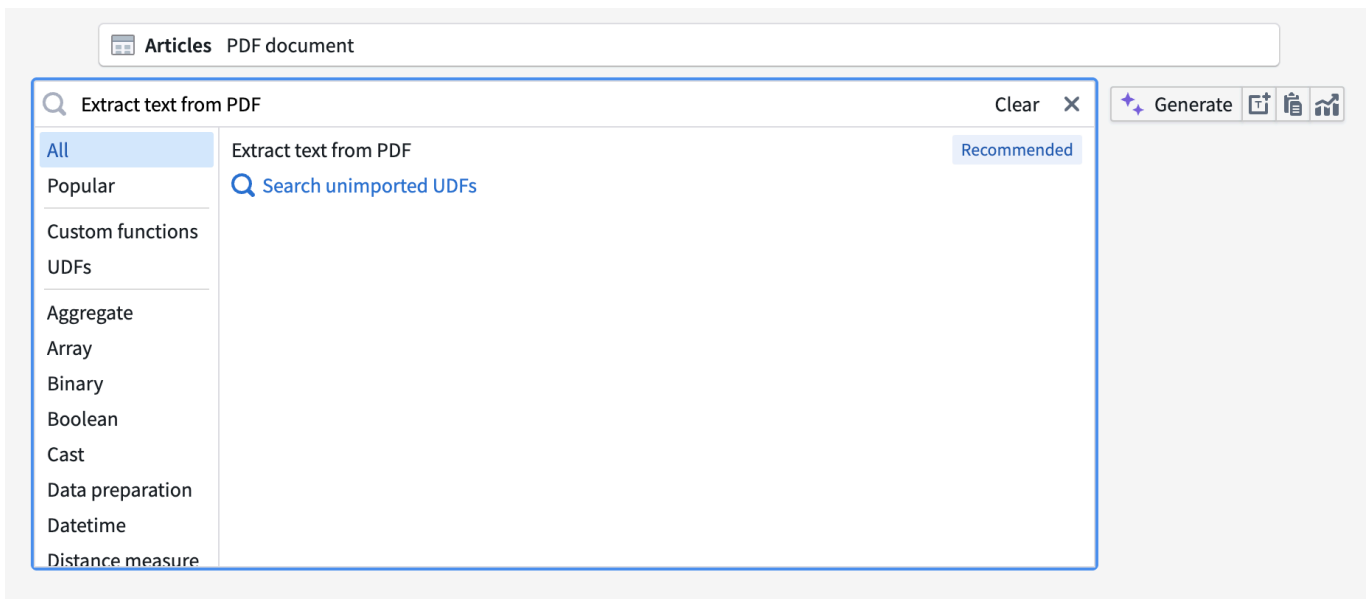


1. Rename the new transform node to "*Process PDFs*" in the top left corner

A screenshot of the Document Processing Transform interface. The top bar shows 'Document Processing Transform' with tabs for 'Graph', 'Proposals', and 'History'. Below the top bar, the 'Process PDFs' node is highlighted. The main area shows a search bar and a list of output PDF documents. On the left, there is a sidebar with 'Articles' and 'Schedules' tabs. The 'Articles' tab is active, showing a list of documents with their titles and file names. The 'Schedules' tab is also visible. The bottom of the page shows a preview of the output documents, including 'Tuberculosis_in_the_United_States_in_2023.pdf', 'Standardized_Treatment_of_Active_T...osis.pdf', 'Programmatic_Management_of_Drug...osis.pdf', 'Impact_of_Alcohol_Abuse_on_the_A...stem.pdf', and 'Comparison_and_Lessons_Learned_fr...osis.pdf'.

You can see a preview of the output of your current transformation at the bottom of the page. It gets updated each time you add a new transform to your node. You can chain transforms by selecting them from the search bar at the bottom of the current chain of operations (empty at the moment). Please keep adding transform blocks, one below the other, until instructed otherwise.

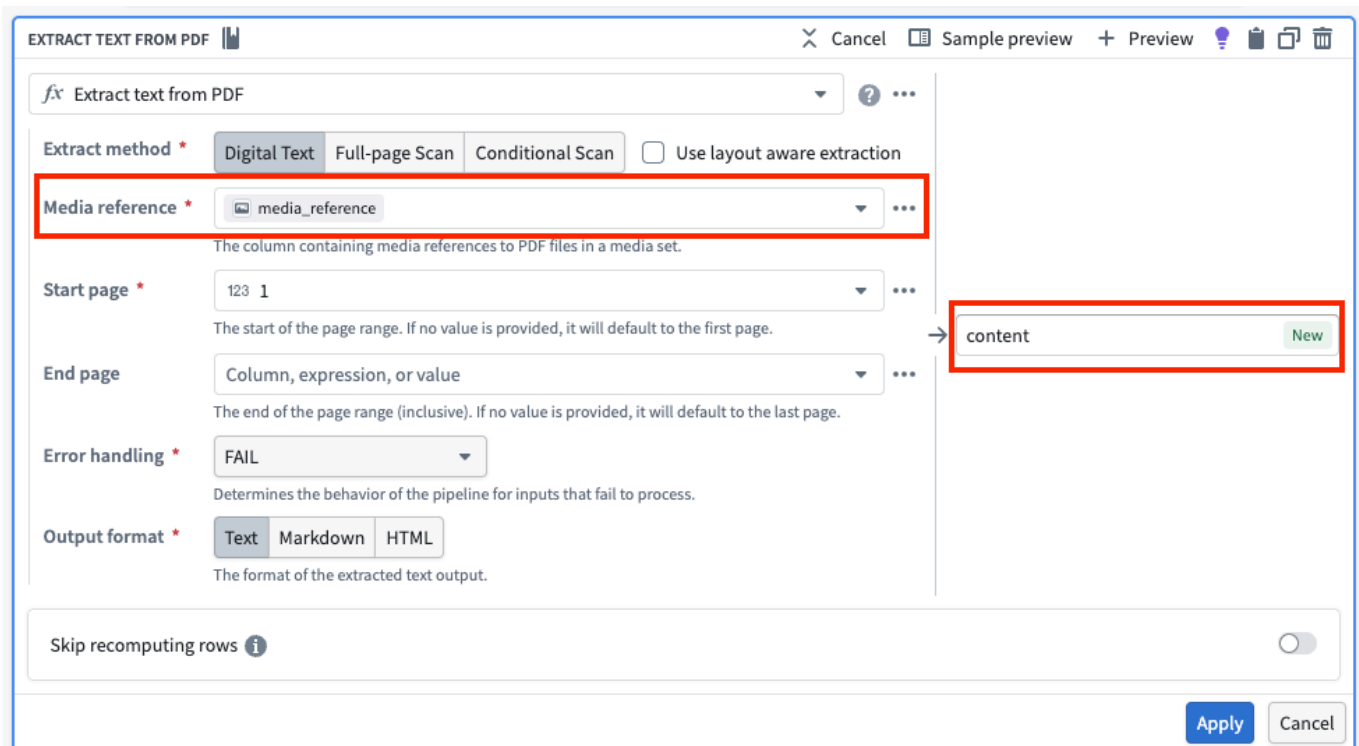
1. Search for and add a *Extract text from PDF* transform block



2. Select the `mediaReference` column in the *Media reference* dropdown

3. Set the output column name to "`content`"

4. Click *Apply*



The "Extract text from PDF" transform block extracts text elements from the PDF documents in the Media Set you uploaded earlier. In some cases, the text is stored as images (for example if the document was photo-scanned). If you are in this situation, you should change the "Extract Method" to Full-page Scan.

1. Search for and add a *Explode Array with Position* transform block

2. Set the `content` column as the array to explode
3. Set the output column name to "`content`"
4. Click *Apply*

The screenshot shows the configuration for the 'EXPLODE ARRAY WITH POSITION' transform block. The 'Array' field is set to 'content'. The output column is also 'content', with a 'Replace' flag. The 'Keep empty / null arrays' checkbox is unchecked. The 'Apply' button is highlighted.

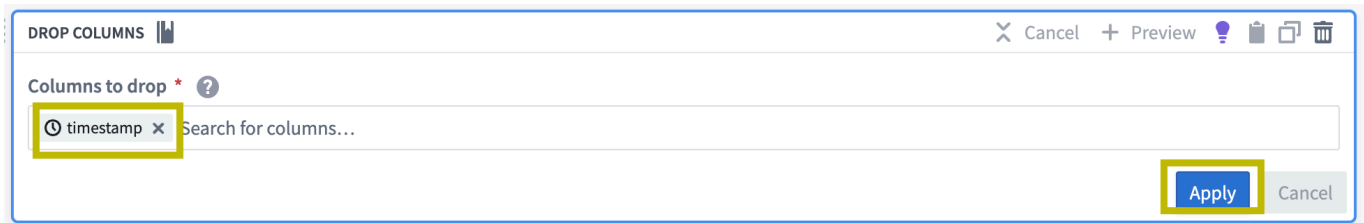
For each row, the "Explode Array with Position" transform block takes the input array and creates a new row for each element of the array. Each new row is identical to the original, except the output column ("`content`" in the case above), which contains the corresponding element of the input array and its index in a struct.

1. Search for and add the *Extract Many Struct Fields* transform block
2. Set the `content` column as the struct to process
3. Set the first locator to `position` and extract that property into a new column called "`pageNumber`"
4. Click on *Add item*
5. Set the second locator to `element` and extract that property into the existing "`content`" column
6. Click *Apply*

The screenshot shows the configuration for the 'EXTRACT MANY STRUCT FIELDS' transform block. The 'Struct' field is set to 'content'. There are two locators: 'position' mapped to 'pageNumber' (marked 'New') and 'element' mapped to 'content' (marked 'Replace'). The 'Add item' button is visible below the locators. The 'Apply' button is highlighted.

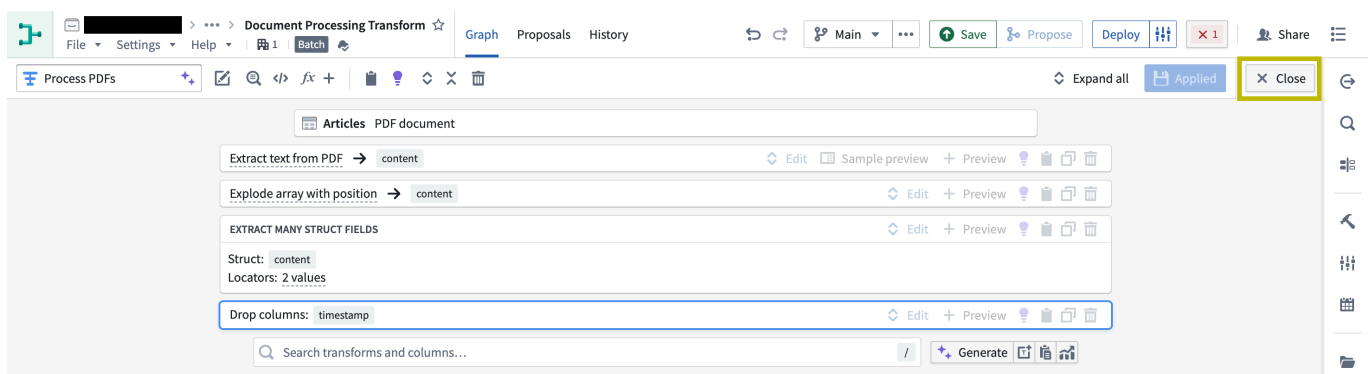
You can see a "New" or "Replace" flag next to the output(s) of your transform block. These highlight if you are overwriting an existing column of the same name.

1. Search for and add the *Drop Columns* transform block
2. Select the **timestamp** column to drop
3. Click *Apply*



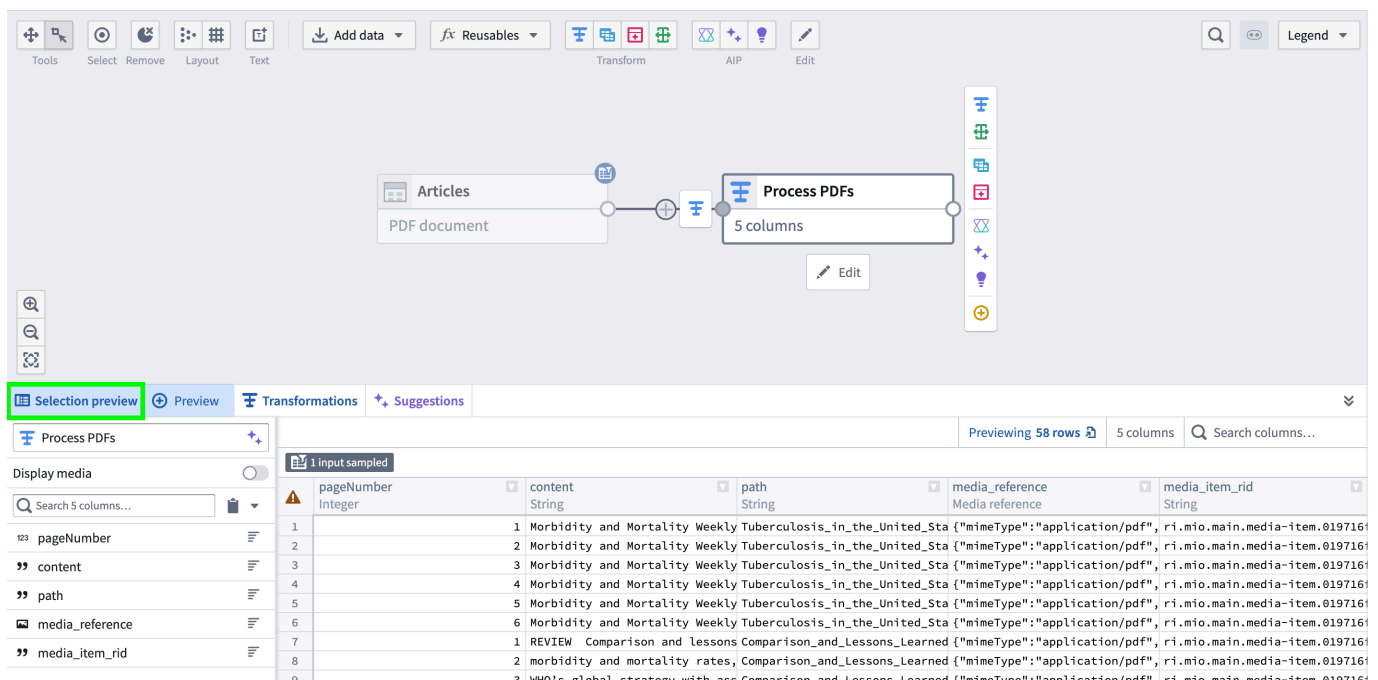
This column is added by default when the PDFs are automatically converted into rows. It is not necessary for this example, so we remove it here.

1. Click *Close* in the top right corner



The text you extracted from the PDF files is simpler to interact with than the raw files you've uploaded. This makes it possible to easily run data transformations and AIP interactions on it.

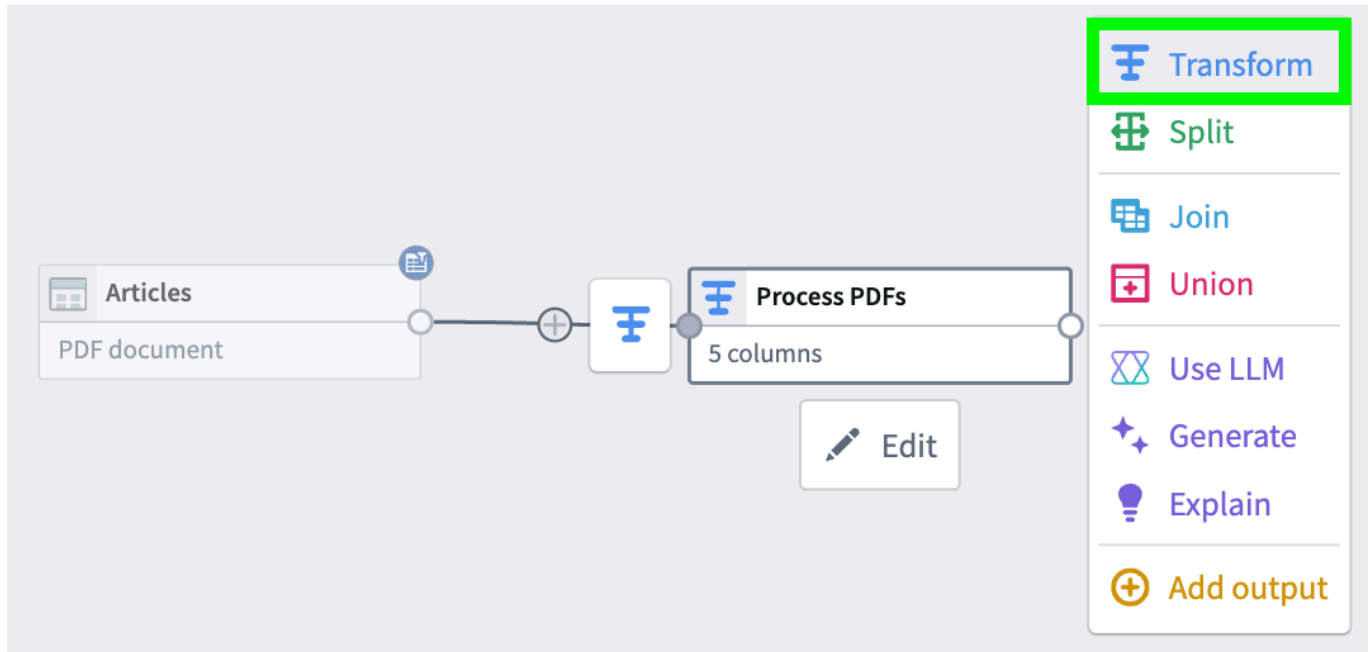
If you close out of the transform screen we've been working in to get to the below screen, take a look at the *Selection preview* tab at the bottom of your screen, you should see something like this:



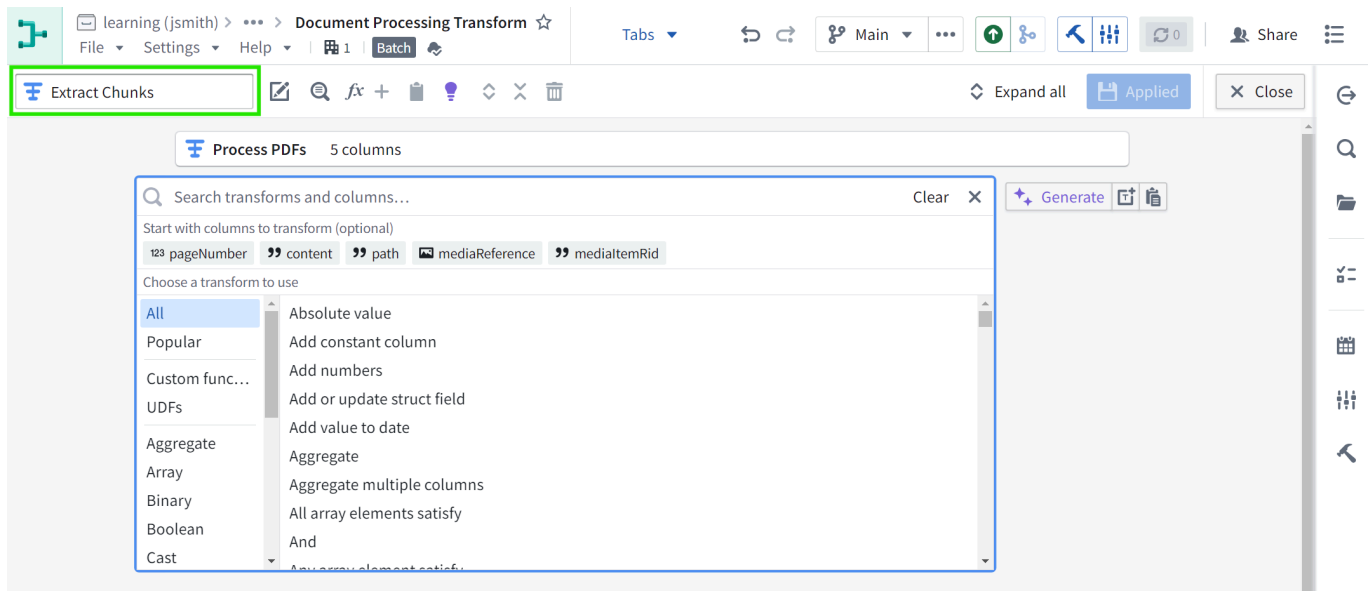
Extract Chunks

Lesson 15 / 51 · [原文链接](#)

1. Select the *Process PDFs* transform node
2. Click the *Transform* button



3. Rename the new transform node to "*Extract Chunks*" in the top left



1. Search for and add a *Chunk string* transform block
2. Set the `content` column as the input expression
3. Set *Chunk size* to 512
 - Smaller values lead to more specific information in the chunks but much longer execution time. 512 strikes a nice balance in this use-case.
4. Set the output column name to "*content*," which will replace the existing column

5. Click *Apply*

CHUNK STRING

fx Chunk string

Expression * `content`

The column containing the document text to chunk.

Chunk size * `512`

Create chunks of size around this number. Must be greater than 0.

Chunk overlap * `0`

Allow chunks to overlap in content by around this number. Must be greater than or equal to 0 and less than the selected chunk size.

Separators *

- `\n\n`
- New line character (`\n`)
- 1 space
- Empty string

[+ Add value](#) [Clear all](#)

Chunk string on these provided separators. The default separators has the effect of trying to keep all paragraphs, and then sentences, and then words together for as long as possible.

☒ Keep separator

Preview: `content` [Replace](#)

[Apply](#) [Cancel](#)

The "Chunk String" transform block splits a string into similar length segments, possibly with overlapping. It's particularly useful when trying to create small blocks of text, each of which contains a single argument without accidentally chopping off too much related content (hence the overlap).

1. Search for and add an *Explode array with position* transform block
2. Set the `content` column as the array to explode
3. Set the output column name to "`content`," once again replacing the existing column
4. Click *Apply*

EXPLODE ARRAY WITH POSITION

fx Explode array with position

Array * `content`

Array of values to explode.

☐ Keep empty / null arrays

Preview: `content` [Replace](#)

[Apply](#) [Cancel](#)

1. Search for and add an *Extract many struct fields* transform block
2. Set the `content` column as the struct to process
3. Set the first locator to `position` and extract that property into a new column called "`chunkNumber`"
4. Click in *Add item*
5. Set the second locator to `element` and extract that property into the existing "`content`" column
6. Click *Apply*

EXTRACT MANY STRUCT FIELDS

Cancel Preview

Struct * ?

content

Locators * ?

position chunkNumber New

element content Replace

Add item

Apply Cancel

Search transforms and columns...

Generate

7. Click *Close* in the top right corner

learning (jsmith) > ... > Document Processing Transform ☆

File Settings Help 1 Batch

Extract Chunks

Expand all Applied Close

Process PDFs 5 columns

Chunk string → content Edit Preview

Explode array with position → content Edit Preview

EXTRACT MANY STRUCT FIELDS Edit Preview

Struct: content

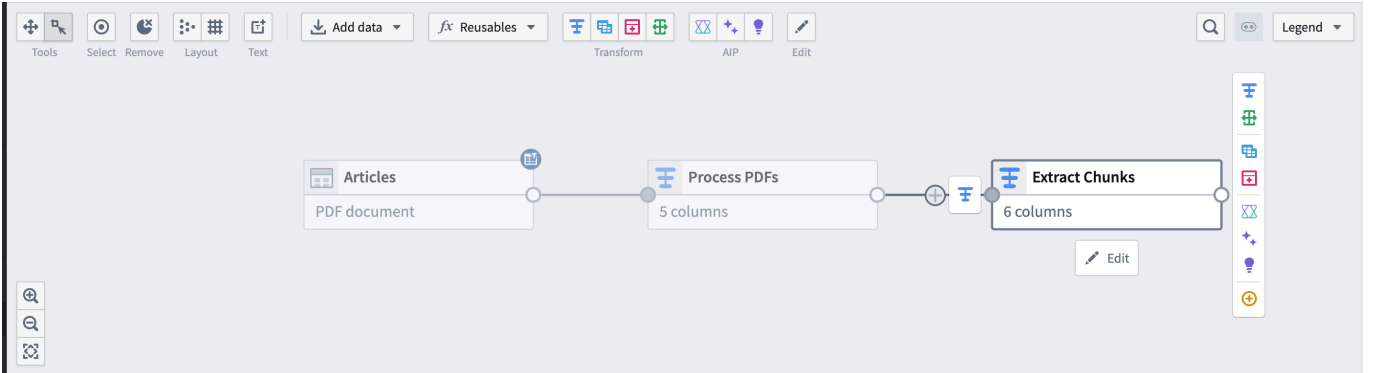
Locators: 2 values

Search transforms and columns...

Generate

Chunking the text creates segments short enough that you can assume they contain single bits of information. By filtering these chunks, you can find all information contained in your data asset related to a particular topic. This filtered collection will be used to constrain LLMs to answer your questions using only real information.

If you take a look at the *Selection preview* tab at the bottom of your screen, you should see something like this:



Selection preview Preview Transformations Suggestions

Extract Chunks

Previewing 500 rows 6 columns Search columns...

Display media

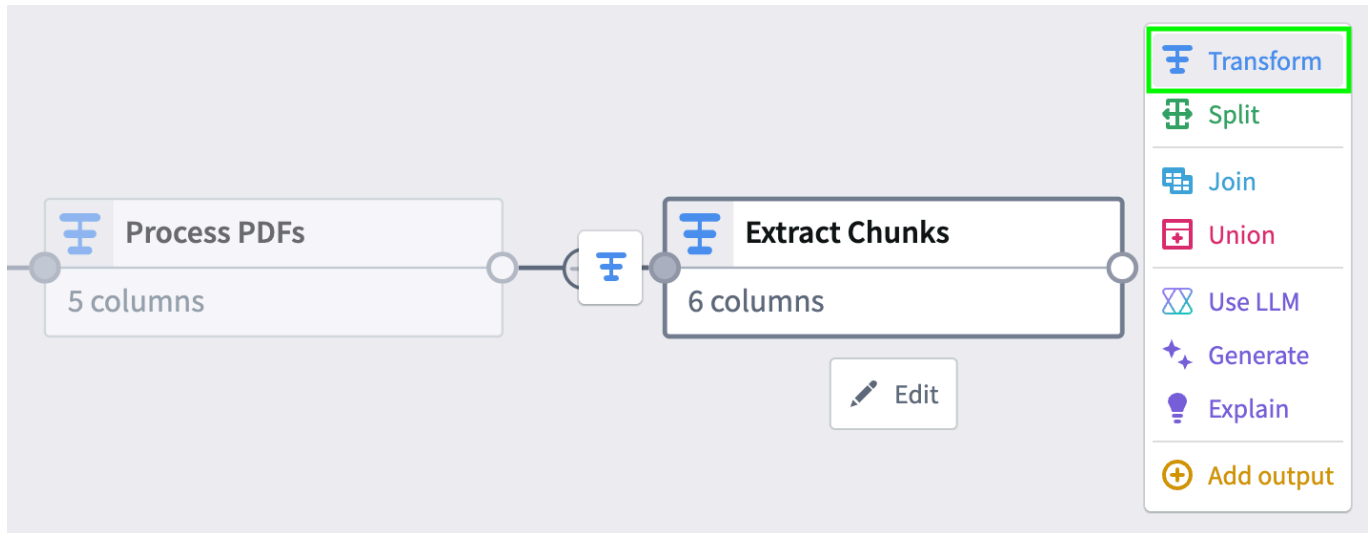
Search 6 columns...

	chunkNumber Integer	content String	pageNumber Integer	path String	media_reference Media reference	media_item_rid String
1		1 Morbidity and Mortality W		1 Tuberculosis_in_the_Unit	{ "mimeType": "application/	ri.mio.main.media-item.0
2		2 2020, coinciding with the		1 Tuberculosis_in_the_Unit	{ "mimeType": "application/	ri.mio.main.media-item.0
3		3 an increase of 1,295 case		1 Tuberculosis_in_the_Unit	{ "mimeType": "application/	ri.mio.main.media-item.0
4		4 global morbidity and mort		1 Tuberculosis_in_the_Unit	{ "mimeType": "application/	ri.mio.main.media-item.0
5		5 (TB) remains one of the w		1 Tuberculosis_in_the_Unit	{ "mimeType": "application/	ri.mio.main.media-item.0
6		6 declined considerably in		1 Tuberculosis_in_the_Unit	{ "mimeType": "application/	ri.mio.main.media-item.0
7		7 Surveillance System (NTSS		1 Tuberculosis_in_the_Unit	{ "mimeType": "application/	ri.mio.main.media-item.0
8		8 TB case counts and rates		1 Tuberculosis_in_the_Unit	{ "mimeType": "application/	ri.mio.main.media-item.0
9		9 Self-reported race and et		1 Tuberculosis_in_the_Unit	{ "mimeType": "application/	ri.mio.main.media-item.0
10		10 as of February 17, 2024		1 Tuberculosis_in_the_Unit	{ "mimeType": "application/	ri.mio.main.media-item.0

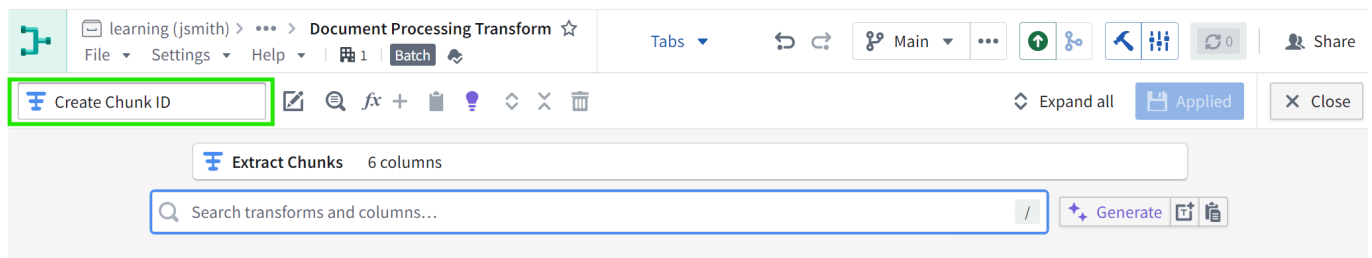
Create Chunk ID

Lesson 16 / 51 · [原文链接](#)

1. Select the *Extract Chunks* transform node
2. Click the *Transform* button



3. Rename the new transform node to "Create Chunk ID" in the top left



1. Search for and add a *Concatenate strings* transform block
2. Set the separator as an underscore "_"
3. Set the first element in the concatenation to be the `media_Item_Rid` column
4. Set the second element in the concatenation to be a *Cast* expression with type *String* applied to the `pageNumber` column
5. Click in *Add item*
6. Set the third element in the concatenation to be a *Cast* expression with type *String* applied to the `chunkNumber` column
7. Name the new column "*chunkId*"
8. Click *Apply*

CONCATENATE STRINGS

fx Concatenate strings

Separator *

Expressions *

media_item_rid

fx Cast to String

Expression *

Type * String

fx Cast to String

Expression *

Type * String

chunkId New

Apply Cancel

9. Click *Close*

learning (jsmith) > Document Processing Transform

File Settings Help 1 Batch

Create Chunk ID

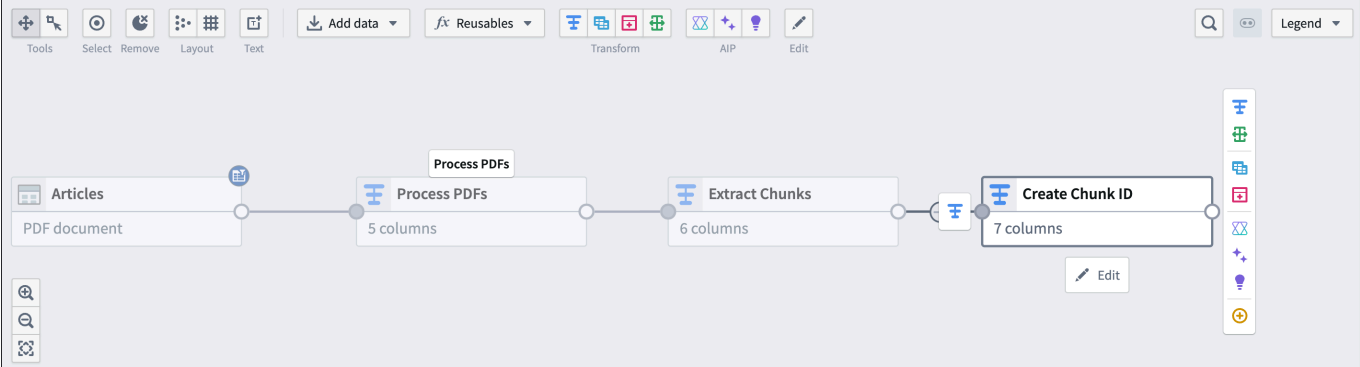
Extract Chunks 6 columns

Concatenate strings → chunkId

Close

ChunkId will be used as the primary key to uniquely reference the Chunk objects that you will create in the Ontology later on. It is very important to design a primary key that is unique (and not null) when planning the architecture of your application. When possible, you should use a combination of other columns rather than a randomly generated or hashed code. This makes it easier to debug any potential duplication.

If you take a look at the *Selection preview* tab at the bottom of your screen, you should see something like this:



Selection preview Preview Transformations Suggestions

Previewing 500 rows 7 columns Search columns...

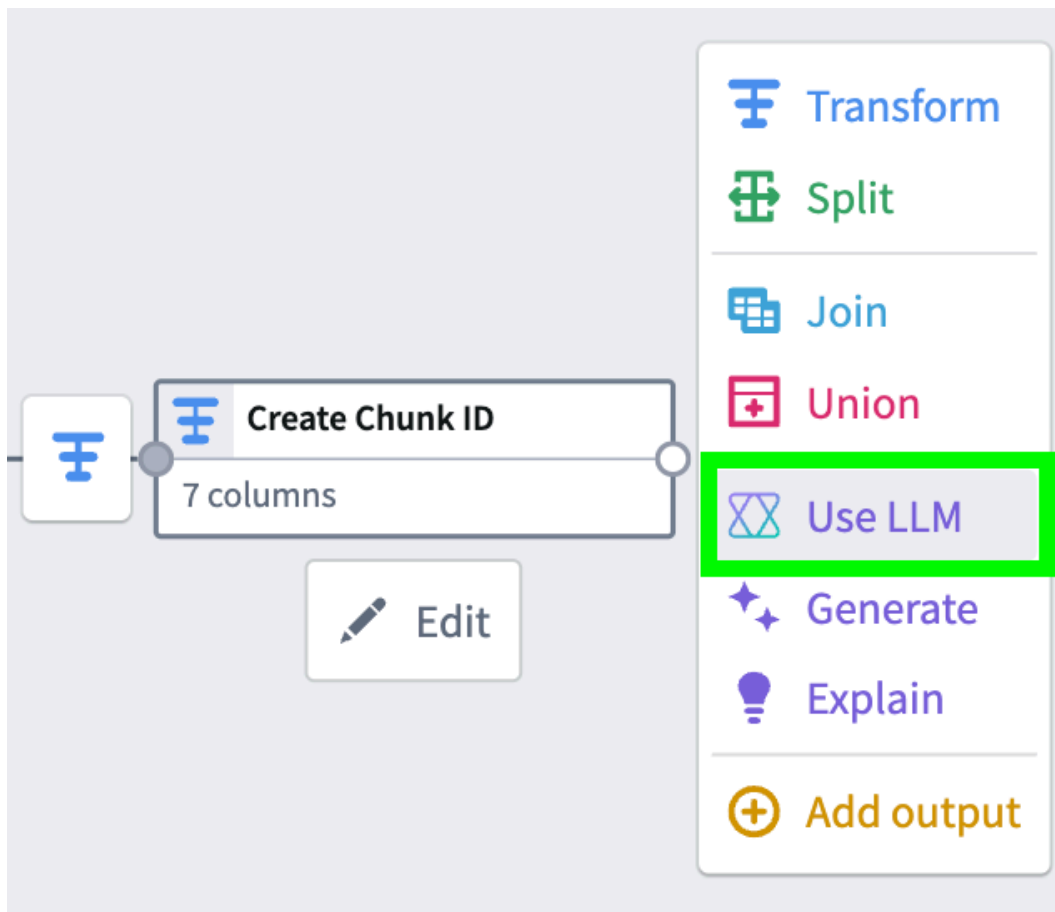
1 input sampled							
	chunkId	chunkNumber	content	pageNumber	path	media_reference	media_item_rid
	String	Integer	String	Integer	String	Media reference	String
1	ri.mio.main.media-ite		1 Morbidity and Mortali		1 Tuberculosis_in_the_U {"mimeType":"applicat	ri.mio.main.media-ite	
2	ri.mio.main.media-ite		2 2020, coinciding with		1 Tuberculosis_in_the_U {"mimeType":"applicat	ri.mio.main.media-ite	
3	ri.mio.main.media-ite		3 an increase of 1,295		1 Tuberculosis_in_the_U {"mimeType":"applicat	ri.mio.main.media-ite	
4	ri.mio.main.media-ite		4 global morbidity and		1 Tuberculosis_in_the_U {"mimeType":"applicat	ri.mio.main.media-ite	
5	ri.mio.main.media-ite		5 (TB) remains one of t		1 Tuberculosis_in_the_U {"mimeType":"applicat	ri.mio.main.media-ite	
6	ri.mio.main.media-ite		6 declined considerably		1 Tuberculosis_in_the_U {"mimeType":"applicat	ri.mio.main.media-ite	
7	ri.mio.main.media-ite		7 Surveillance System (		1 Tuberculosis_in_the_U {"mimeType":"applicat	ri.mio.main.media-ite	
8	ri.mio.main.media-ite		8 TB case counts and ra		1 Tuberculosis_in_the_U {"mimeType":"applicat	ri.mio.main.media-ite	
9	ri.mio.main.media-ite		9 Self-reported race an		1 Tuberculosis_in_the_U {"mimeType":"applicat	ri.mio.main.media-ite	
10	ri.mio.main.media-ite		10 as of February 17, 20		1 Tuberculosis_in_the_U {"mimeType":"applicat	ri.mio.main.media-ite	
11	ri.mio.main.media-ite		11 Estimates were used f		1 Tuberculosis_in_the_U {"mimeType":"applicat	ri.mio.main.media-ite	

Use LLM

Lesson 17 / 51 · [原文链接](#)

1. Select the *Create Chunk ID* transform node
2. Click the *Use LLM* button

Note: If you do not see the 'Use LLM' option in your Foundry instance, you may not have access to AIP features. Contact your organization's Foundry administrator or your Palantir representative for assistance.



1. Click on *Empty prompt*

Create a prompt

Select a template for guided prompting or create one from scratch

 Classification Classify data into given categories	 Sentiment analysis Rate data on a scale based on sentiment
 Summarization Summarize data to a given brevity	 Entity extraction Extract relevant entities given a set of categories
 Translation Translate data to another language	 Empty prompt Create prompt from scratch

1. Keep the default name "*Use LLM*"
2. Add the following prompt to the instructions field:

Your task is to summarize a short text snippet from a scientific article, and extract a list of entities mentioned.

The list of entities could be of any one of the following categories:

- "biological specimens" - For example: neuron, liver, human, etc...
- "harmful agents" - For example: e coli, coronavirus, alcohol, etc...
- "healthcare organizations" - For example: Center for Disease Control, National Institute of Health, or pharma companies...
- "treatments" - For example: paracetamol, penicillin, etc...
- "symptoms" - For example: cough, fever, nausea, etc ...

Please do not include any entities that don't fall in any of the above categories. If the text contains no relevant entities, return an empty list.

Make sure each item in the list of entities refers to only one entity in the singular form without any adjective prefixes or explanation.

So I expect the entity to be "T Cell" instead of "inactive T Cells and B Cells".

It is very important that there are no duplicates in your final list of entities!

3. Type forward-slash "/" in the input data field and pick the `content` column
4. Set the output type to be a struct with one string property called "*summary*" and one array of strings called "*entities*"
5. Rename the output to "*response*"

learning (jsmith) > ... > Document Processing Transform ☆

File Settings Help | 1 Batch

GraphProposalsHistory

Use LLMDescriptionExamples

Instructions

Describe the role the model will play and outline the task it will perform.

Your task is to summarize a short text snippet from a scientific article, and extract a list of entities mentioned.

The list of entities could be of any one of the following categories:

- "biological specimens" - For example: neuron, liver, human, etc...
- "harmful agents" - For example: e coli, coronavirus, alcohol, etc...
- "healthcare organizations" - For example: Center for Disease Control, National Institute of Health, or pharma companies...
- "treatments" - For example: paracetamol, penicillin, etc...
- "symptoms" - For example: cough, fever, nausea, etc ...

Please do not include any entities that don't fall in any of the above categories. If the text contains no relevant entities, return an empty list.

Make sure each item in the list of entities refers to only one entity in the singular form without any adjective prefixes or explanation. So I expect the entity to be "T Cell" instead of "inactive T Cells and B Cells".

It is very important that there are no duplicates in your final list of entities!

Provide input data

* This task will run for each row using the values in the referenced string columns.

content

Reference column values or parameters with the "/" key.

Model

GPT-4o Show configurations ▶

Output type *

Struct

☐ Include errors

Name

summary

...

Type

String

⋮

🗑

Name

entities

...

Type

Array

⋮

🗑

String

▼

+

Add field

Output column *

response

New

Advanced settings

Hide all settings ▶

Skip recomputing rows ⓘ

Input table

Output table

▶ Trial run

Errors

Pipeline Builder will choose a model by default to produce a response. In this example it was GPT-4o. If you wish, you can choose alternative models such as GPT 4 or Mixtral 7B, by clicking on the *Show configurations* button in the *Model* section of the UI.

Instructions

Describe the role the model will play and outline the task it will perform.

Your task is to summarize a short text snippet from a scientific article, and extract a list of entities mentioned.

The list of entities could be of any one of the following categories:

- "biological specimens" - For example: neuron, liver, human, etc...
- "harmful agents" - For example: e coli, coronavirus, alcohol, etc...
- "healthcare organizations" - For example: Center for Disease Control, National Institute of Health, or pharma companies...
- "treatments" - For example: paracetamol, penicillin, etc...
- "symptoms" - For example: cough, fever, nausea, etc ...

Please do not include any entities that don't fall in any of the above categories. If the text contains no relevant entities, return an empty list.

Make sure each item in the list of entities refers to only one entity in the singular form without any adjective prefixes or explanation. So I expect the entity to be "T Cell" instead of "inactive T Cells and B Cells".

It is very important that there are no duplicates in your final list of entities!

Provide input data

★ This task will run for each row using the values in the referenced string columns.

content

Reference column values or parameters with the "/" key.

Model

GPT-4o

Show configurations ▶

Output type ★

Struct ▼

☐ Include errors

Sometimes the LLM may return duplicates in the entities array. If this happens add a deduplication step to make sure the entities are unique.

remove duplicate entitie

Expand all

Applied

Use LLM

8 columns

Get struct field → entities

Array distinct → entities

Search transforms and columns...

Generate

1. Click on *Input table* in the bottom left corner
2. Select a couple of rows by dragging your cursor over the row numbers
3. Click *Use rows for trial run*

Input table

Output table

Trial run

Errors

Display media

Search 7 columns...

1 input sampled

chunkId

String

chunkNumber

Integer

content

String

pageNumber

Integer

path

String

mediaReference

Media reference

mediaItemRid

String

123	chunkNumber	1	ri.mio.main.media-it	0	REVIEW Comparison a	0	Comparison_and_Lesso	{ "mimeType": "applica	ri.r
2	content	2	ri.mio.main.media-it	1	funding have lagged	0	Comparison_and_Lesso	{ "mimeType": "applica	ri.r
3	pageNumber	3	ri.mio.main.media-it	2	diseases (NTDs) and	0	Comparison_and_Lesso	{ "mimeType": "applica	ri.r
4	path	4	ri.mio.main.media-it	3	of NTDs include unde	0	Comparison_and_Lesso	{ "mimeType": "applica	ri.r
5	mediaReference	5	ri.mio.main.media-it	4	as well as zoonotic	0	Comparison_and_Lesso	{ "mimeType": "applica	ri.r
6	mediaItemRid	6	ri.mio.main.media-it	5	a heavy underlying b	0	Comparison_and_Lesso	{ "mimeType": "applica	ri.r
7		7	ri.mio.main.media-it	6	neglect. Multiple in	0	Comparison_and_Lesso	{ "mimeType": "applica	ri.r
8		8	ri.mio.main.media-it	7	10.1371/journal.pgph	0	Comparison_and_Lesso	{ "mimeType": "applica	ri.r
9		8	ri.mio.main.media-it	8	Competing interests:	0	Comparison_and_Lesso	{ "mimeType": "applica	ri.r

1. Click *Run* in the top right corner.

2. Check the example results

3. If you are unhappy with the response, feel free to modify the prompt

The screenshot shows the 'Trial runs' section of the Pipeline Builder. At the top, there are tabs for 'Input table', 'Output table', 'Trial run' (selected), and 'Errors'. Below the tabs, a message states: 'Trial runs Test your transform on a few examples. Select input rows to autogenerate. Approximately 858 LLM tokens were used and 3.7 seconds taken to compute this trial run.' A green box highlights the 'Run' button. Below this, there are two example runs. Each run shows an input 'content' field and the resulting JSON output. The first example input is 'diseases (NTDs) and highlights shared characteristics.' and the output is a JSON object with a 'summary' and 'entities' field. The second example input is 'of NTDs include underlying burden of disease, influen' and the output is a similar JSON object. At the bottom, there are buttons for 'Add trial run' and 'Select from input table'.

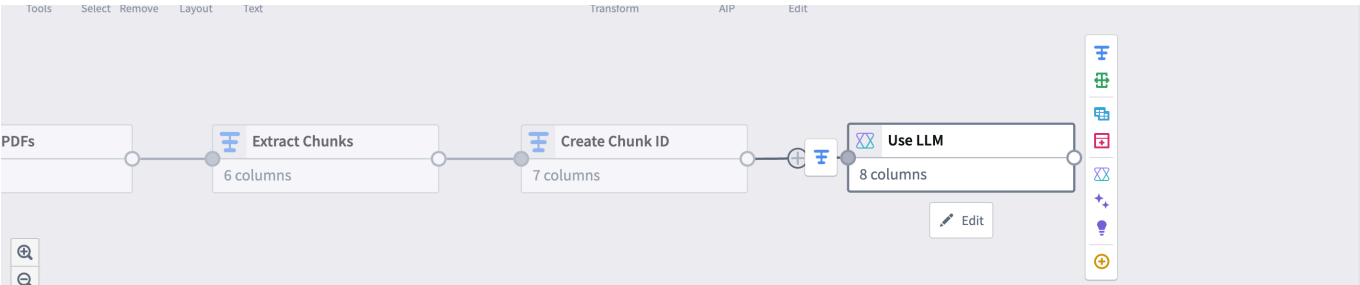
1. Click *Apply*

2. Click *Close*

The screenshot shows the 'Document Processing Transform' configuration screen. At the top, there are tabs for 'Graph', 'Proposals', and 'History'. Below the tabs, there are buttons for 'Use LLM', 'Description', and 'Examples'. A green box highlights the 'Apply' button and another green box highlights the 'Close' button. Below these buttons, there is a large text area for 'Instructions'. The instructions text is: 'Describe the role the model will play and outline the task it will perform. Your task is to summarize a short text snippet from a scientific article, and extract a list of entities mentioned. The list of entities could be of any one of the following categories: - "biological specimens" - For example: neuron, liver, human, etc... - "harmful agents" - For example: e coli, coronavirus, alcohol, etc... - "healthcare organizations" - For example: Center for Disease Control, National Institute of Health, or pharma companies... - "treatments" - For example: paracetamol, penicillin, etc... - "symptoms" - For example: cough, fever, nausea, etc... Please do not include any entities that don't fall in any of the above categories. If the text contains no relevant entities, return an empty list.' Below the instructions, there are tabs for 'Input table', 'Output table', 'Trial run' (selected), and 'Errors'. Below the tabs, a message states: 'Trial runs Test your transform on a few examples. Select input rows to autogenerate. Approximately 858 LLM tokens were used and 3.7 seconds taken to compute this trial run.' A green box highlights the 'Run' button. Below this, there are two example runs. Each run shows an input 'content' field and the resulting JSON output. The first example input is 'diseases (NTDs) and highlights shared characteristics.' and the output is a JSON object with a 'summary' and 'entities' field. The second example input is 'of NTDs include underlying burden of disease, influen' and the output is a similar JSON object. At the bottom, there are buttons for 'Add trial run' and 'Select from input table'.

Pipeline Builder takes your prompt template, resolves the placeholders (adds the actual text from the content column in this case), and sends it to AIP for each row of input. Under the hood, it takes care of optimising performance by issuing parallel calls, while respecting the various models' rate limits.

If you take a look at the *Selection preview* tab at the bottom of your screen, you should see something like this:



Selection preview Preview Suggestions

Use LLM

Truncated preview · ~4.5K total tokens used · ~21 seconds to compute

Previewing 10 rows

8 columns

Search columns...

1 input sampled

	response Struct	chunkId String	chunkNumber Integer	content String	pageNumber Integer	path String	media_reference Media reference	media_ String
1	{"summary":"The text ri.mio.main.media-it		2	2020, coinciding wit	1	Tuberculosis_in_the_	{"mimeType":"applica ri.mio	
2	{"summary":"The repo ri.mio.main.media-it		1	Morbidity and Mortal	1	Tuberculosis_in_the_	{"mimeType":"applica ri.mio	
3	{"summary":"The incr ri.mio.main.media-it		4	global morbidity and	1	Tuberculosis_in_the_	{"mimeType":"applica ri.mio	
4	{"summary":"In 2023, ri.mio.main.media-it		3	an increase of 1,295	1	Tuberculosis_in_the_	{"mimeType":"applica ri.mio	
5	{"summary":"Tubercul ri.mio.main.media-it		5	(TB) remains one of	1	Tuberculosis_in_the_	{"mimeType":"applica ri.mio	
6	{"summary":"The text ri.mio.main.media-it		7	Surveillance System	1	Tuberculosis_in_the_	{"mimeType":"applica ri.mio	
7	{"summary":"The text ri.mio.main.media-it		6	declined considerabl	1	Tuberculosis_in_the_	{"mimeType":"applica ri.mio	
8	{"summary":"The text ri.mio.main.media-it		9	Self-reported race a	1	Tuberculosis_in_the_	{"mimeType":"applica ri.mio	
9	{"summary":"The CDC ri.mio.main.media-it		10	as of February 17, 2	1	Tuberculosis_in_the_	{"mimeType":"applica ri.mio	
10	{"summary":"The arti ri.mio.main.media-it		8	TB case counts and r	1	Tuberculosis_in_the_	{"mimeType":"applica ri.mio	

Display media

Search 8 columns...

response

chunkId

chunkNumber

content

pageNumber

path

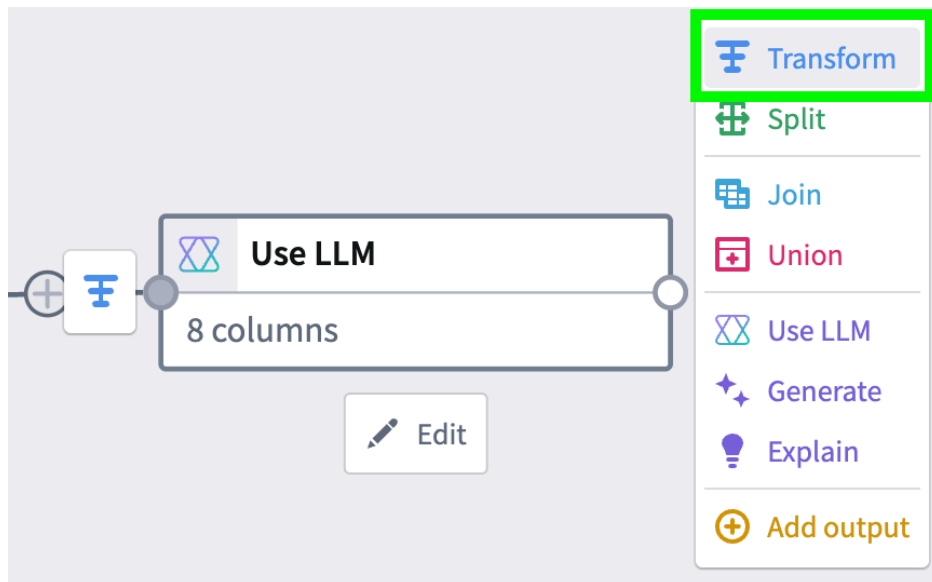
media_reference

media_item_id

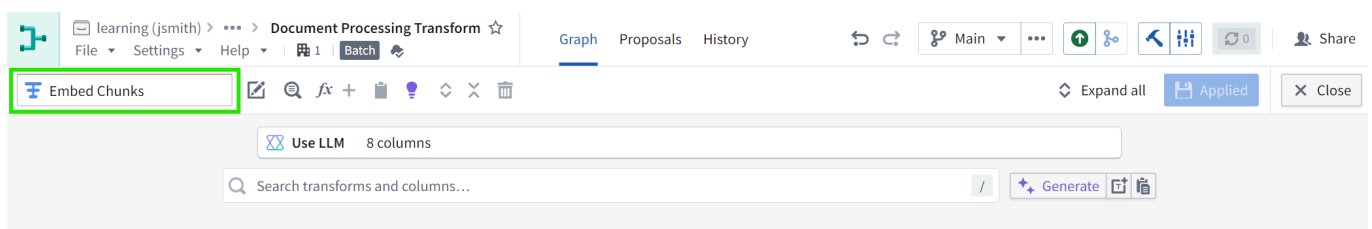
Embedding

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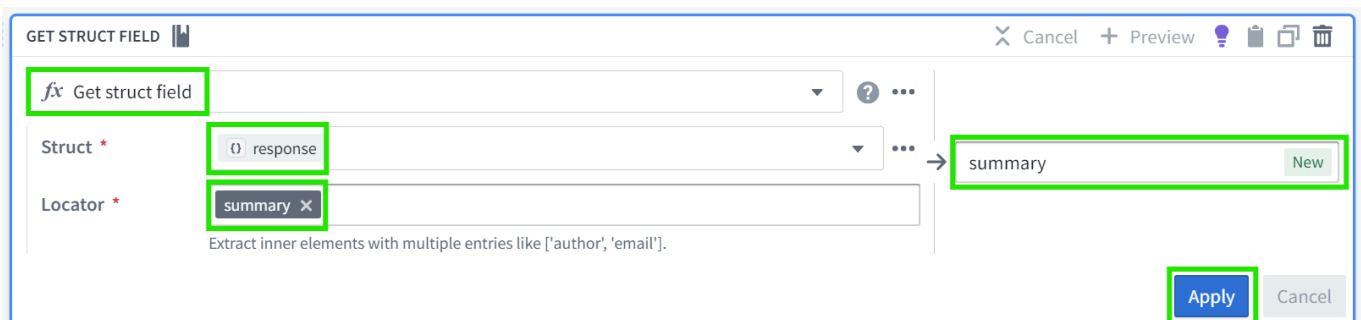
1. Select the *Use LLM* transform node
2. Click the *Transform* button



3. Rename the transform node to "*Embed Chunks*" in the top left



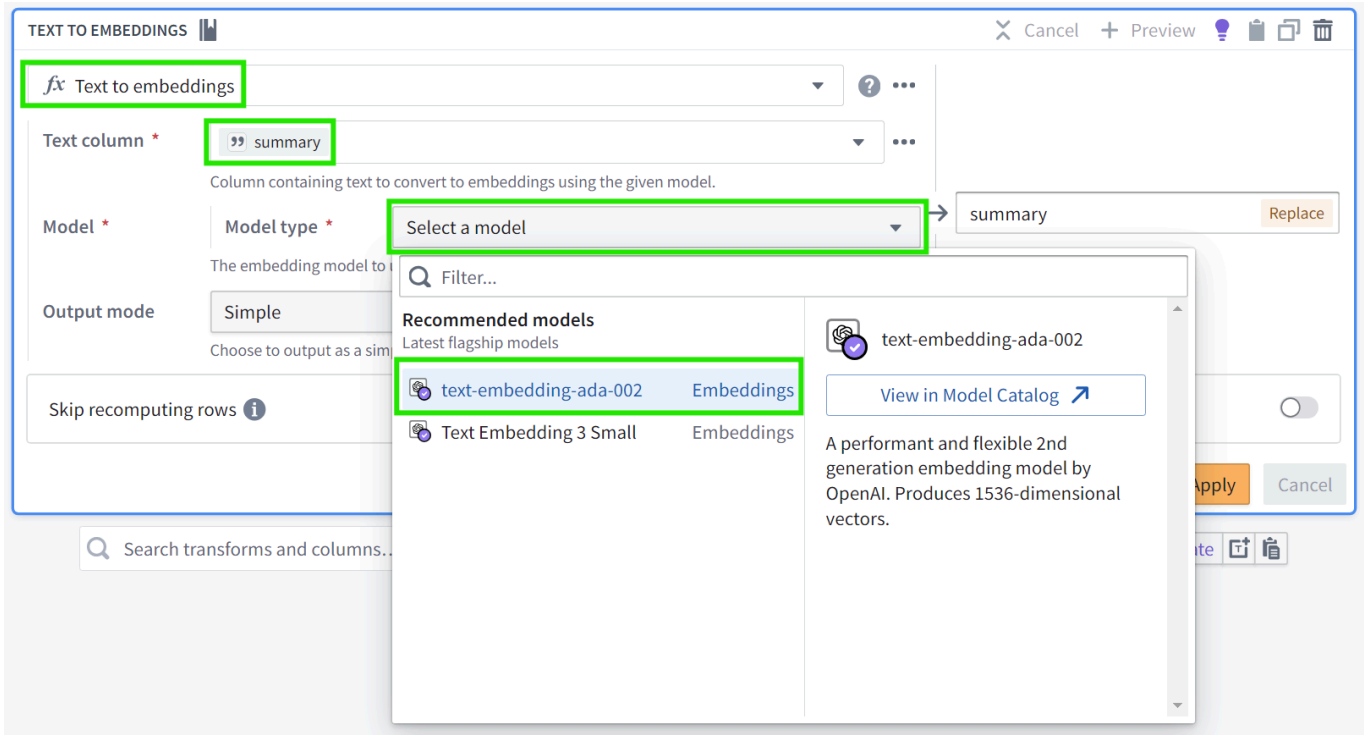
1. Search for and add a *Get struct field* transform block
2. Set the response column as the struct to process
3. Set summary as the property to extract into a new column called "*summary*"
4. Click *Apply*



1. Search for and add the *Text to embeddings* transform block
2. Set *summary* as the column to embed
3. Next to *Model type*, click on *Select a model*

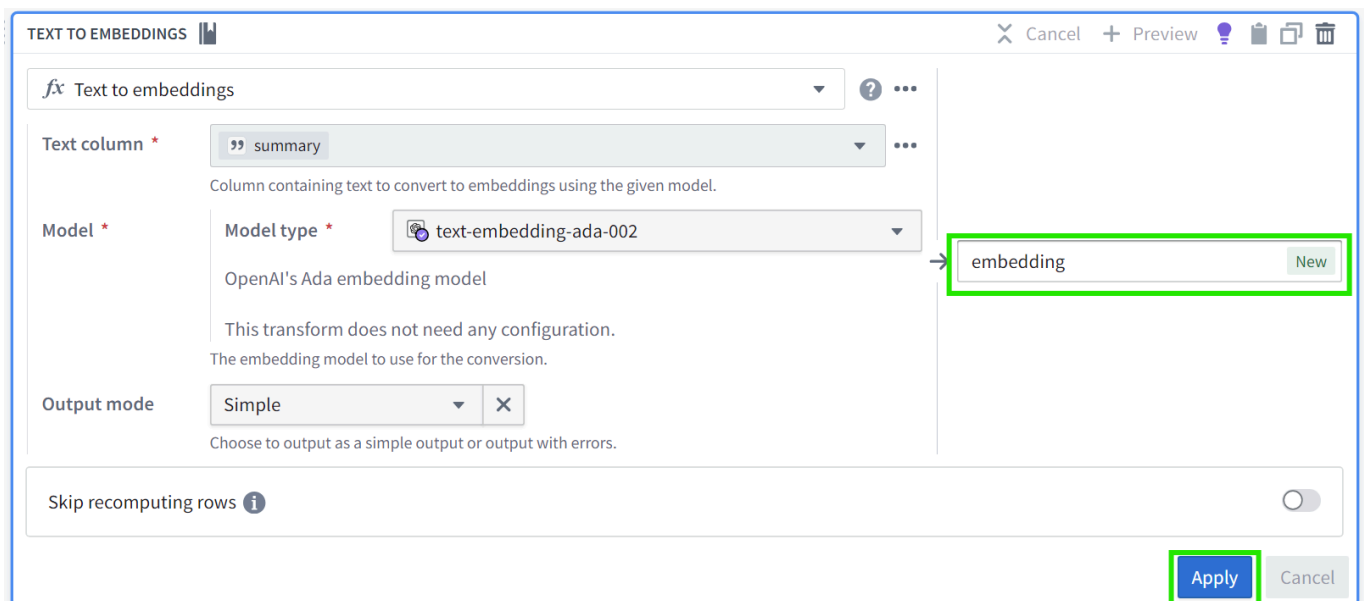
4. Select one of the recommended models

- In this example, we use *text-embedding-ada-002*



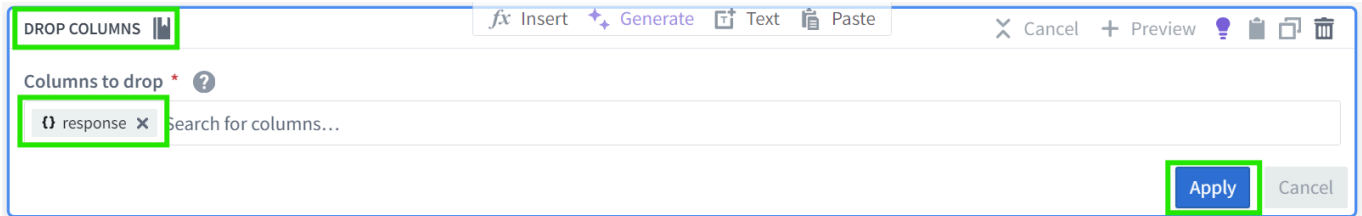
5. Set the output column name to "embedding"

6. Click Apply

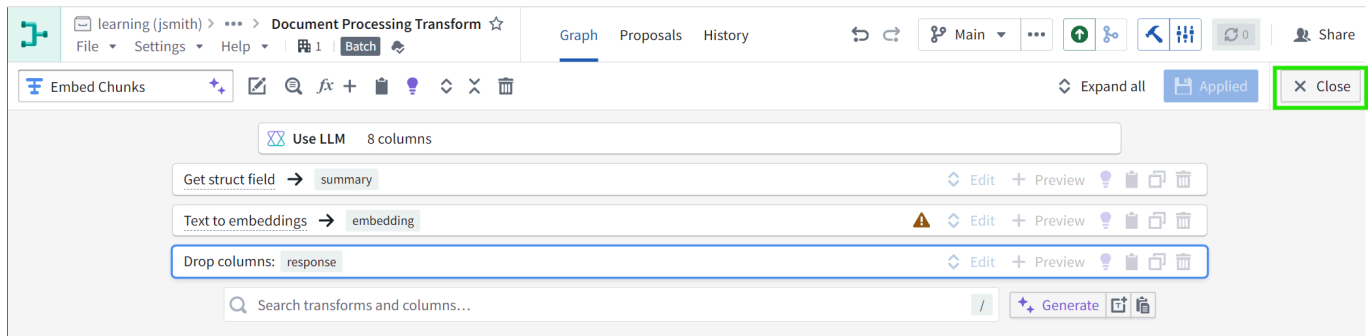


The "Text to Embedding" transform block creates a vector representation of the subject of the input. When compared to other rows, the strings with embeddings nearest each-other will contain similar topics. (In a simplified case, "hot" might be embedded as [0.8], "cold" as [0.2], and "freezing" as [0.1]. "Cold" and "freezing" are similar, therefore their embeddings are near too.)

1. Search for and add the *Drop columns* transform block
2. Select response as the column to drop
3. Click *Apply*



4. Click *Close*

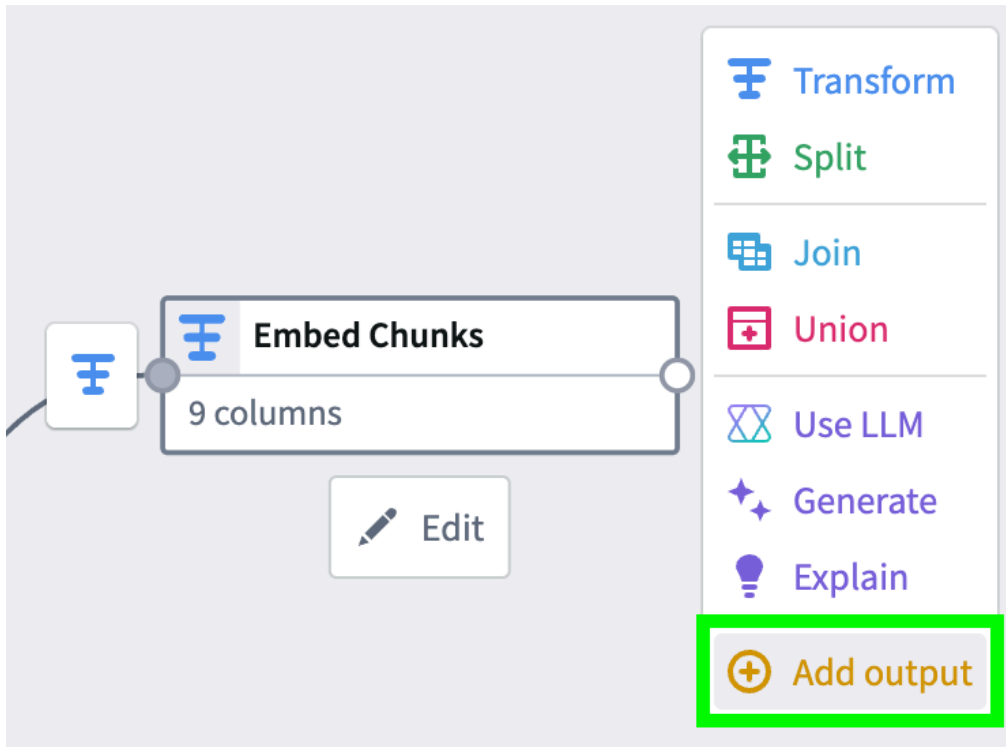


Embedding represents text as a point in a high dimensional space, such that similar concepts end up near each-other. This property can be used to carry out semantic search. Specifically, it allows you to find text chunks that cover similar topics to your question, which can be used to formulate an answer.

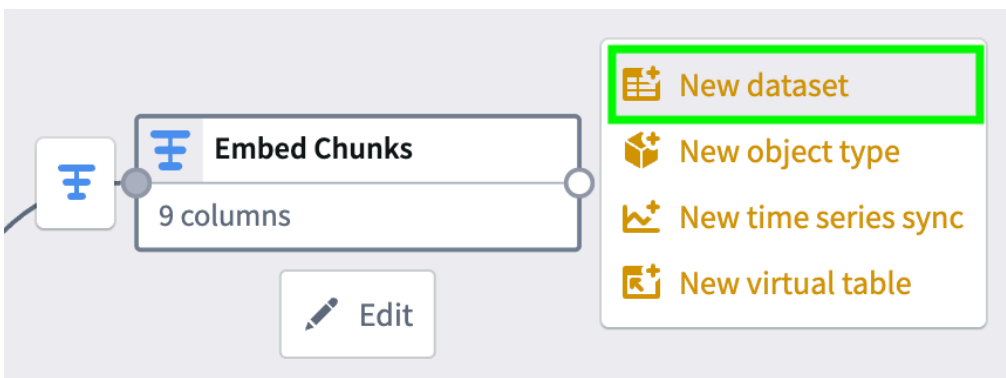
Add Output for Chunks

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1. Select the *Embed Chunks* transform node
2. Click the *Add output* button



3. Click the *New dataset* button



1. Rename the output dataset to "*Chunks*" in the configuration panel on the right

[← Back to outputs](#)

Chunks

Output will be created after first build

✓ 9/9 columns mapped

Configure expectations

Configure write mode

Configure write format

Configure markings

☐ Show errors only

	[1.0]	embedding	✓	×
	”	summary	✓	×
	”	chunkId	✓	×
	123	chunkNumber	✓	×
	”	content	✓	×
	123	pageNumber	✓	×
	”	path	✓	×
		mediaReference	✓	×
	”	mediaItemRid	✓	×

Use upstream schema

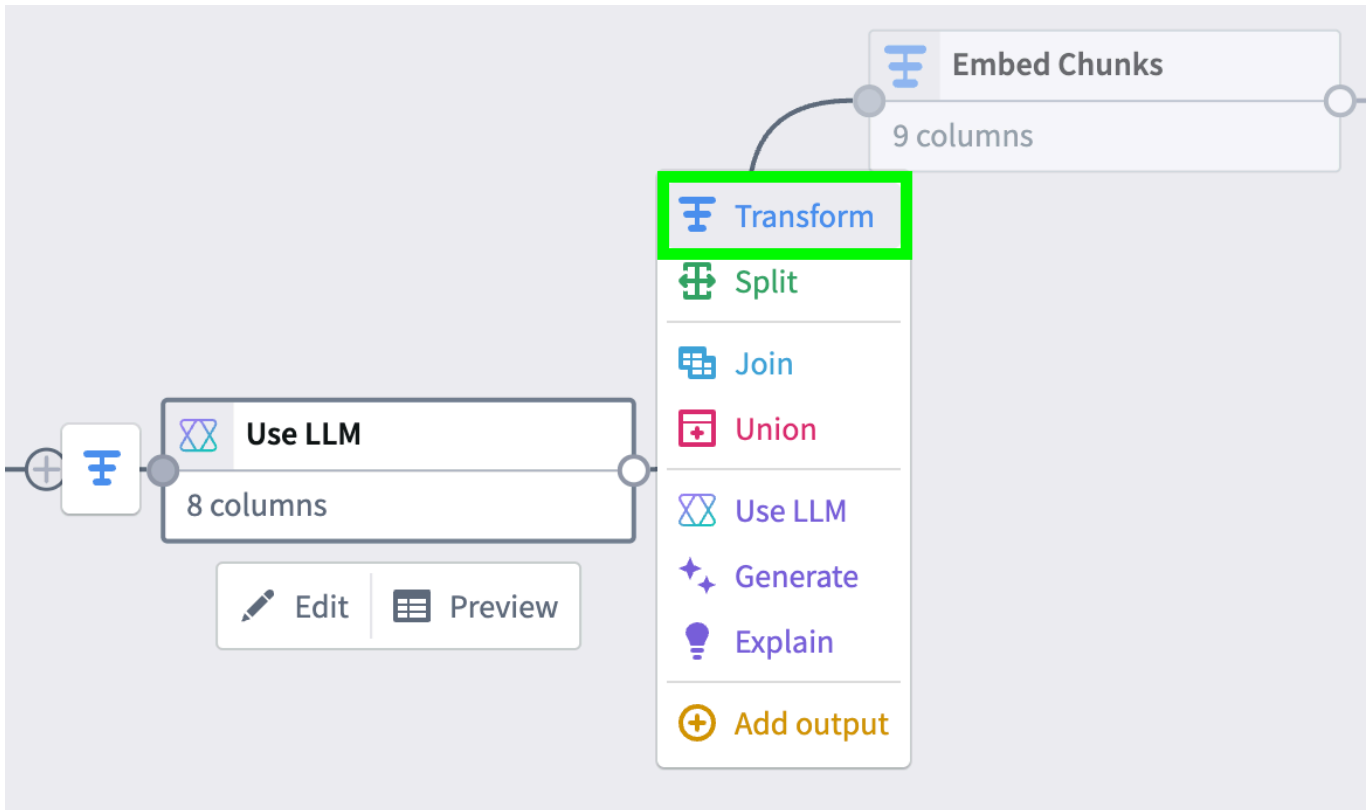
Add column

The output configuration tab also allows you to set data expectations, such as "no nulls allowed", "column value must be between N and K", or "column values must be unique". These allow you to uphold high data quality and build robust, production pipelines.

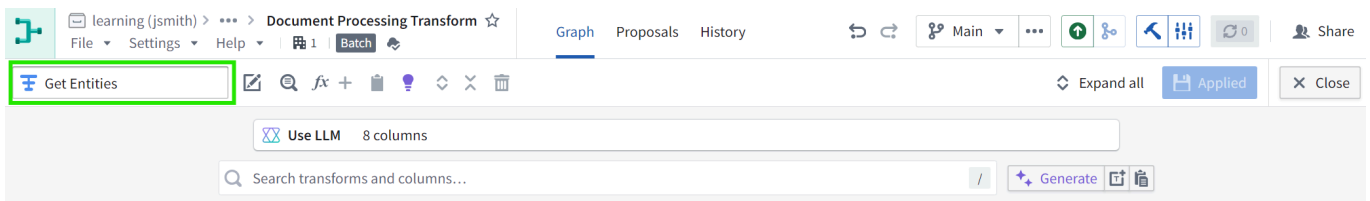
Get Entities

Lesson 20 / 51 · [原文链接](#)

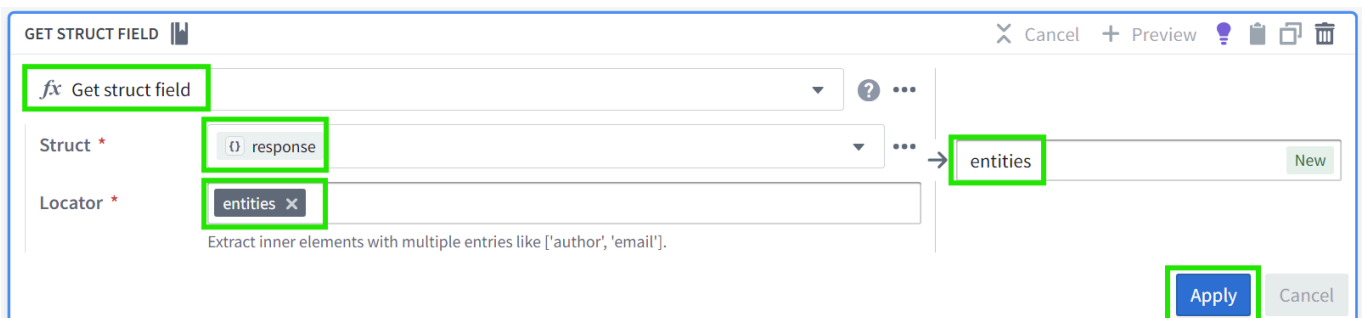
1. Select the *Use LLM* transform node again
2. Click the *Transform* button to start a second transform path from the node



3. Rename the transform node to "*Get Entities*" in the top left



1. Search for and add a *Get struct field* transform block
2. Set the response column as the struct to process
3. Set entities as the property to extract into a new column called "*entities*"
4. Click *Apply*



1. Search for and add an *Explode array* transform block
2. Set the `entities` column as the array to explode
3. Set the output column name to "`entityName`"
4. Click *Apply*

The screenshot shows the 'EXPLODE ARRAY' configuration window. The title bar says 'EXPLODE ARRAY'. At the top right are buttons for 'Cancel', '+ Preview', and icons for help, save, and delete. The main area has a dropdown menu set to '.fx Explode array'. Below it, the 'Expression' field is set to 'entities' with a tooltip that says 'The array to explode.' To the right, the output column name is 'entityName' with a 'New' button next to it. At the bottom left is a toggle for 'Keep empty / null arrays' which is currently off. At the bottom right are 'Apply' and 'Cancel' buttons.

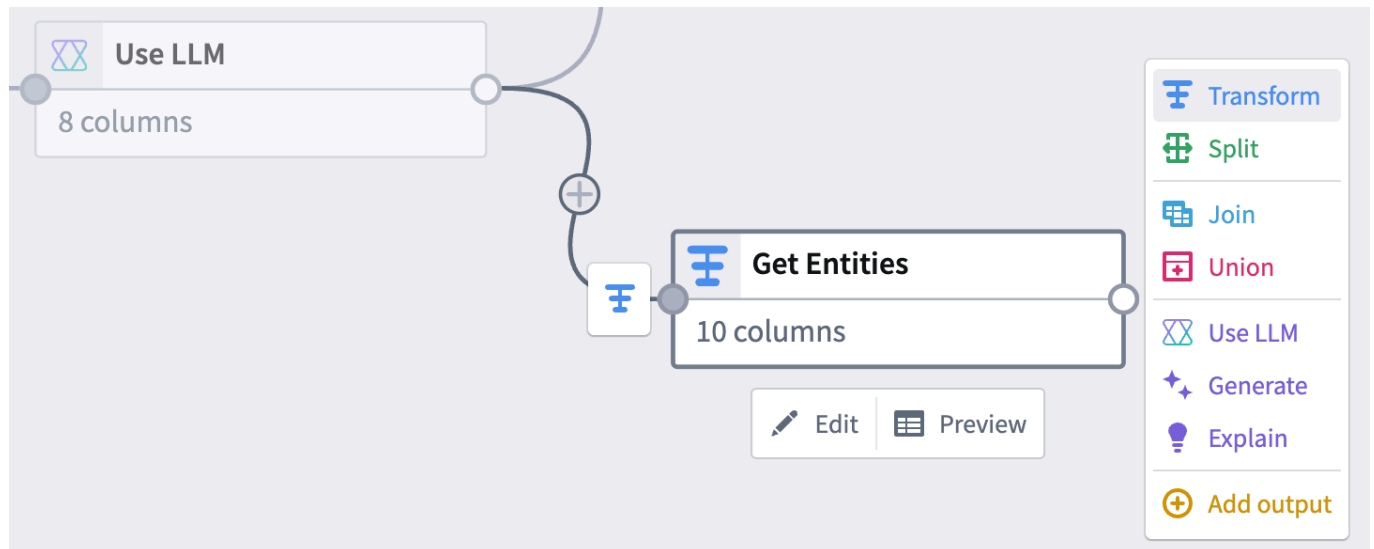
5. Click *Close*

The screenshot shows the 'Document Processing Transform' interface. The top bar includes a menu (File, Settings, Help), a 'Batch' button, and tabs for 'Graph', 'Proposals', and 'History'. On the right are buttons for 'Main', 'Expand all', 'Applied', and 'Close' (which is highlighted with a green box). The main workspace shows a transform graph with two blocks: 'Get struct field' pointing to 'entities' and 'Explode array' pointing to 'entityname'. A search bar at the bottom says 'Search transforms and columns...'.

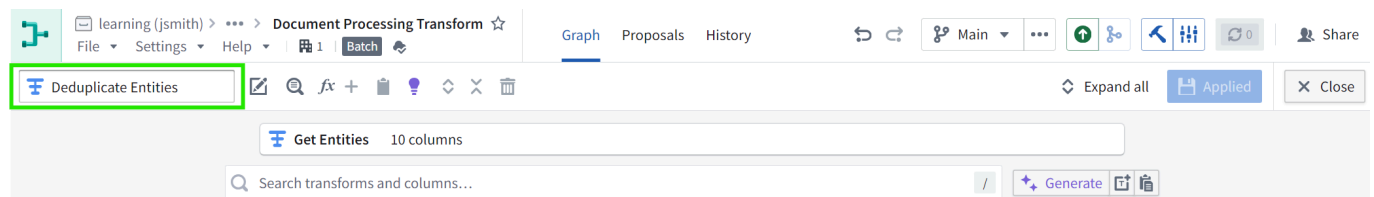
Resolve Entities

Lesson 21 / 51 · [原文链接](#)

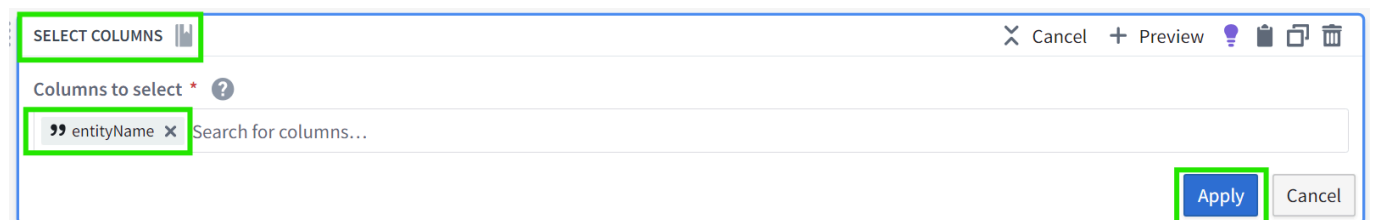
1. Select the *Get Entities* transform node
2. Click the *Transform* button



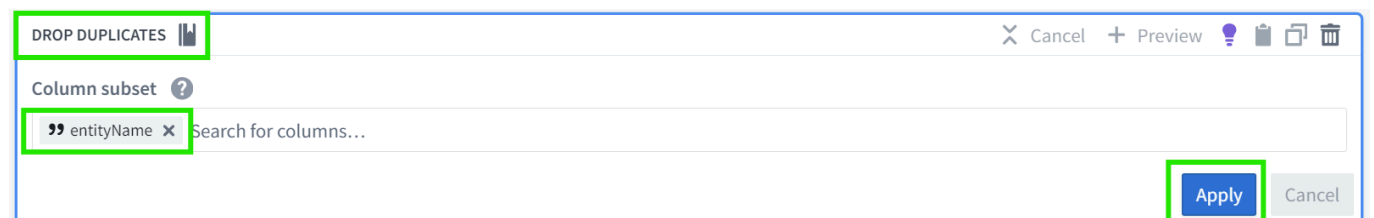
3. Rename the transform node to "*Deduplicate Entities*" in the top left



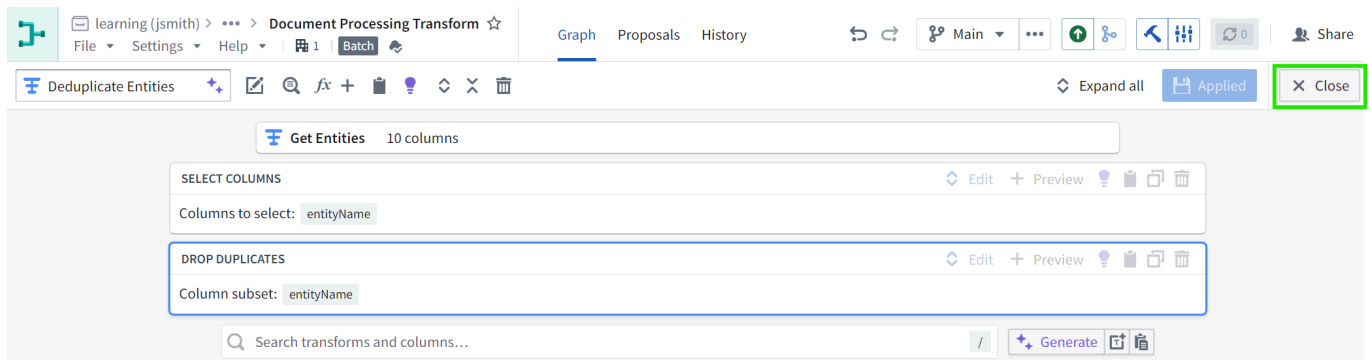
1. Search for and add a *Select columns* transform block
2. Select the *entityName* column to keep
3. Click *Apply*



1. Search for and add a *Drop duplicates* transform block
2. Select *entityName* to be the basis of de-duplication
3. Click *Apply*



4. Click *Close*

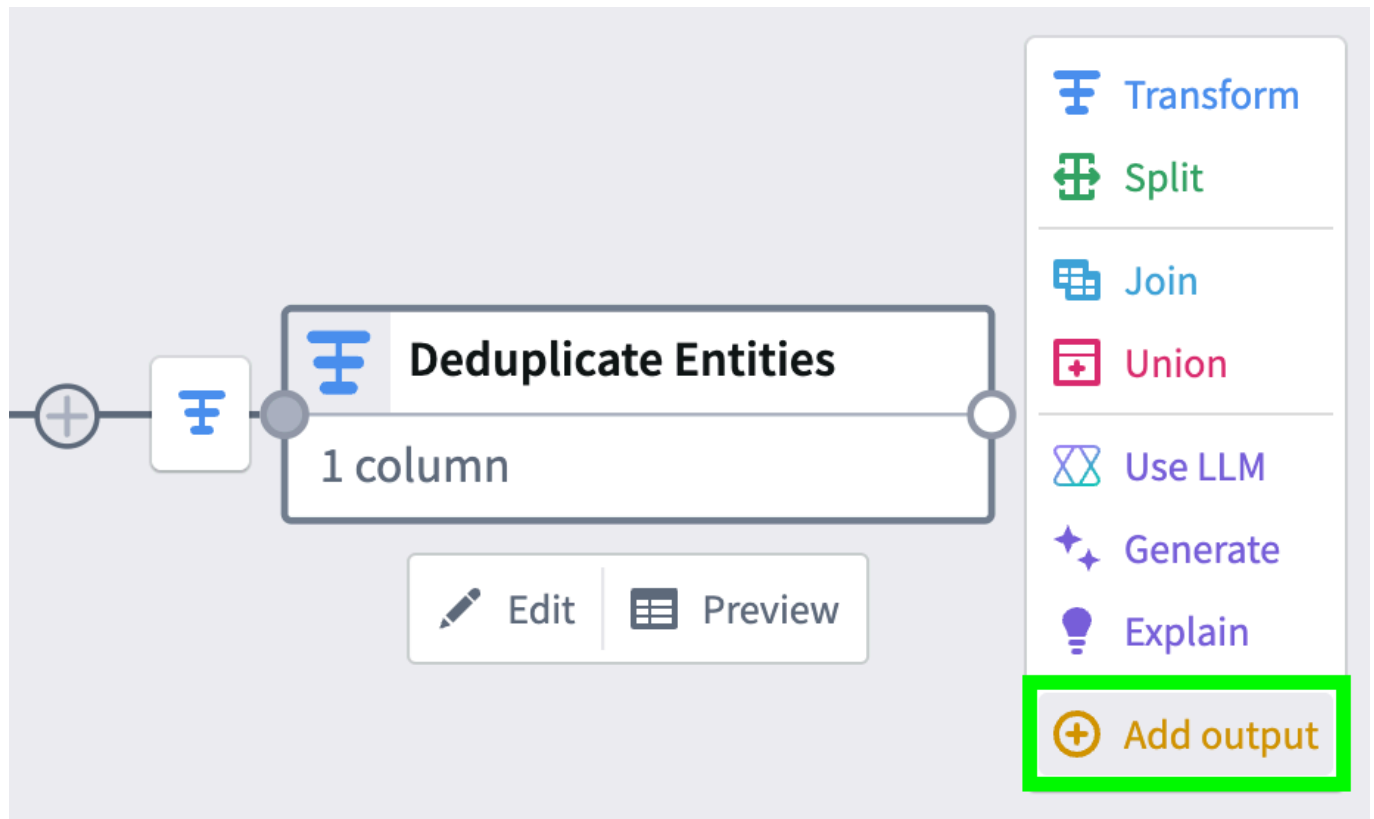


The entities extracted from the document chunks allow you to discover and understand relationships between the most significant concepts in your data asset.

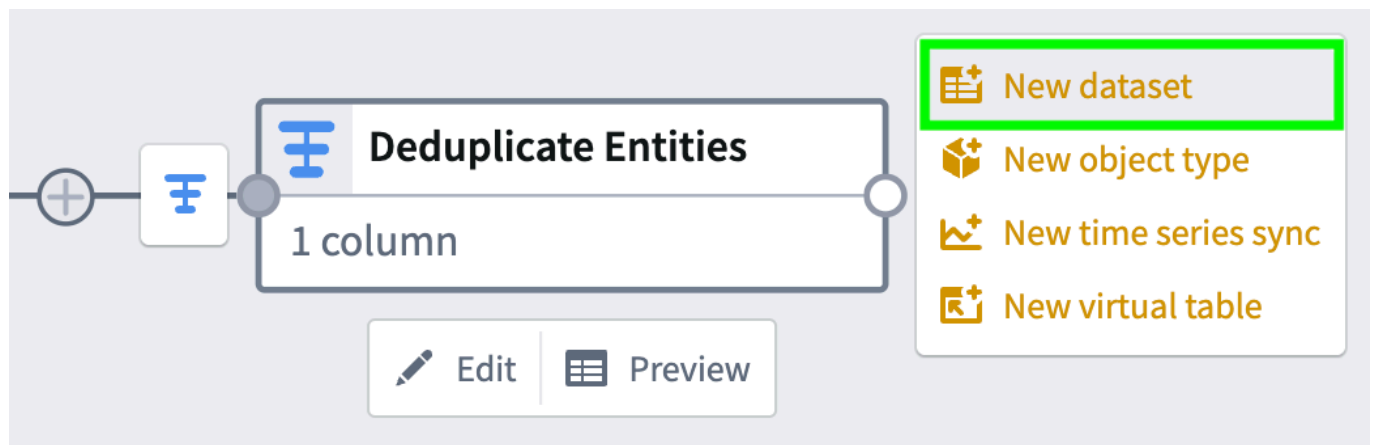
Add Output for Entities

Lesson 22 / 51 · [原文链接](#)

1. Select the *Deduplicate Entities* transform node
2. Click the *Add output* button



3. Click the *New dataset* button



1. Rename the output dataset to "*Entities*" in the configuration panel on the right

← Back to outputs



Entities

Output will be created after first build

✓ 1/1 column mapped



Configure expectations



Configure write mode



Configure write format



Configure markings



Search columns...



Show errors
only



entityName



Use upstream schema



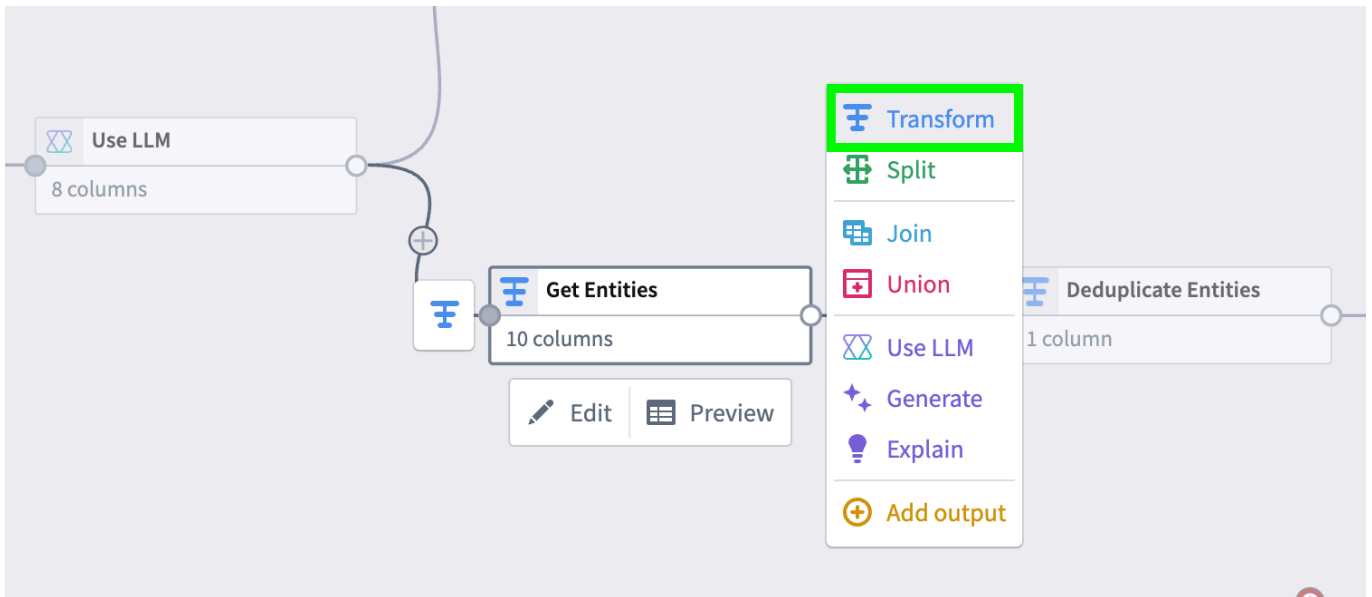
Add column

Define Join Table

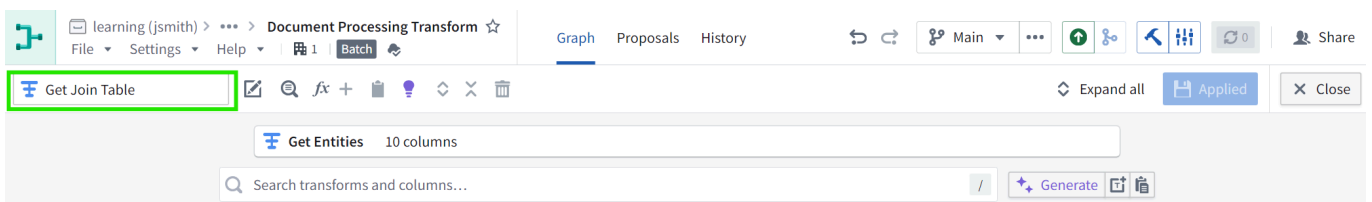
Lesson 23 / 51 · [原文链接](#)

The relationship between Chunks and Entities is many-to-many. That is, one Chunk can contain many Entities, and an Entity can be present in many Chunks. To map this relationship, a column is not enough; we need a join table.

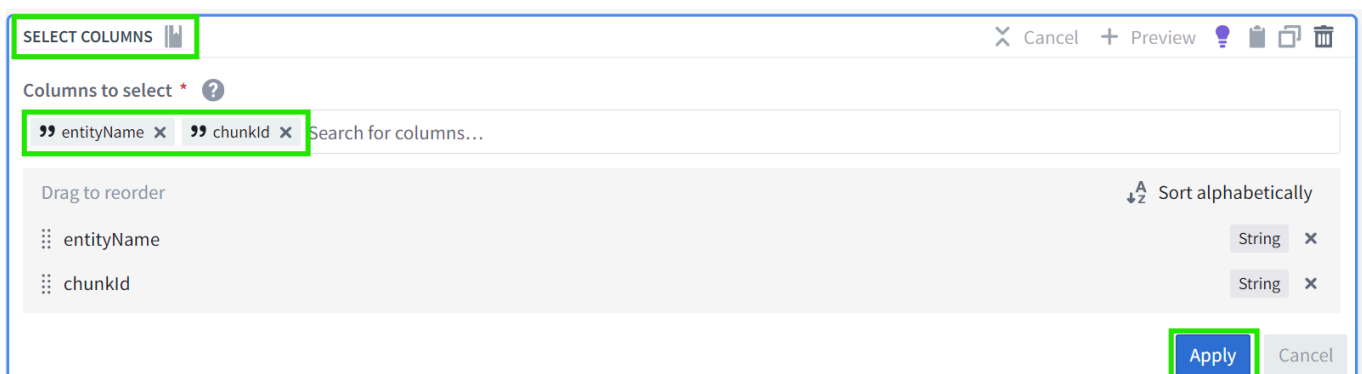
1. Select the *Get Entities* transform node
2. Click the *Transform* button



3. Rename the transform node to "Get Join Table" in the top left



1. Search for and add a *Select columns* transform block
2. Select the *entityName* and *chunkId* columns to keep
3. Click *Apply*



4. Click *Close*

learning (jsmith) > ... > Document Processing Transform ☆

File ▾ Settings ▾ Help ▾

1 | Batch

GraphProposalsHistory

↶↷

Main ▾ ...

↶↷

0

Share

Get Join Table

Expand all

Applied

Close

Get Entities10 columns

SELECT COLUMNS

Columns to select: entityName and chunkId

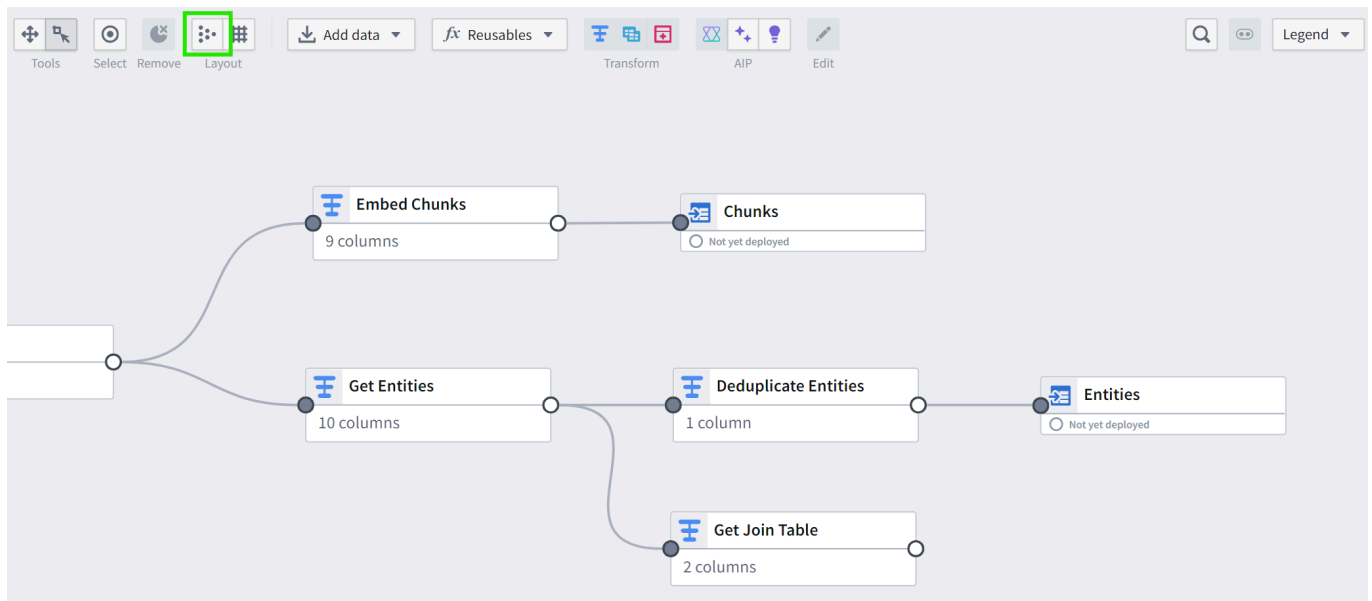
Search transforms and columns...

Generate

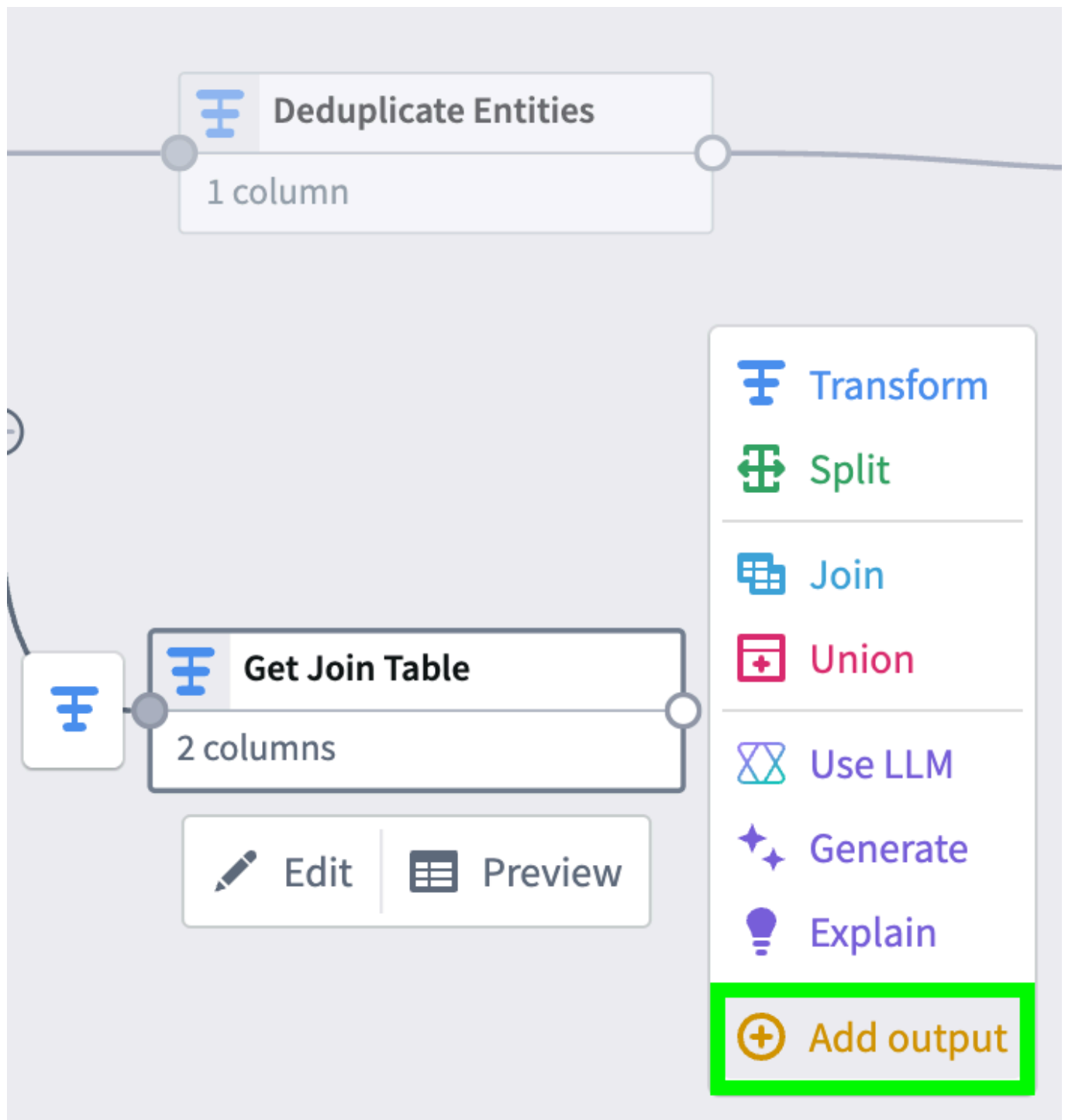
Add Output for the Join Table

Lesson 24 / 51 · [原文链接](#)

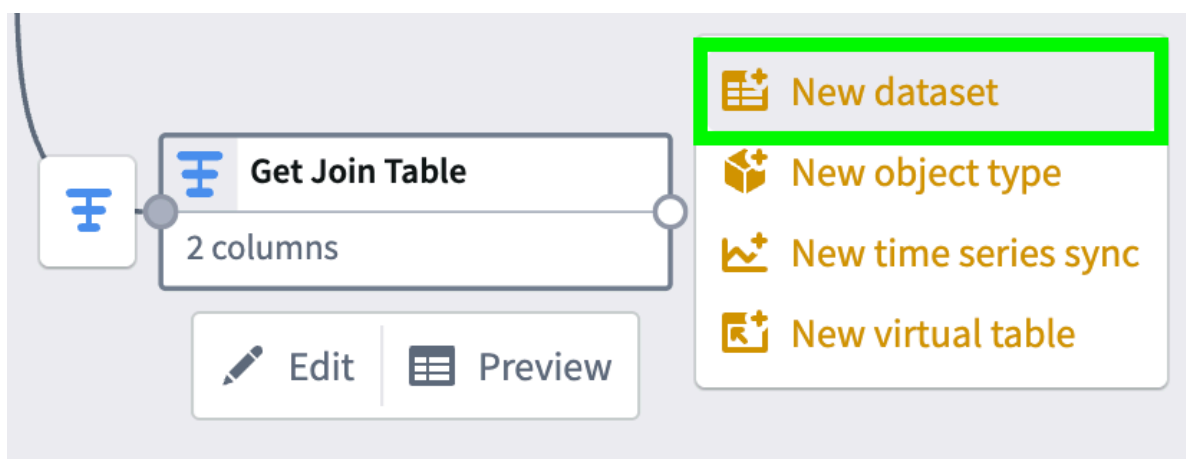
1. Select the *Layout nodes* button in the top left to automatically clean-up your graph



2. Select the *Get Join Table* transform node
3. Click the *Add output* button



4. Click the *New dataset* button



1. Rename the output dataset to "Join Table" in the configuration panel on the right

[← Back to outputs](#)  



Output will be created after first build

 2/2 columns mapped

 Configure expectations

 Configure write mode

 Configure write format

 Configure markings

☐ Show errors only

















 Use upstream schema

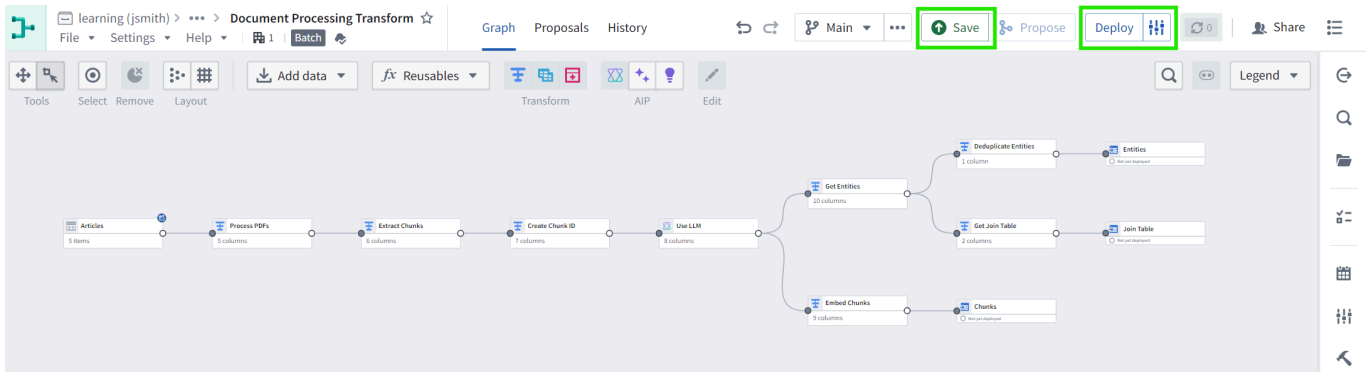
 Add column

A single chunk might mention multiple entities, and a single entity might be mentioned in multiple chunks. Therefore, this connection cannot be described using simple primary and foreign key columns on the source datasets. Instead, we create a list of all chunk-entity pairs, called a join table.

Deploy Pipeline

Lesson 25 / 51 · [原文链接](#)

1. Click *Save* in the top right corner
2. Click *Deploy* in the top right corner



3. In *Select outputs to build*, make sure all datasets are selected
4. Click *Deploy pipeline* in the bottom right corner

Deploy this pipeline

Update pipeline logic and build target outputs.

Last deployment

None

Deploy settings

Errors

Select outputs to build

Deploying the pipeline will update logic for all outputs, but only the selected outputs will be built.

▼

☒ All

▼

☒ Single jobs

☒ ☐  Chunks

☒ ☐  Entities

☒ ☐  Join Table

 View changes

 Deploy pipeline

1. Click *Import resources* in the warning pop-up



2. Click *Add references* in the import pop-up

To use resources from other projects, you must first add a reference to them in this project. All marking requirements from the original project will still apply. You can view and edit all references through the [main Project page](#).

Resources (2)

 GPT-4o

 text-embedding-ada-002

Cancel

 Add references

3. You may need to click *Deploy pipeline* once more

You need to add the outputs of the pipeline to the same job grouping. Follow the steps below to do this:

1. Click *Deploy Settings* next to the *Errors* tab

Deploy this pipeline

Update pipeline logic and build target outputs.

Last deployment

None

Deploy settings

 Errors 1

 All outputs that share a cached LLM upstream must be placed in the same non-default job grouping. Cache: 7ac14a9d-a283-42fc-af3e-c24251d690ba

2. Click *Configure*

Deploy this pipeline

Update pipeline logic and build target outputs.

Last deployment Succeeded May 28, 2025, 3:44 PM

[Deploy settings](#) ⚠ Warnings 2

Compute profile [Configure](#)

Default Extra small - warm pool

Select outputs to build

Deploying the pipeline will update logic for all outputs, but only the selected outputs will be built.

▼ ☒ All

▼ ☒ Single jobs

- ☒ ☐  Chunks
- ☒ ☐  Entities
- ☒ ☐  Join Table







3. Mouse over the *Chunks* output under *Single jobs* to reveal the *Move to group...* button and then click it

Build settings

Apply

Default compute profile ?

Extra small - war... ▼

Custom job grouping 0 >

Default job grouping 3 ▼

Single jobs

☐

Chunks

Move to group...

☐

Entities

☐

Join Table

Advanced configuration ▼

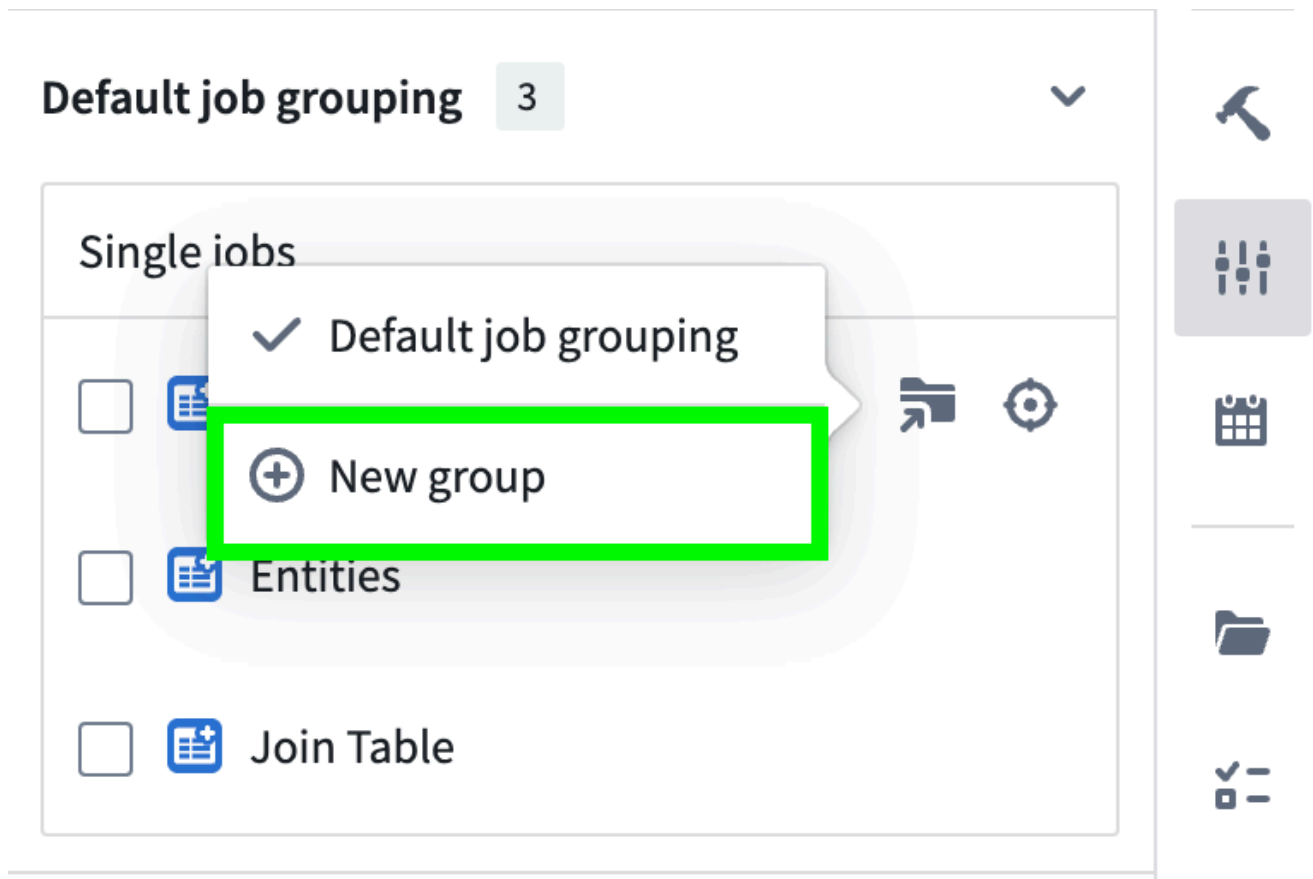
Enable column stats on dataset outputs ⓘ

True

False

No value

4. Click *New Group*, this will move the *Chunks* output to a custom job grouping called *Group 1*



5. Repeat this process for the *Entities* and *Join Table* outputs, however, instead of adding them to a new group, add them to *Group 1*

Build settings

Apply

Default compute profile ?

Extra small - war... ▼

Custom job grouping 1 ▼



Group 1

Add profile



Chunks

Default job grouping 2 ▼

Single



Default job grouping



Group 1



New group

Advanced configuration ▼

Enable column stats on dataset outputs i

True

False

No value

6. After all the outputs have been added to *Group 1*, click *Apply* in the top right of the panel and then *Save*

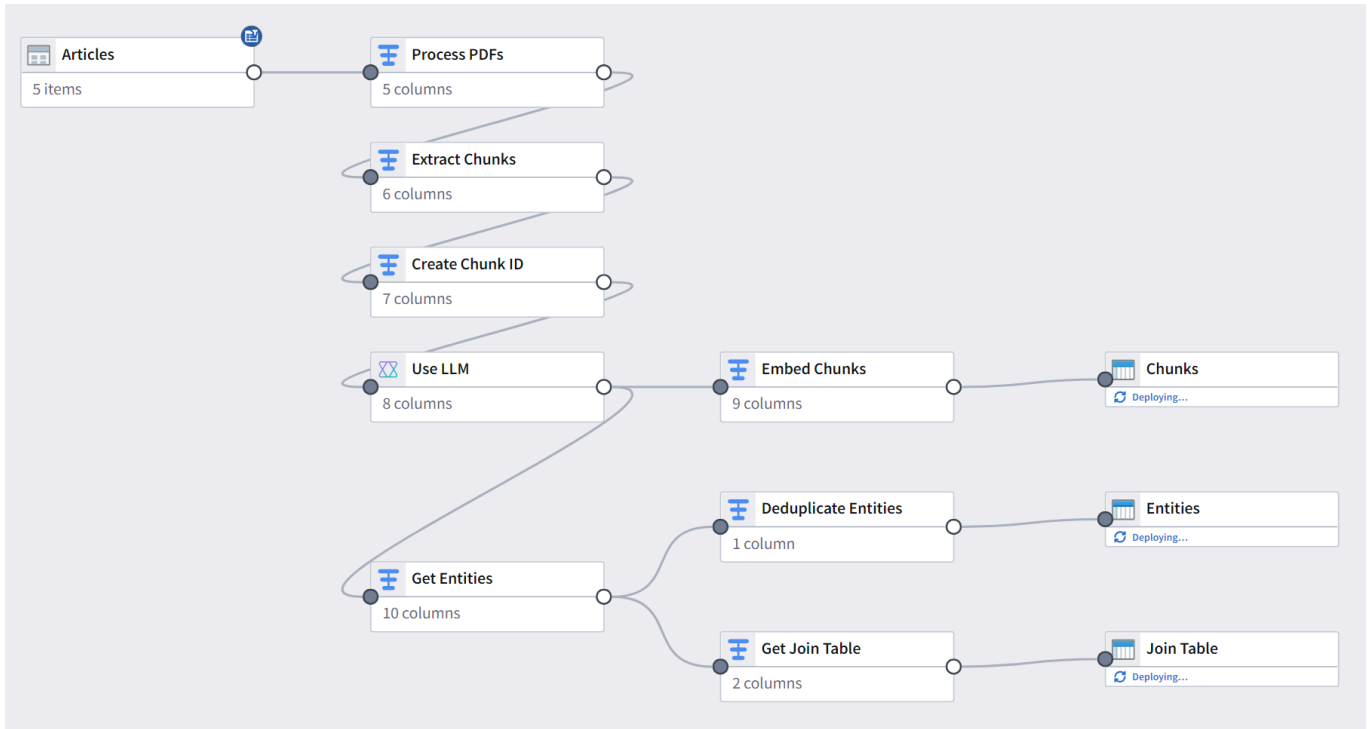
The screenshot displays the Databricks workspace interface. At the top, a navigation bar includes buttons for 'Main', 'Save' (highlighted with a green box), 'Propose', 'Deploy', and 'Share'. Below this, a toolbar shows 'Transform' and 'AIP' tabs. The main workspace area contains a pipeline diagram with three stages: 'Chunks', 'Deduplicate Entities', and 'Get Join Table'. Each stage has a red 'Not yet deployed' status indicator. A legend on the right indicates 'Group 1 (3)'. On the far right, the 'Build settings' panel is open, showing 'Default compute profile' set to 'Extra small - war...', 'Custom job grouping' set to '1', and 'Default job grouping' set to '0'. The 'Advanced configuration' section is also visible, with 'Enable column stats on dataset outputs' set to 'False'. The 'Apply' button in the top right of the 'Build settings' panel is highlighted with a green box.

7. You should now be able to *Deploy pipeline* successfully

Summary

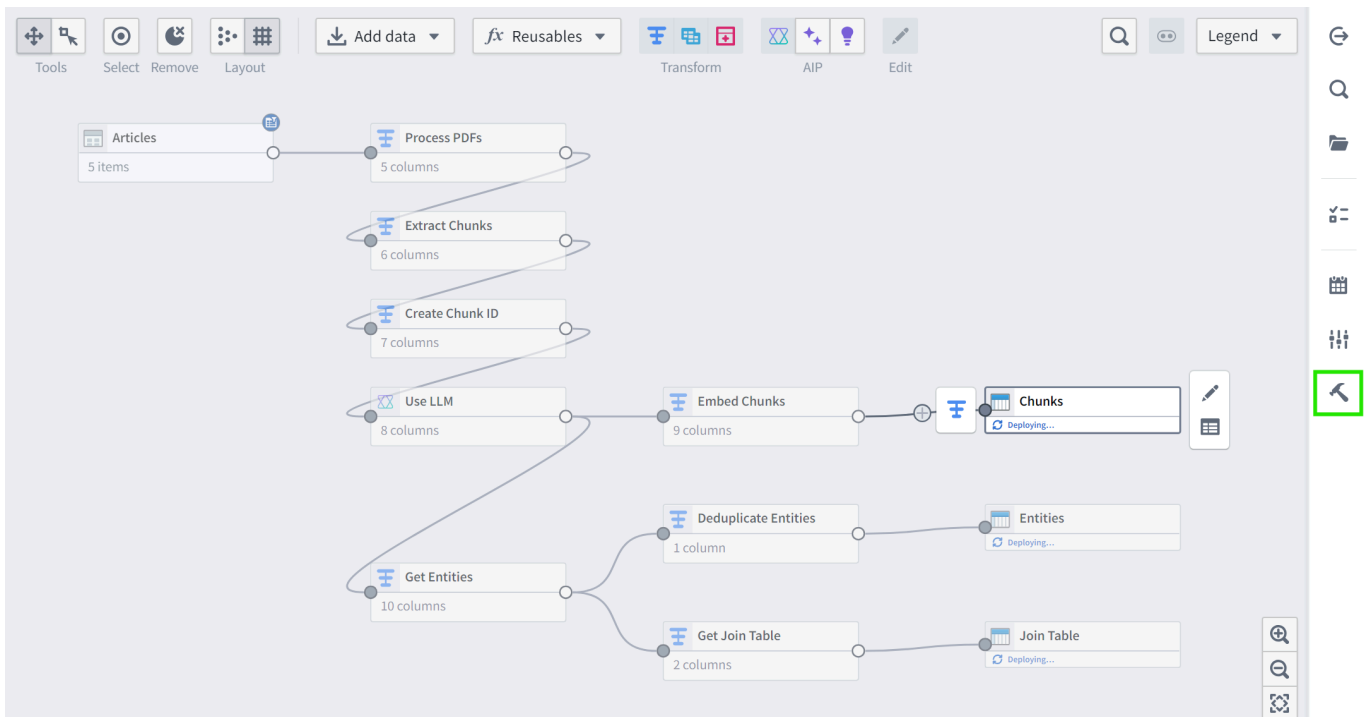
Lesson 26 / 51 · [原文链接](#)

You now have a data transformation taking PDF documents as an input and turning them into chunks, which represent nuggets of information. You are also extracting the most significant entities from these chunks which will allow you to discover relationships between chunks in later steps.



For now, you should take a look at your build's progress:

1. Click on the *Deploy Pipeline* icon in the right-side bar



2. Click *View deployment history* in the right panel that opened up

Running task 1/3

Start time

Nov 14, 2024, 4:18 PM

Duration

4m 23s

Deployment tasks

✓ Deployed logic for 3 targets

• Building outputs (0/3)

Build report

Cancel deployment

View deployment history















3. Click *Build report* in the row corresponding to the current build

Viewing version history for branch: Main
Filter

Deployment
4 minutes ago

✓ Deployed logic for 3 targets

- Building outputs (0/3) Build report

3 saved versions

Your datasets should build in a little over 5 minutes.

Build progress

Gantt chart
Job status
Dataset path...

Queued
Waiting
Running
Succeeded
Failed
Canceled

Job status
Progress details
Critical path

Datasets	Start time	Duration	Status
Join Table	Today at 4:18 PM	5m 40s	Running stage 2 of 3...
Entities	Today at 4:18 PM	5m 40s	Running stage 2 of 3...
Chunks	Today at 4:18 PM	5m 40s	Running stage 2 of 3...

Pipeline Builder is Foundry's no-code solution for authoring data transformations. It comes with hundreds of optimised data operations out of the box, as well as support for core software engineering concepts such as version control, scheduling, and incremental execution. It is well suited for the majority of data-pipeline building tasks, and provides a boilerplate free entry point when rapid development is priority over complex technical configuration.

To learn more about Pipeline Builder you might follow this course up with the [Pipeline Deep Dive](#) course, or by taking a look at the [documentation](#).

Foundry also provides a high-code transform authoring environment called Code Repositories. It allows you to leverage the fine controls of python and java to create performant transforms, and rely on popular libraries for specialized functionality.

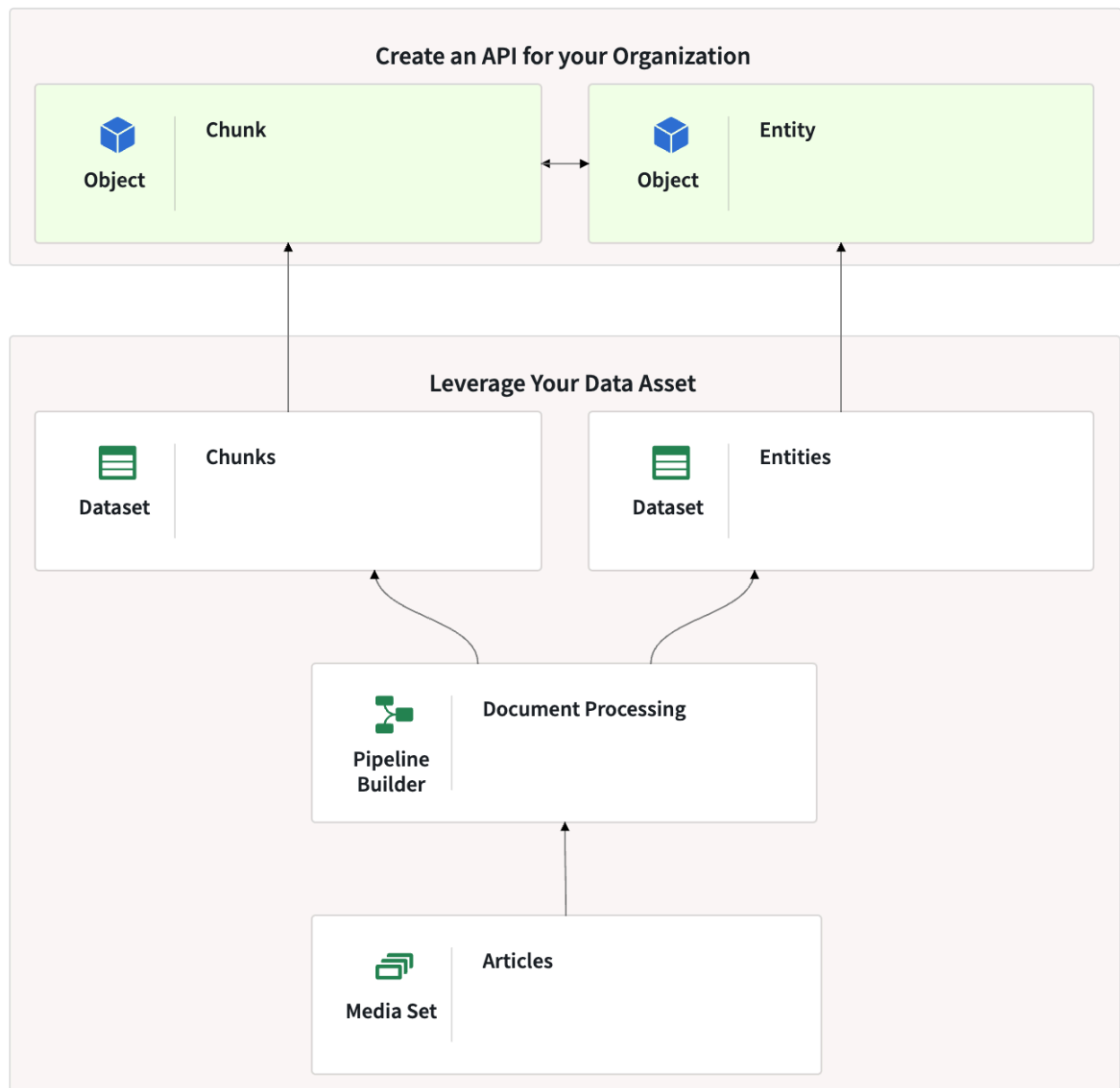
To learn more about Transform authoring in Code Repositories, take a look at the [documentation](#).

Introduction

Lesson 27 / 51 · [原文链接](#)

An Ontology directly describes your data, representing your data assets as well as their relationships. It allows you to define the ways in which operational users can view and interact with your data assets.

In this section you will create Ontology objects, backed by the chunks and entities datasets created in the previous step. These will underpin all interactive, user-facing features that you'll implement in the following sections.



To get started, please open the Ontology Manager app.

Q ontology manager

Clear

X

Q All search results for 'ontology manager'

to see all results

JUMP TO

Highlighted samples from your results

Ontology Manager

Platform app

Apps

HOTKEYS

↑ + J Open Quicksearch

↑ / ↓ Move in list

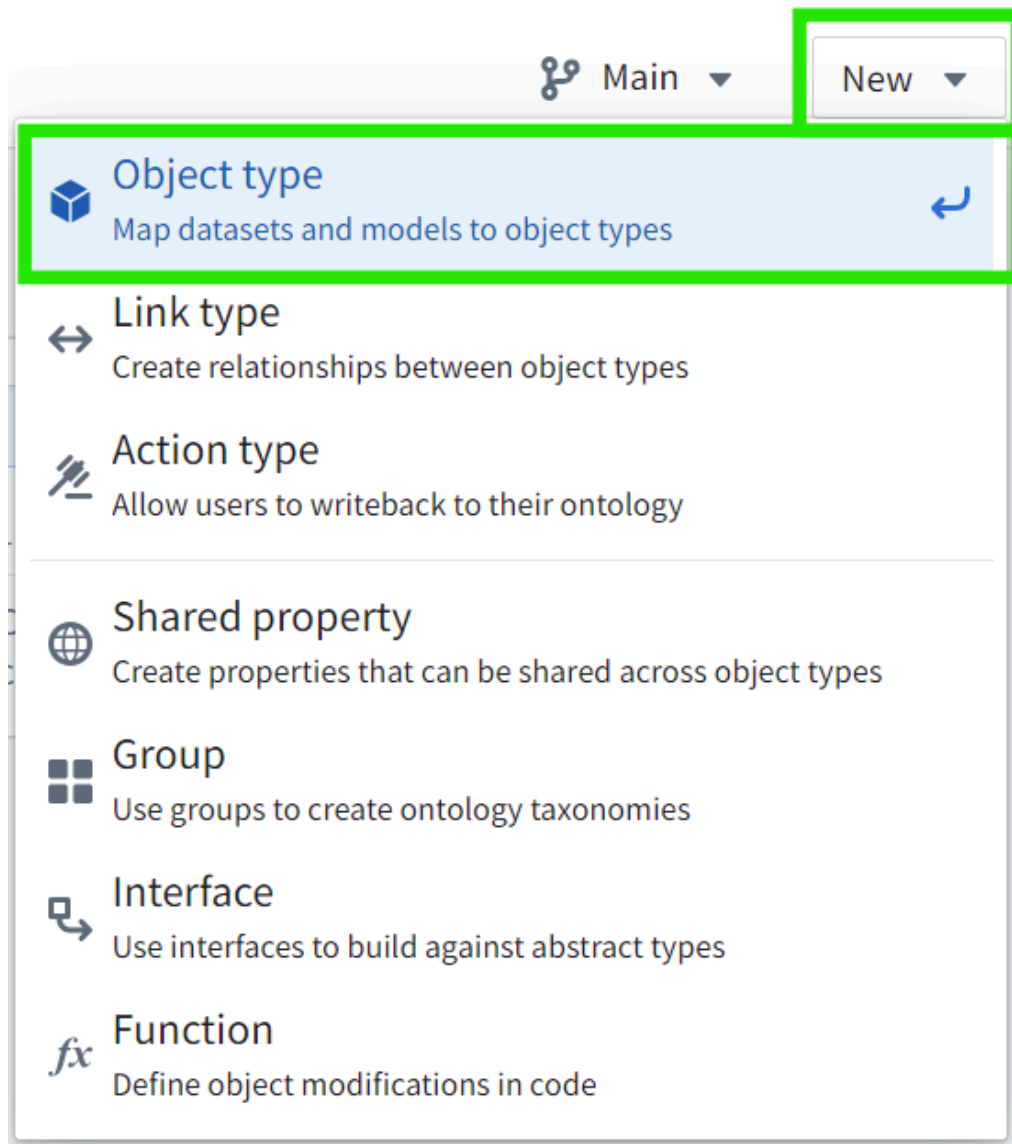
↵ Select item

tab Select next command

Create the Chunk Object Type

Lesson 28 / 51 · [原文链接](#)

1. Click *New* in the top right corner
2. Select *Object type* from the dropdown



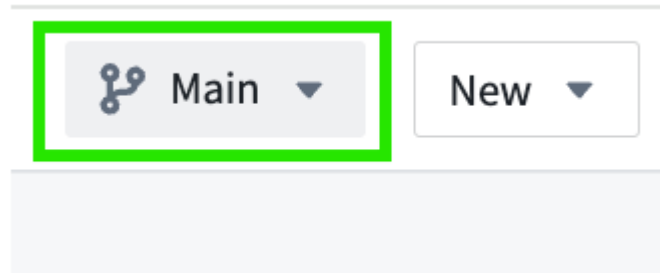
The following steps are only relevant for users where the 'New' button is not selectable. Proceed to Step 2 below if you are able to select the 'New' button.

Note that some Organizations limit edits to the production Ontology via Ontology Proposals. If the 'New' button in the screenshot above is not clickable, you may need to create a custom branch of the Ontology for the following steps and seek approval from your Organization's Administrator for merging in the changes to the Main branch of the Ontology.

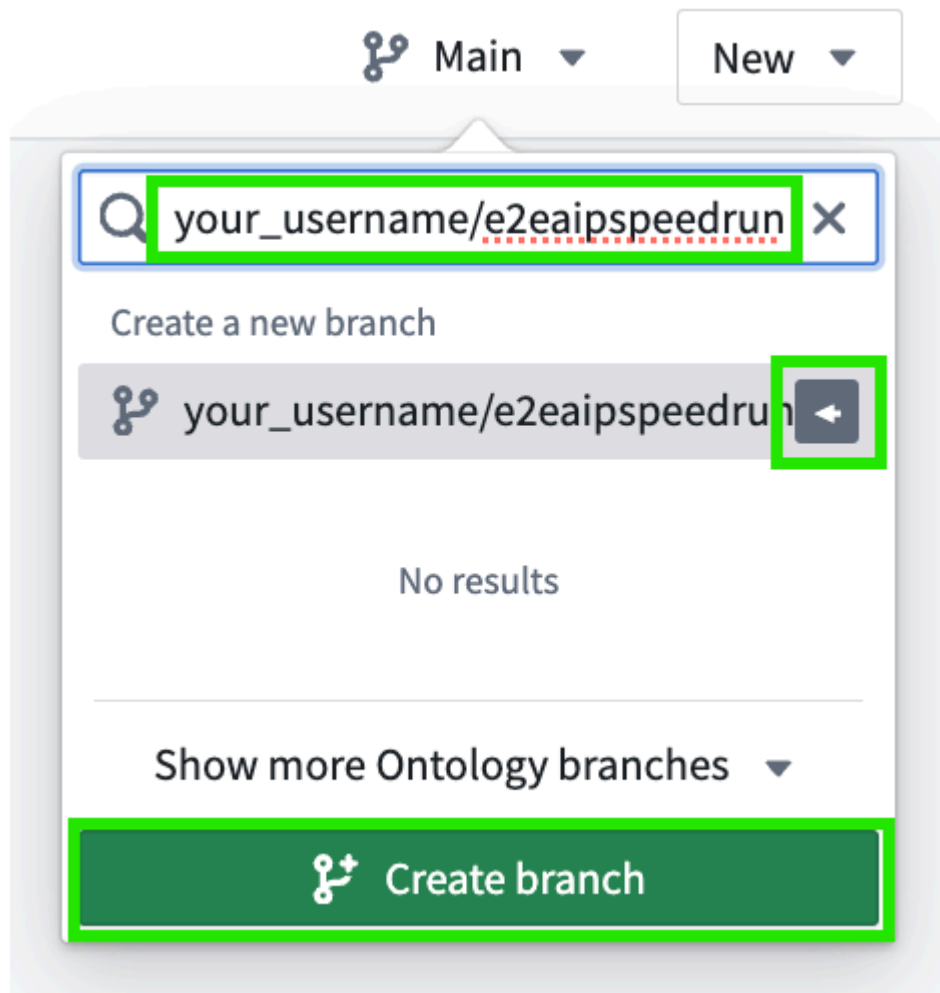
If you can not directly edit the Main Ontology, you can continue by creating a custom branch of the Ontology with the steps below. At the end of the 'Configuring the Ontology' module, you will

need to seek approval for merging in your changes to the Main Ontology before continuing. Further instructions for this can be found in the [Foundry documentation](#).

1. Click the *Main* button in the top right of your screen next to the *New* button



2. Type the name for your custom branch "<your_username>/e2eaipspeedrun"
3. Click the grey arrow to create the branch with that name
4. Click *Create branch*



5. If you have the *Enable indexing on this branch* toggle available on your Foundry instance, select the toggle on and proceed to the next steps.
6. If you do not have the toggle available, you will need to have your branch merged into the Main Ontology branch in order to proceed with the following steps. Contact your organization's Foundry Platform Administrator or Palantir representative for assistance.
7. Click *Create branch*

Create branch

This will create a Branch with your Ontology changes and a draft Proposal to review those changes. Open the Branch to view and add edits. Once all edits are approved, release the Proposal.

Branch name

your_username/e2eaipspeedrun

Enable indexing on this branch

Enable indexing if you need to preview schema changes for modified entities. This will use additional storage and compute. Only supported by Object Storage V2.

Cancel

Create branch

1. In the *Create a new object type* window, under *Datasource*, select *Use existing datasource*
2. Click on the button *Select datasource*

Create a new object type

1 Datasource

2 Metadata

3 Properties

4 Actions

STEP 1

Object type backing

Datasource

Continue without datasource

Generate a dataset for permissions purposes

Use existing datasource

Select a preexisting Foundry datasource

Select a datasource to back this object type

No datasource selected

Select datasource

Next

3. Search and select your *Chunks* dataset within your training project and folder

Select Dataset

←

→

Q Search for anything in your enterprise...

All > All projects > learning (jsmith) > Speedrun: Your First E2E AIP ...

ONLY SHOW

☐

Catalog items

☒

Selectable items

TAGS

Filter by tag

>

FILE TYPE

☐

Dataset

3

Add additional filters

Chunks

Entities

Join Table

Chunks

About

Columns

Preview

No description

Updated46 minutes ago by

Created55 minutes ago by

Location

TypeDataset

Size9 columns • 578 rows • 2 files • 5.2MB

RID5b0-adc9-51f104490bfb

Updated viaDocument Processing Transform

Cancel

Select

4. Click *Next*

Create a new object type

1 Datasource

2 Metadata

3 Properties

4 Actions

STEP 1

Object type backing

Datasource

Continue without datasource

Generate a dataset for permissions purposes

Use existing datasource

Select a preexisting Foundry datasource

✓

Select a datasource to back this object type

Chunks

/learning (jsmith)/Speedrun: Your First E2E AI...

Replace

...

Next

1. Name your Object as "<username> Chunk", where <username> is your username. The plural name should auto-populate.
2. Select an icon that you think helps describe the Chunk object
3. Click *Next*

Create a new object type ×

1 Datasource

2 Metadata

3 Properties

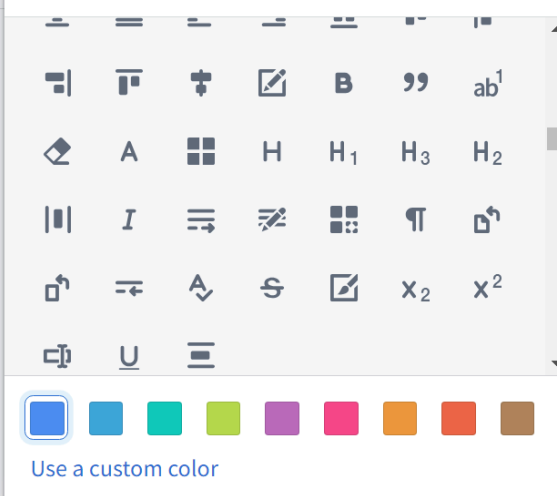
4 Actions

STEP 2

Configure object type metadata

Icon	Name	Plural name
	<username> Chunk ×	<username> Chunks ×

Q Search for an icon...



Back

Next

1. Select *Chunk Id* as the *Primary key* of the new Object; this serves as each Object's unique identifier
2. Select *Summary* as the *Title* of the new Object; this serves as the name of the Object that will be displayed across the platform
3. Click *Next*

1 Datasource

2 Metadata

3 Properties

4 Actions

STEP 3

Create properties

Source	Property	
 > embedding	 Embedding	
 > summary	 Summary	
 > chunkId	 Chunk Id	
 > chunkNumber	 123 Chunk Number	
 > content	 Content	
 > pageNumber	 123 Page Number	

Primary key ?

 Chunk Id

Title ?

 Summary

Back

Next

Don't worry if there is a warning next to *Embedding*, as shown in the screenshot above. This is just to let you know that it is an advanced type and you may need to edit the configuration of this property later. For this course, the default settings are sufficient for what we need.

The *Actions* tab allows you to automatically create common Actions that can be taken upon the new Object Type and select what users are allowed to make these Actions. You can also make these Actions or others manually through Ontology Manager after creating an Object Type.

1. Click *Create* on the bottom right corner to create your Object Type

1 Datasource

2 Metadata

3 Properties

4 Actions

STEP 4

Generate action types

Select action types to generate ?

☐  **Create (<username>) Chunk**
Set Embedding, Summary, Chunk Id, and 6 more properties

☐  **Modify (<username>) Chunk**
Modify Embedding, Summary, Chunk Number, and 5 more properties

☐  **Delete (<username>) Chunk**
Allows deleting object instances and all of their properties

Select who can execute these action types ?

Search users or groups...

▼

Back

Create

2. Save the object to your **Speedrun: Your First AIP Workflow** folder

STEP 5

Save location

Select a location to save this object type to

 Speedrun: Your First AIP Workflow

▼

Browse

▼

 Permissions

▼

You'll now edit some of the properties of the Object to allow them to be properly used in the Ontology.

1. Select the *Media Reference* property from the list
2. Change the *Base type* to *Media Reference* in the right panel

Properties

9 of 9 Columns mapped

Properties (9)

Column mapping

Property name	Status	Visibility	Base formatter	Column
☐ Chunk Id Primary key	Experimental	Normal	No formatting	chunkId
☐ Summary Title	Experimental	Normal	No formatting	summary
☐ Embedding	Experimental	Normal		embedding
☐ Chunk Number 123	Experimental	Normal	No formatting	chunkNumber
☐ Content	Experimental	Normal	No formatting	content
☐ Page Number 123	Experimental	Normal	No formatting	pageNumber
☐ Path	Experimental	Normal	No formatting	path
☐ Media Reference	Experimental	Normal	No formatting	mediaRefere...
☐ Media Item Rid	Experimental	Normal	No formatting	mediaItemRid

Media Reference

General

Source

Display

Interaction

Details

Base type

String

Filter...

Geohash

Geoshape

Attachment

Store uploaded files

Time series

Store historic timestamp values

Media reference

References to media items

Required

Shared property

Assign (optional)

Select a shared property

Convert to a shared property

3. On the *Select media source* window, select *Media Set*

4. Then click in *Select existing*

Select media source

×

A media reference property value is a reference to a media item contained in a media set.

Type of the media source

Media Set

Dataset Deprecated

No media set selected

Create new

Select existing

Cancel

Set as media reference property

5. Navigate and select the *Articles* media set imported at the beginning of this exercise.

6. Click *Select*

Select media source

All > All projects > learning (jsmith) > Speedrun: Your First E2E AIP ...

Search...

This folder

Articles

Cancel

Select

7. Click *Set as media reference property*

Select media source

A media reference property is backed by references to media. These referenced media items are contained in a media source, which can be either a media set or a dataset.

What is the Type of the media source to be added

Media Set Recommended

Dataset

Articles

Change

Cancel

Set as media reference property

Your Object is now created, but needs to be saved to the Ontology

1. Click *Save* in the top right of the screen
2. On the following pop-up window, click *Save to Ontology*
 - Object index will start after changes are saved to the ontology, which can take a few minutes

Ontology Manager

Search resources...Ctrl + K

66 edits

Save

New

Discover

<username>
0 objects

Overview

Properties

Security

Datasources

Capabilities

Object views

Interfaces

Materialization

Automations

Usage

Review edits

All edits (66)

Warnings

Errors

Migrations

Conflicts

<username> Chunk

Created

GENERAL INFORMATION

ID

new-object-type-3cf901

klajmxcj.lt-username-gt-chunk

Display name

<username> Chunk

API name

NewObjectType

_UsernameChunk

Plural display name

<username> Chunks

Primary key

chunk_id

Title property

summary

Backing datasources

Chunks

Icon color

#4600FF

#35A2DA

Discard

Save to ontology

Chunks

Show mapped columns

Automap all

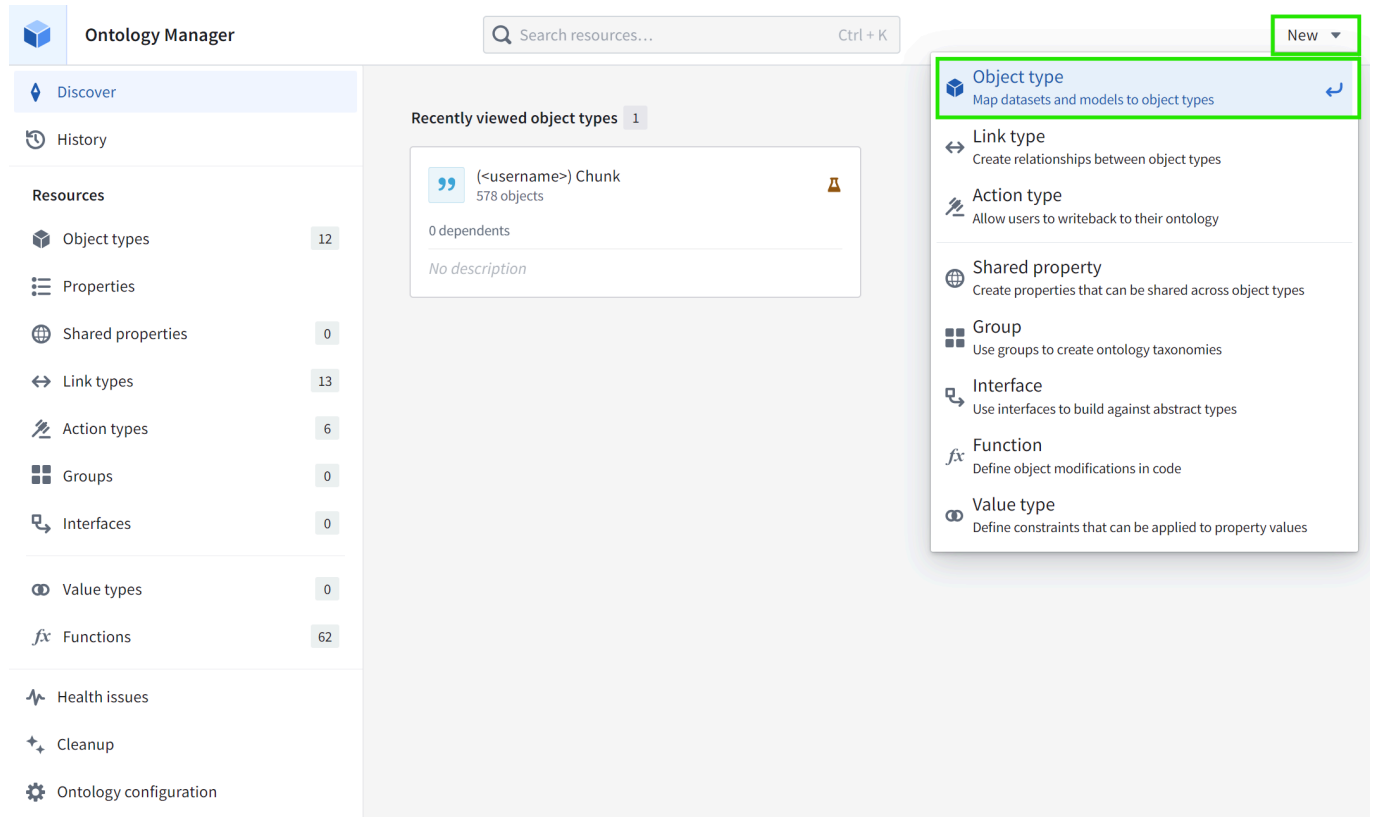
Preview objects

Preview table

Create the Entity Object Type

Lesson 29 / 51 · [原文链接](#)

1. Click *New* in the top right corner
2. Select *Object type* from the dropdown



1. In the *Create a new object type* window, in the *Datasource* step, pick *Use existing datasource*
2. Click on *Select datasource*

Create a new object type



1 Datasource

2 Metadata

3 Properties

4 Actions

STEP 1

Object type backing

Datasource

 Continue without datasource
Generate a dataset for permissions purposes

 Use existing datasource
Select a preexisting Foundry datasource

✓

Select a datasource to back this object type

 No datasource selected

 Select datasource

Next

3. Search and select your *Entities* dataset within your training project and folder

Select Dataset



← →

Search for anything in your enterprise...

All > All projects > learning (jsmith) > Speedrun: Your First E2E AIP ...

ONLY SHOW

☐ Catalog items

☒ Selectable items

TAGS

Filter by tag

FILE TYPE

☐ Dataset 3

Add additional filters

Chunks

Entities

Join Table

Entities

Report issue

About Columns Preview

No description

Updated Nov 14, 2024, 4:26 PM by 

Created Nov 14, 2024, 4:18 PM by 

Location ...

Type Dataset

Size One column • 298 rows • 2 files • 2.1MB

RID 2b-b59b-037be28c0572 

Updated via  Document Processing Transform

Cancel

Select

4. Click *Next*

Create a new object type ×

1 Datasource

2 Metadata

3 Properties

4 Actions

STEP 1

Object type backing

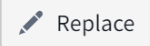
Datasource

 Continue without datasource
Generate a dataset for permissions purposes

 Use existing datasource
Select a preexisting Foundry datasource ✓

Select a datasource to back this object type

 Chunks
[redacted]/learning (jsmith)/Speedrun: Your First E2E AI...

 Replace ...

Next

1. Name your Object as "*(<username>) Entity*", where *<username>* is your username. The plural name should auto-populate.
2. Select an icon that you think helps describe the Entity object
3. Click *Next*

Create a new object type ×

1 Datasource

2 Metadata

3 Properties

4 Action

STEP 2

Configure object type metadata

Icon	Name	Plural name
	(<username>) Entity ×	(<username>) Entities ×

Search for an icon...

action











Use a custom color

Back

Next

1. Select *Entity Name* as both the *Primary key* and the *Title*
2. Click *Next*

Create a new object type

1 Datasource

2 Metadata

3 Properties

4 Actions

STEP 3

Create properties

Primary key ?

” Entity Name x ▾

Title ?

” Entity Name x ▾

Source

Property

entityName ▾

” ▾

Entity Name

Primary key

Title

+ Add property

Back

Next

1. Select *Create* and *Delete* Action Types
 - You are unable to select *Modify* as the only field is also the Primary Key which can not be modified via an Action
2. Select only your own user to be able to execute the Action types

Create a new object type ×

1 Datasource

2 Metadata

3 Properties

4 Actions

STEP 4

Generate action types

Select action types to generate ?

☒  Create (<username>) Entity
Set Entity Name

☐  Modify (<username>) Entity
Modify properties of object instances

☒  Delete (<username>) Entity
Allows deleting object instances and all of their properties

Select who can execute these action types ?

Back

Create

3. Save the object to your **Speedrun: Your First AIP Workflow** folder

STEP 5

Save location

Select a location to save this object type to

 Speedrun: Your First AIP Workflow

Browse ▼

 Permissions

▼

4. Click *Create* to create your Object Type

Your Object is now created, but needs to be saved to the Ontology

1. Click *Save* in the top right of the screen
2. On the following window, click *Save to ontology*

Ontology Manager

Search resources...Ctrl + K

19 edits

Save

New

Discover

(<username> Entity

0 objects

Overview

Properties

Security

Datasources

Capabilities

Object views

Interfaces

Materialization

Automations

Usage

Review edits

All edits (19)WarningsErrorsMigrationsConflicts

<username> Entity

Created

GENERAL INFORMATION

IDnew-object-type-f00f42klajmxcj.lt-username-gt-entity

Display name(<username> Entity

API nameNewObjectTypeUsernameEntity

Plural display name(<username>) Entities

Primary keyentity_name

Title propertyentity_name

Backing datasourcesEntities

Icon color

Discard

Save to ontology

No action types using this object type

<username> Entity

Preview objects

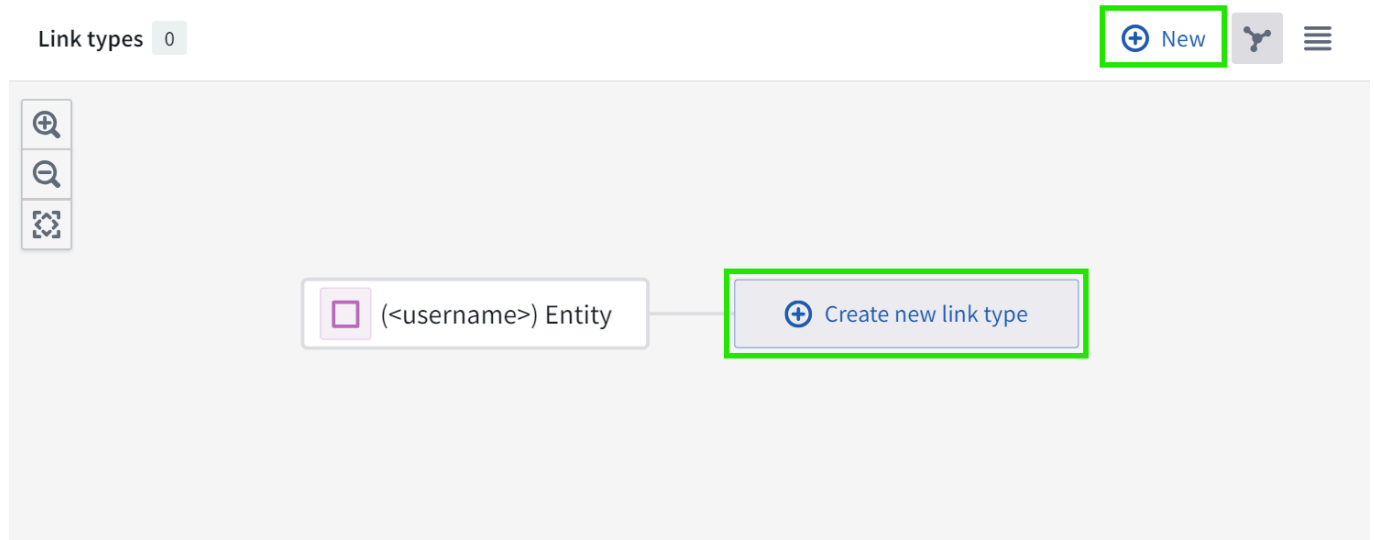
Preview table

Configure the Object Relation

Lesson 30 / 51 · [原文链接](#)

In order to establish the link between Chunks and Entities, we can use the join table we created in Pipeline Builder.

1. On the *Overview* page of the new Entity object type, scroll down to *Link types*
2. Click *Create new link type*
 - Alternatively, you can click *New* in the top right corner and select *Object Relation*



1. Select the *Join table dataset* as the relationship type

2. Click *Next*

Create a new link type

×

1 Relationship type

2 Link resources

3 Link type names

STEP 1

Relationship type

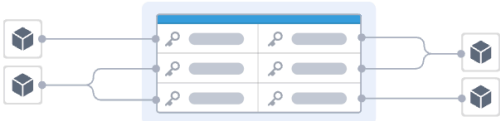
Choose method for defining links between instances

☐ Object type foreign keys

Link object types using a **foreign key** property.
Used for **one-to-one** and **one-to-many** link types.

☒ Join table dataset

Link object types using a dataset containing columns with object type primary keys.
Used for **many-to-many** link types.



☐ Intermediary backing object type New

Link object types and define properties using a middle backing object type.
Used for **many-to-many** link types that **include properties**.

Skip

Next

1. Select (<username>) *Chunk* as the other end of the relation.

2. In the *Select join table dataset* section, click *Select join table*

Create a new link type

1 Relationship type

2 Link resources

3 Link type names

STEP 2

Link resources

Choose the object types you wish to link.

Entity

Entity Name

Chunk

Chunk Id

Select join table dataset

No join table selected

Select join table

Generate join table

Skip

Back

Next

3. Navigate to the *Join Table* in your Project and select it

Select Dataset

← → 🔍 Search for anything in your enterprise...

All > All projects > learning (jsmith) > Speedrun: Your First E2E AIP ...

ONLY SHOW

☐ Catalog items
 ☒ Selectable items

TAGS

Filter by tag >

FILE TYPE

☐ Dataset

3

Add additional filters

Chunks

Entities

Join Table

Join Table

About

Columns

Preview

No description

Updated

Nov 14, 2024, 4:27 PM by [redacted]

Created

Nov 14, 2024, 4:18 PM by [redacted]

Location

[redacted]...

Type

Dataset

Size

2 columns • 1.2k rows • 2 files • 1.9MB

RID

ec7-96ae-1eb589a7f3e2

Updated via

Document Processing Transform

Cancel

Select

4. Pick `entityName` as the primary key column for the Entity end of the relation

5. Pick `chunkId` as the primary key column for the Chunk end of the relation

6. Click *Next*

1 Relationship type

2 Link resources

3 Link type names

STEP 2

Link resources

Choose the object types you wish to link.



Entity () Entity

Entity Name

Chunk () Chunk

Chunk Id

Select join table dataset ⓘ

Join Table

Undo

Select a column which matches the primary key of () Entity

Column matching () Entity ⓘ

entityName

Column matching () Chunk ⓘ

chunkId

Skip

Back

Next

1. Here you will see an overview of the link you have created. The *API Name* should be auto-generated.

Create a new link type

1 Relationship type

2 Link resources

3 Link type names

Each () Entity has many () Chunks

Name

() Chunks

API Name

() Entity.

usernameChunks

.all()

() Chunk → () Entity

Each () Chunk has many () Entities

Name

() Entities

API Name

() Chunk.

usernameEntities

.all()

Skip

Back

Submit

2. Click *Submit*

Your Object Relation is now created, but needs to be saved to the Ontology

1. Click *Save* in the top right of the screen
2. On the following window, click *Save to ontology*

Ontology Manager

View all related resourcesCtrl + K

15 edits

Save

New

<username> Entity

<username>

<username> Entity

<username>

<username> Chunk

<username> Chunk

Overview

Security

Datasources

Usage

Review edits

All edits (15)WarningsErrorsMigrationsConflicts

<username> Entities

<username> Chunks

Created

GENERAL INFORMATION

IDlink-typeklajmxcj.lt-username-gt-chunk-lt-username-gt-entity1

Relation typeoneToManymanyToMany

LINK TO<USERNAME> CHUNK

Display name<username> Chunk

API name_usernameChunk

Plural display name<username> Chunks

Object typeklajmxcj.lt-username-gt-chunk

Selectable columnselected

Discard

Save to ontology

one<username> Chunk has one

<username> Entity

one<username> Entity has one

<username> Chunk

Summary

Lesson 31 / 51 · [原文链接](#)

Your Ontology now describes two Object Types - Chunks and Entities. These represent major, high-quality, data assets in your organization as well as their relations. In a way, this setup acts like an API for your app builders and business users who no longer need to re-derive data features, nor understand the potential complexities of data engineering pipelines. You are ready to start building interactive features on top of your existing groundwork.

Beyond just being an indexed data-store and knowledge graph, the Ontology also allows you to define (and restrict) the ways in which operational users can update your data assets. Once such Actions are configured, app builders only need to curate the various Object views and interactive capabilities, speeding up their development process. Furthermore, the Ontology is shared across your organization, allowing you to benefit from Object Types and Actions already configured by your colleagues.

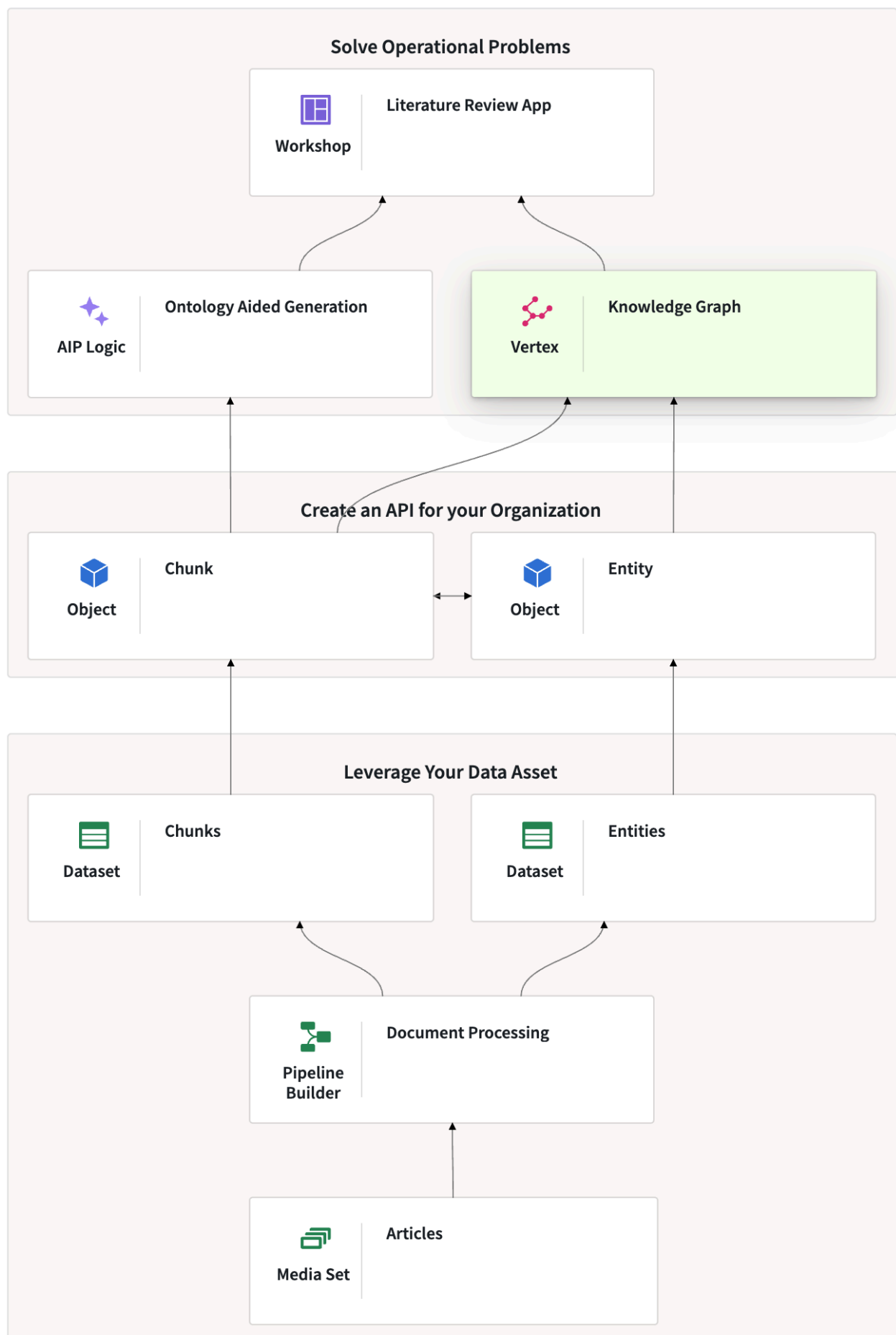
To learn more about Ontology building, you might follow this course up with the [Ontology Deep Dive](#), or by taking a look at the [documentation](#).

Introduction

Lesson 32 / 51 · [原文链接](#)

In this section you will visualise the relationship between the Chunks and Entities derived from your documents. This will allow you to identify which entities connect different chunks of text.

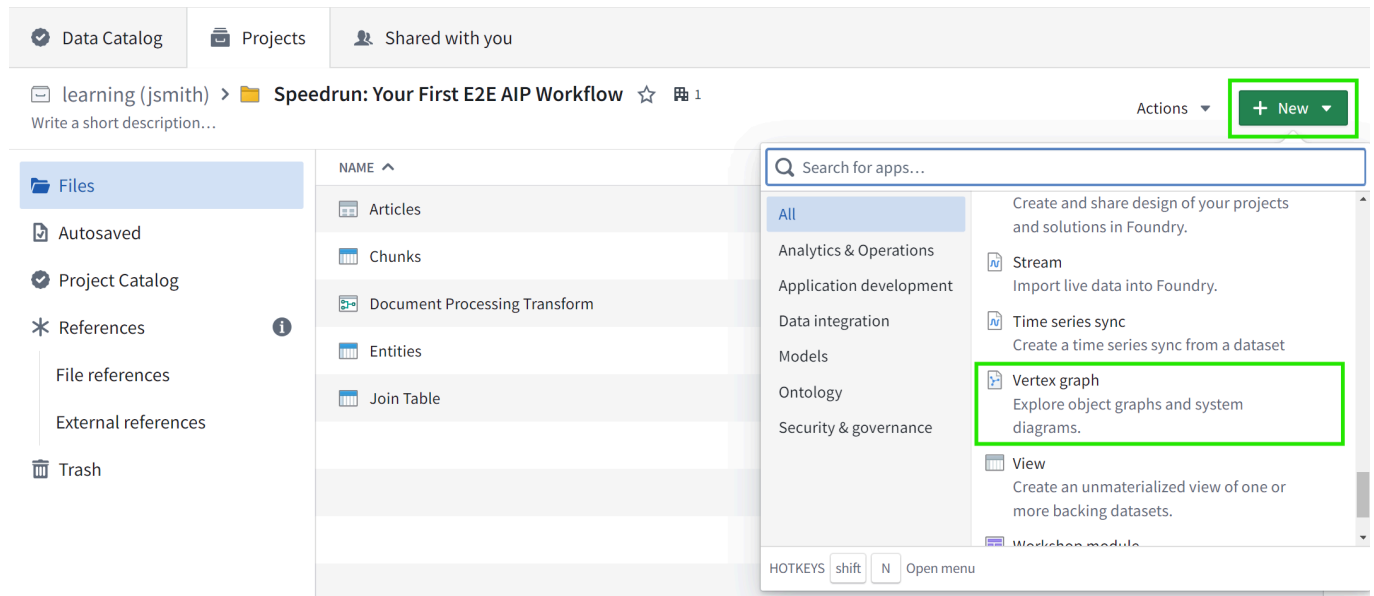
For this task you will use Vertex, Foundry's graph and simulation tool.



Create a Vertex Graph

Lesson 33 / 51 · [原文链接](#)

1. Navigate to your training folder inside your learning project
2. Click *New* in the top right corner
 - Alternatively, you can right-click in your folder and pick *New*
3. Select *Vertex graph* from the dropdown

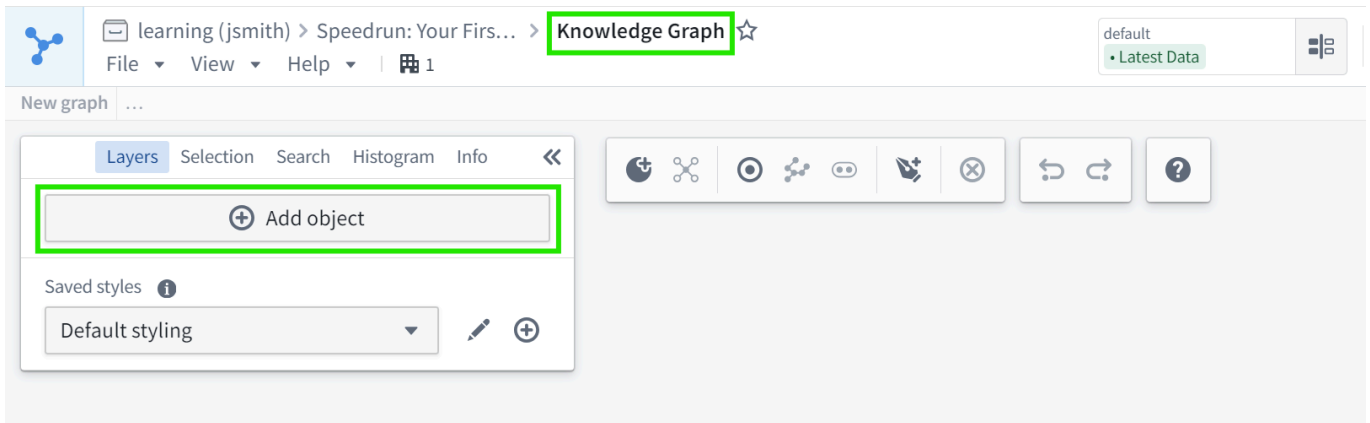


Visualize Object Relations

Lesson 34 / 51 · [原文链接](#)

1. Rename the Graph as *Knowledge Graph*

2. Click *Add object* in the top left corner



3. Search for your object types by typing your *<username>* in the search box

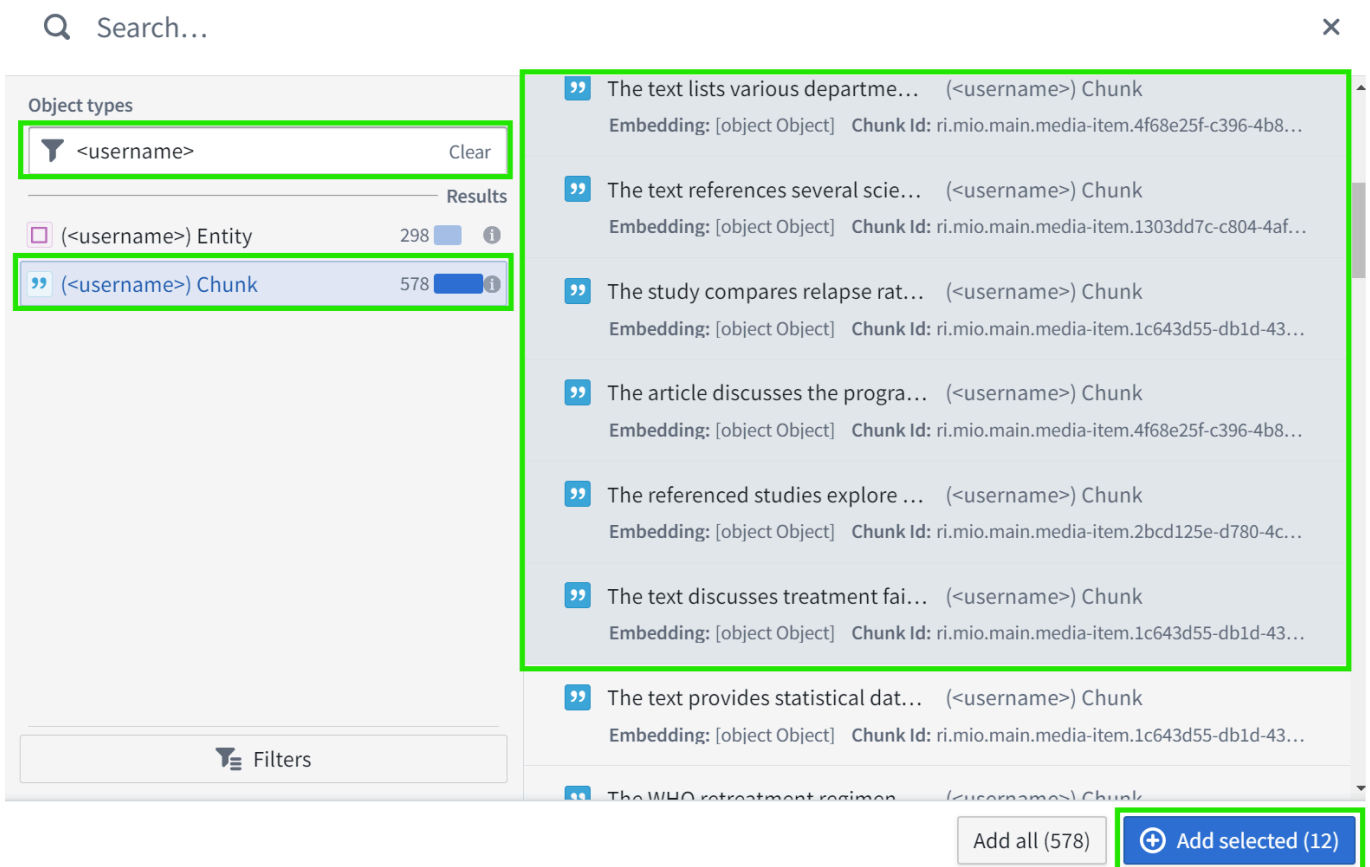
4. Select your *Chunk* object from the list on the left

5. Select a dozen objects from the list on the right

- You can do this by holding down shift and clicking on multiple rows

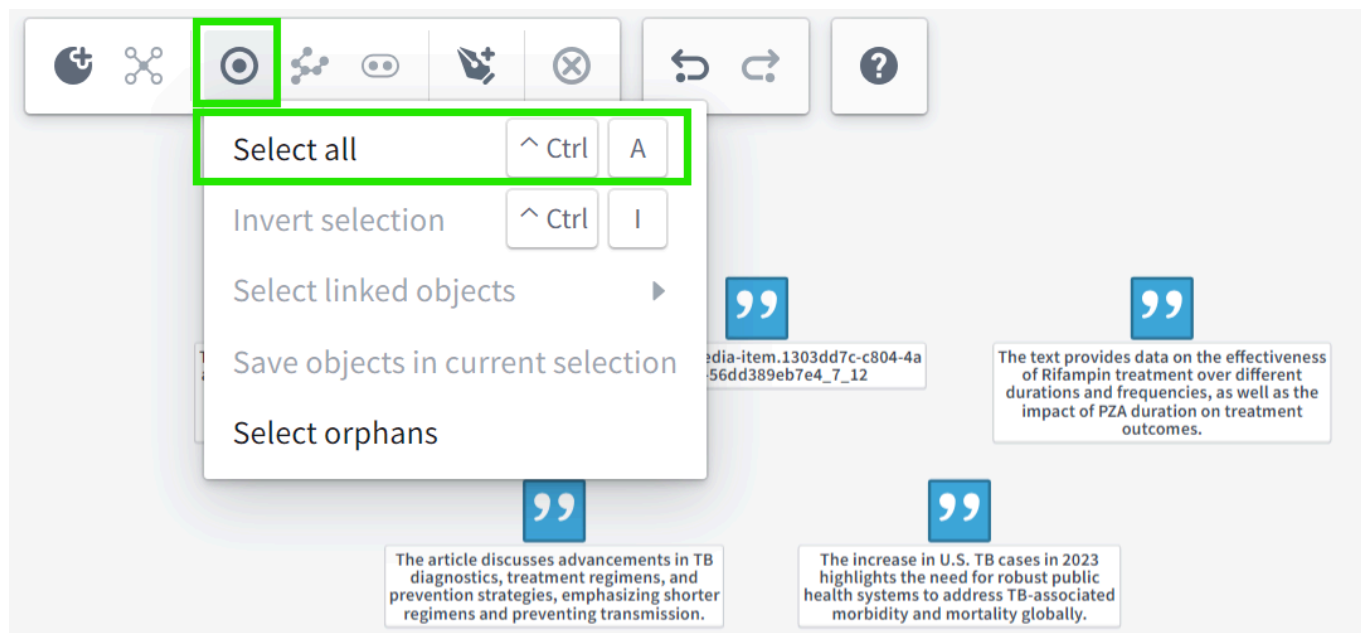
6. Click *Add selected*

7. Click anywhere in the graph to drop the Chunk objects

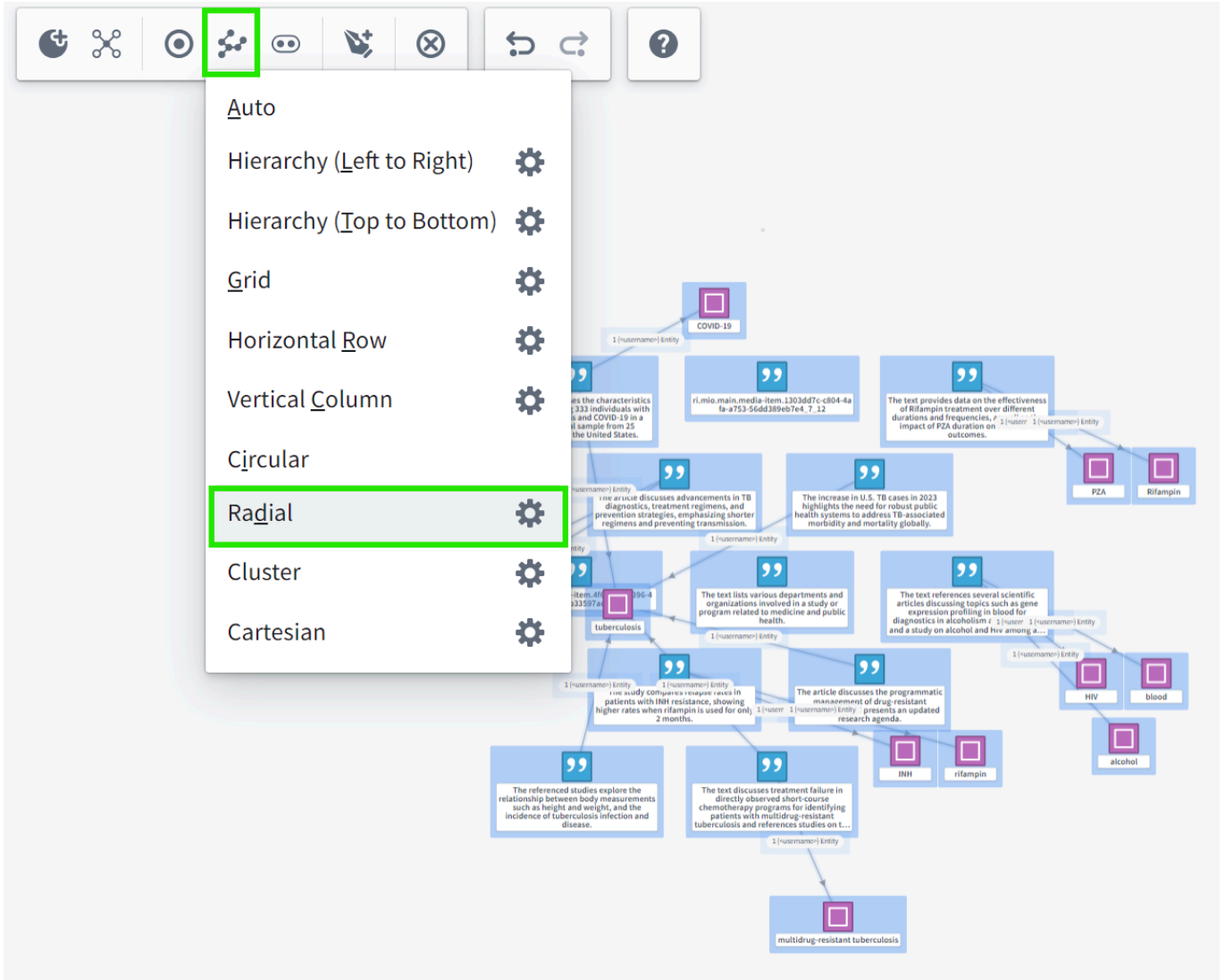


1. Select all objects on the Vertex canvas

- You can either press **Ctrl + A** or press the *Select all* button



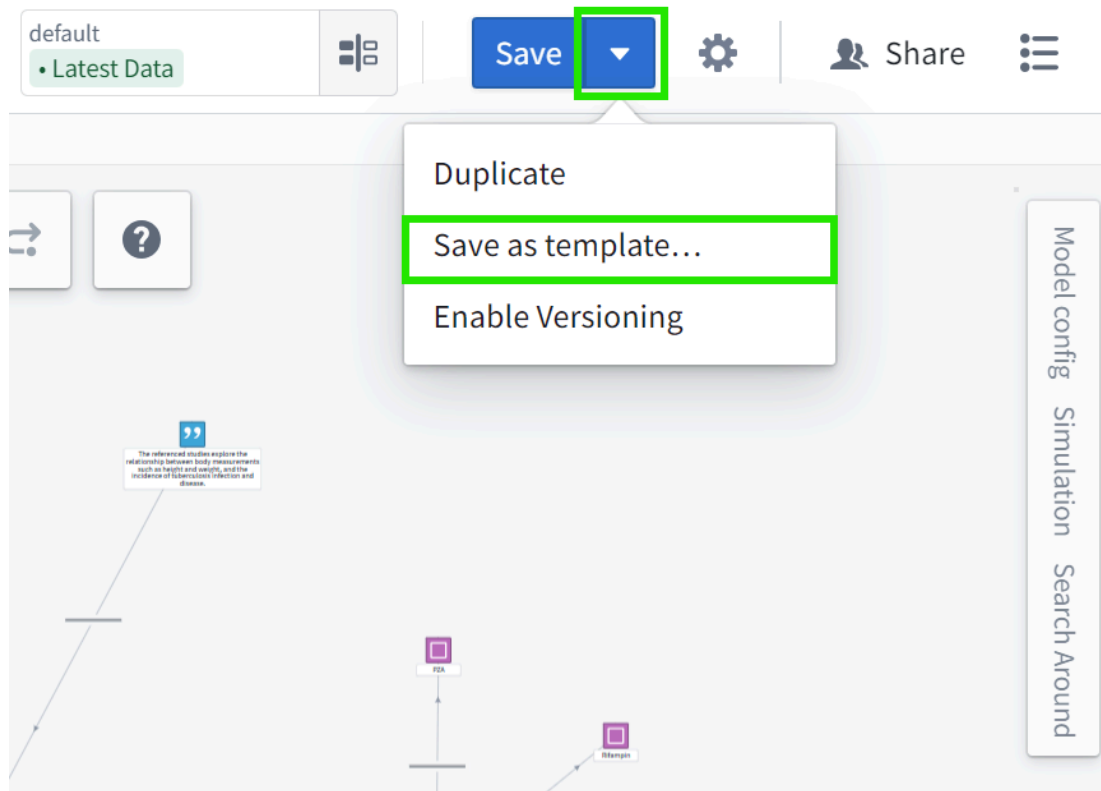
2. Click on the *Search Around* button in the toolbar on the top
3. Click on the relationship to the *Entity* object



Configure a Vertex Template

Lesson 35 / 51 · [原文链接](#)

1. Click on the dropdown next to the *Save* button and select *Save as template...*



Vertex templates allow you to procedurally recreate Object relationship graphs, enabling context-relevant displays.

1. Click *Add parameter* in the Object parameters section
2. Select your *Chunk* object type
3. Click *Continue*

Save as template

×

Turn this Graph into a Template by adding object parameters.

1

Configure parameters

Object parameters

Add parameters of different object types to your template to customize the generated graph based on user input objects.

(<username>) Chunks

×

+ Add description

Required

Multiple objects

Type

(<username>) Chunk

Experimental

▼

+ Add parameter

Non-object parameters

Continue ↓

2

Configure Search Arounds

3

Configure layers

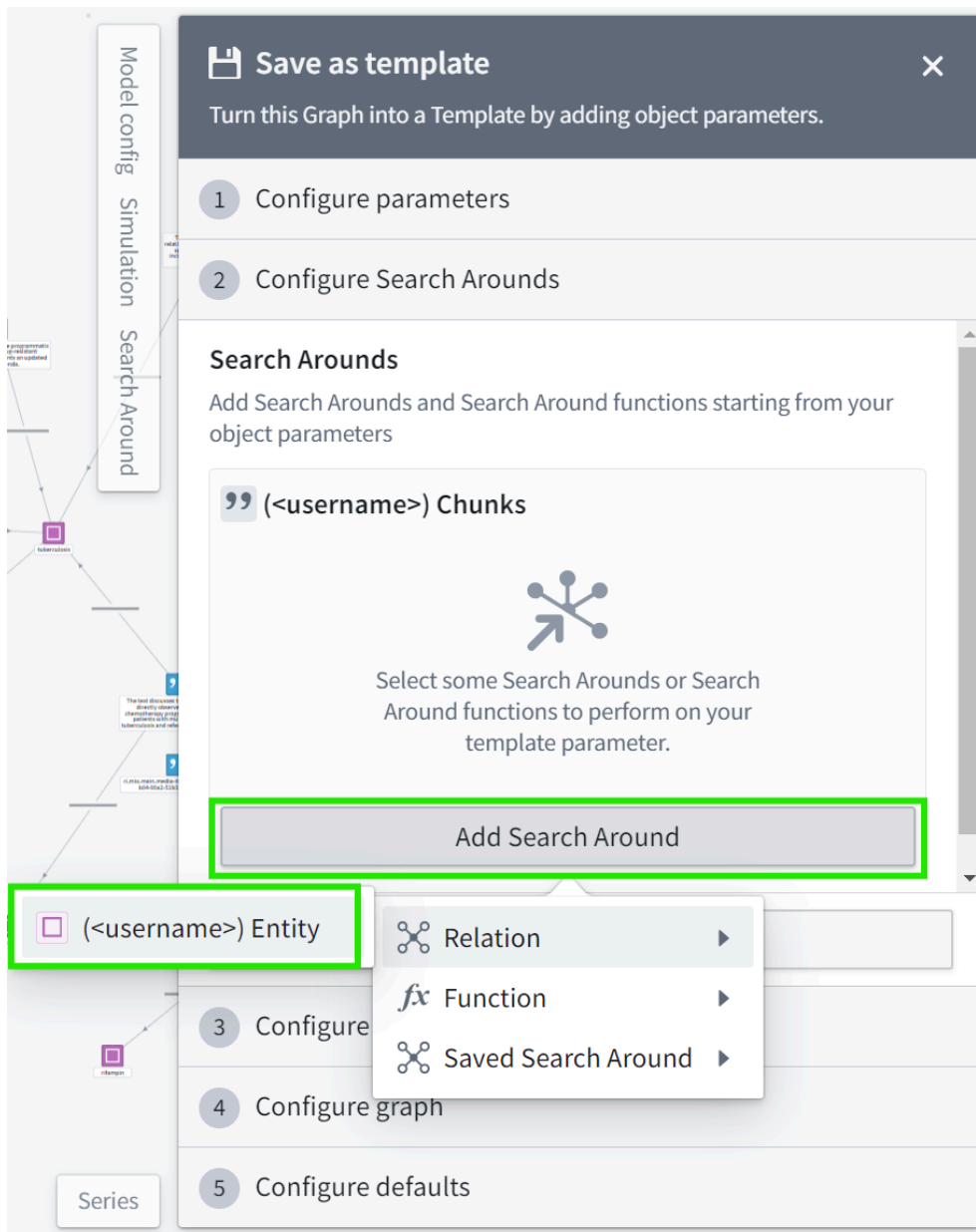
4

Configure graph

5

Configure defaults

1. Click *Add Search Around* below your *Chunks* object
2. Select the relationship to your *Entity* object
3. Click *Continue*



1. Click *Continue*

- This summarises any style changes made to the nodes and edges; they are fine as they are

Save as template

Turn this Graph into a Template by adding object parameters.

1

Configure parameters

2

Configure Search Arounds

3

Configure layers

Node layers

Retain node layers styling or exclude them.

” (<username>) Chunk

Include as Styling

□ (<username>) Entity

Include as Styling

Edge layers

Retain edge layers styling or exclude them

□ — ” (<username>) Entity ↔ (<username>) Chu...

Continue ↓

4

Configure graph

5

Configure defaults

1. Name your Vertex template *Knowledge Graph Template*
2. Set the layout to *Radial* and leave the default configuration as it is
3. Click *Continue*

 Save as template

Turn this Graph into a Template by adding object parameters.

1 Configure parameters

2 Configure Search Arounds

3 Configure layers

4 Configure graph

Name

Knowledge Graph Template

Description (optional)

Layout

Choose which layout to apply once all graph generation functions and Search Arounds have been run

Radial ▾

Layout parameters

Configure layout parameters

Continue ↓

5 Configure defaults

1. Click *View summary*

Save as template



Turn this Graph into a Template by adding object parameters.

- 1 Configure parameters
- 2 Configure Search Arounds
- 3 Configure layers
- 4 Configure graph
- 5 Configure defaults

Settings

- ☐ Show grid lines in viewing mode
- ☐ Show grid lines in editing mode
- ☐ Expand labels in editing mode
- ☐ Generate a saved selection for each object parameter input
- ☐ Show color borders for generated saved selections by default

View summary 

2. Click *Save template*



Save as template



Turn this Graph into a Template by adding object parameters.

Template summary

Parameters

[Edit](#)

(<username>) Chunks

Layers

[Edit](#)

(<username>) Chunk

Styling



(<username>) Entity

Styling



— ” (<username>) Entity ↔ (<username>) Ch...

Styling

Graph

[Edit](#)

Name

Knowledge Graph Template

Layout

Radial

Edit template

Save template

3. Pick your training folder, on your learning Project, as the save location

4. Click *Save*

Save



File name



Knowledge Graph Template



Location



Speedrun: Your First E2E AIP Workflow



Browse

Cancel

Save

Summary

Lesson 36 / 51 · [原文链接](#)

You now have a reusable template to display subsets of your Knowledge Graph. This will be used in the last section of this course to visualise the knowledge AIP is using to answer your questions.

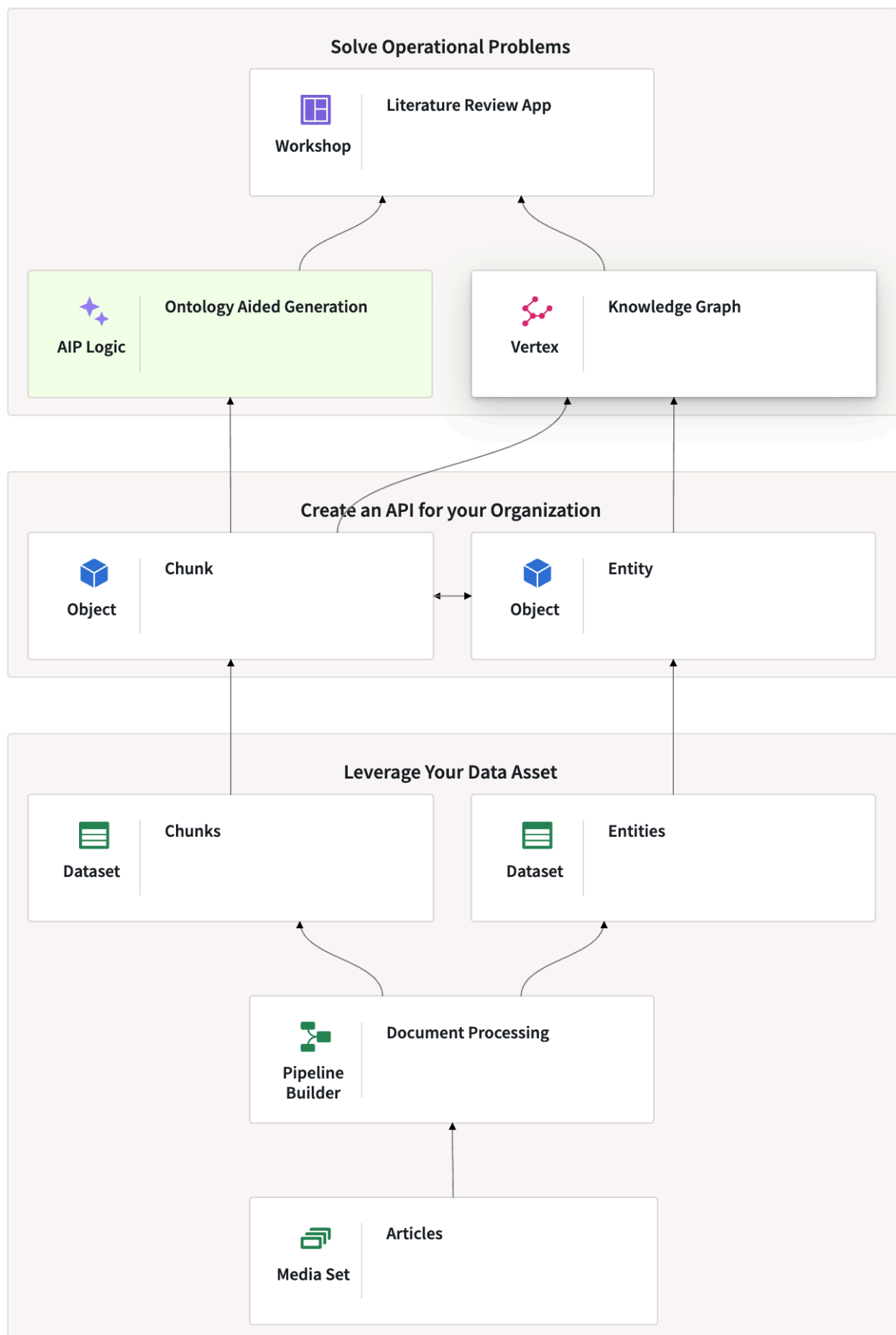
Beyond being a tool to visualise Ontology Objects and their Links, Vertex also allows you to model complex systems. You can chain statistical or machine learning models to simulate how changes to one Object might affect properties of others.

To learn more about Vertex and its advanced modelling capabilities, please take a look at the [documentation](#).

Introduction

Lesson 37 / 51 · [原文链接](#)

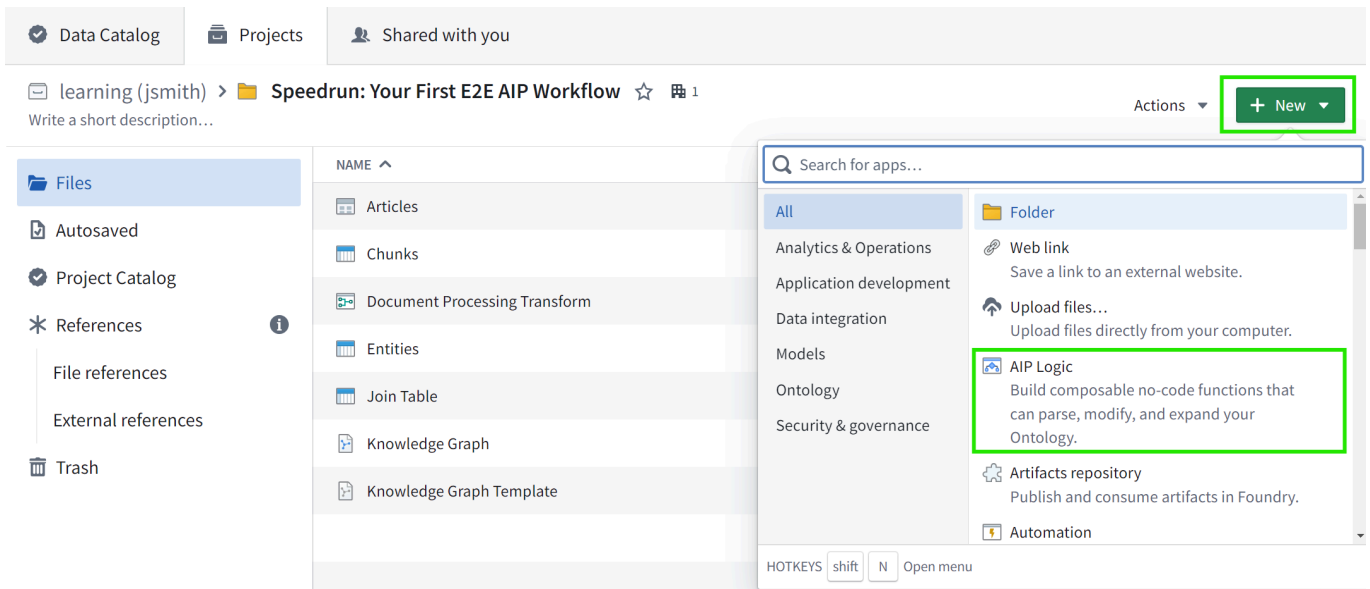
In this section you will develop an interactive AIP function to answer questions based on the knowledge base you created earlier. Leveraging the Ontology in such a way enables you to deliver solutions specific to your work without having to retrain LLMs or worry about hallucinations.



Create a Logic Function

Lesson 38 / 51 · [原文链接](#)

1. Navigate to your training folder
2. Click *New* in the top right corner
 - Alternatively, you can right-click in your folder and pick *New*
3. Select *AIP Logic* from the dropdown



1. Rename your Logic function "*Ontology Augmented Generation*"
2. Ensure you are saving to your training folder
3. Click *Save*

Choose Logic file location



File name



Ontology Augmented Generation



Location



Speedrun: Your First E2E AIP Workflow

Browse

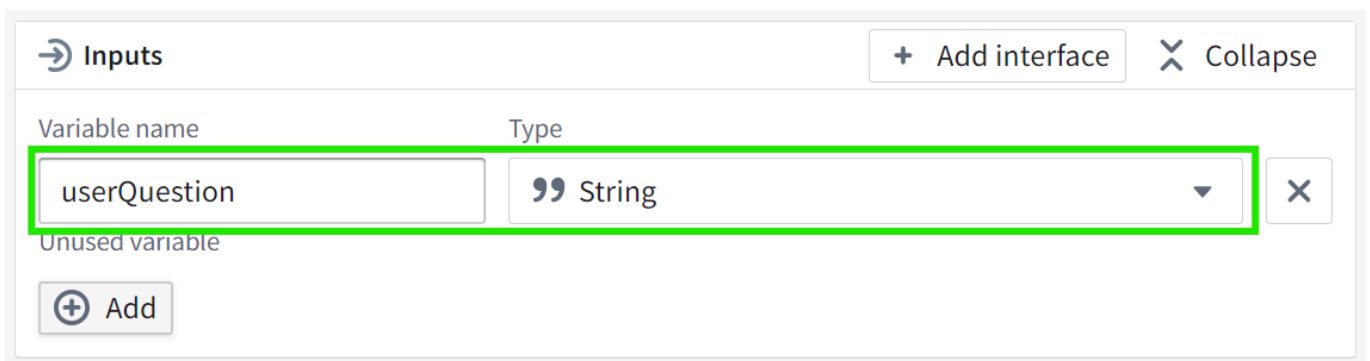
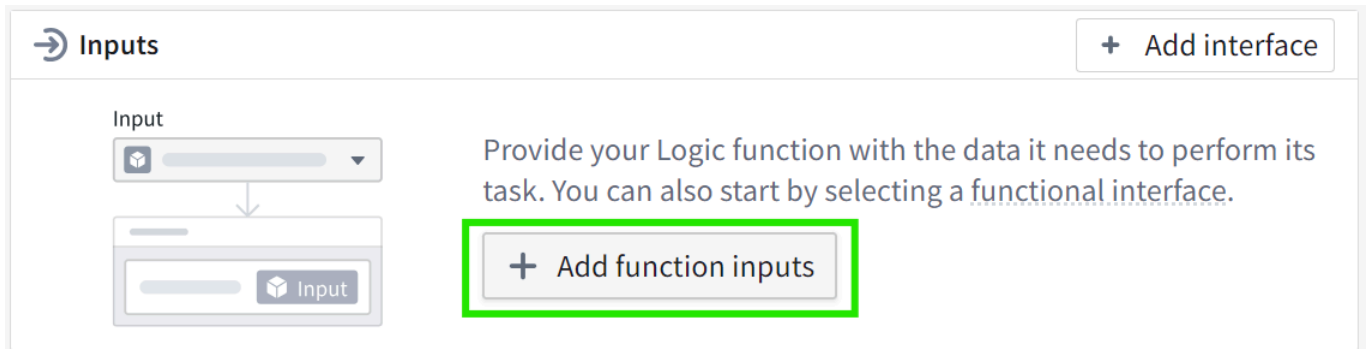
Cancel

Save

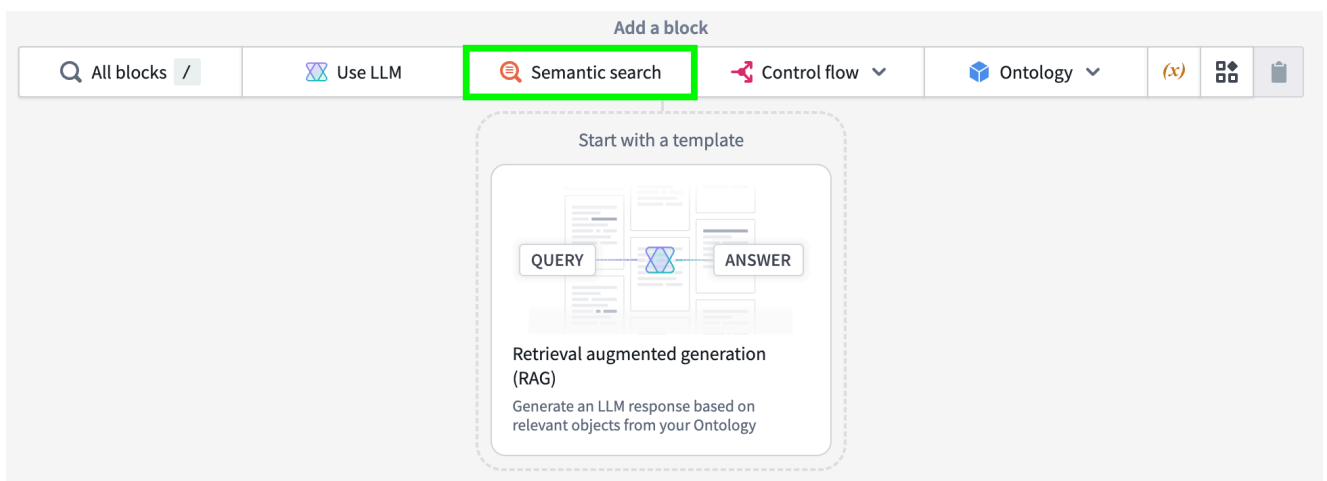
Compose Your Function

Lesson 39 / 51 · [原文链接](#)

1. Click *Add function inputs*
2. Name the new input "*userQuestion*"
3. Select *String* as its type



1. Click the magnifying glass icon to add a *Semantic search* function block



2. Click on the *Object set* dropdown
3. Select *Create new object set* and then your *Chunk* object type

Semantic search: Object

Object set

Select a variable...

() Chunk

Create new object set

Incompatible variables

userQuestion

4. Select *Embedding* as the *Property to search*
5. Set the *Number of returned objects* to 10
6. Set the *userQuestion* variable as the *Query*
 - A new variable Searched All (<username>) Chunks will be created by this block

Searched All () Chunks: () Chunks

Object set

All () Chunks

Property to search

Embedding

Number of returned objects

Integer 10

Query

userQuestion

1. Click the AIP icon to insert a *Use LLM* function block

Add a block

All blocks / Use LLM Semantic search Control flow Ontology (x) [Icons]

2. Add the following prompt in the *Instructions (System prompt)* field

Your task is to answer the user's question using only the reference text provided. If the references don't allow you to answer the question, say so.

It is very important that you don't use any prior knowledge, and only rely on the references provided.

3. Add the below prompt in the *Provide input data (Task prompt)* field

Please answer the following question: [user question]

To compose your response, you may only depend on the following information:
[semantic search]

4. Replace "[user question]" with the corresponding variable
 1. Type "/" (forward slash)
 2. Select *userQuestion* in the variable picker

5. Replace "[semantic search]" with the corresponding variable

1. Type "/" (forward slash)

2. Select Searched All (<username>) Chunks in the variable picker

3. Select *Content* in the property picker

4. Click *Add 1 property*

- A new variable Searched All (<username>) Chunks – formatted will be created and added to the *Task prompt*

6. Change the *Prompting strategy* to *Single completion*

The screenshot displays the configuration interface for an AI Platform (AIP) task. The top section is labeled 'Use LLM: String'. Below this, the 'Prompt' section includes a 'Show help' link and 'Instructions (System prompt)' which state: 'Your task is to answer the user's question using only the reference text provided. If the references don't allow you to answer the question, say so. It is very important that you don't use any prior knowledge, and only rely on the references provided.' The 'Provide input data (Task prompt)' section contains two lines: 'Please answer the following question: userQuestion' and 'To compose your response, you may only depend on the following information: Searched All () Chunks - formatted'. A note below states 'Press "/" to insert a variable'. A warning message indicates 'Tools are not supported when using the "Single completion" prompting strategy'. The 'Output type' is set to 'String'. The 'Model' is 'GPT-4o'. The 'Prompting strategy' is set to 'Single completion'. The 'Prompting strategy' dropdown shows 'Single completion' selected, with a description: 'Use a model to return a single completion. This only adds return type instructions to your prompt. This adds the least token overhead among all prompting strategies and so will be faster.' The 'Output enforcement' section is titled 'Improve output reliability at the cost of additional execution time' and has a toggle for 'Allow corrections' which is currently turned on, with a description: 'Re-prompt the model for a valid response when its output doesn't match the required format. This reduces function failures at the cost of increased token usage.'

When writing prompts for AIP, it might help to imagine that the prompt is a message to the newest member on your team. It will only be able to provide reliable responses if the task you give is clearly described without the use of any jargon, and if it has access to all the data required to solve your problem.

1. Check that the output variable is set to *Use LLM*

Output

Logic functions only support one output. Select ontology edits to return all ontology edits proposed in a logic run.

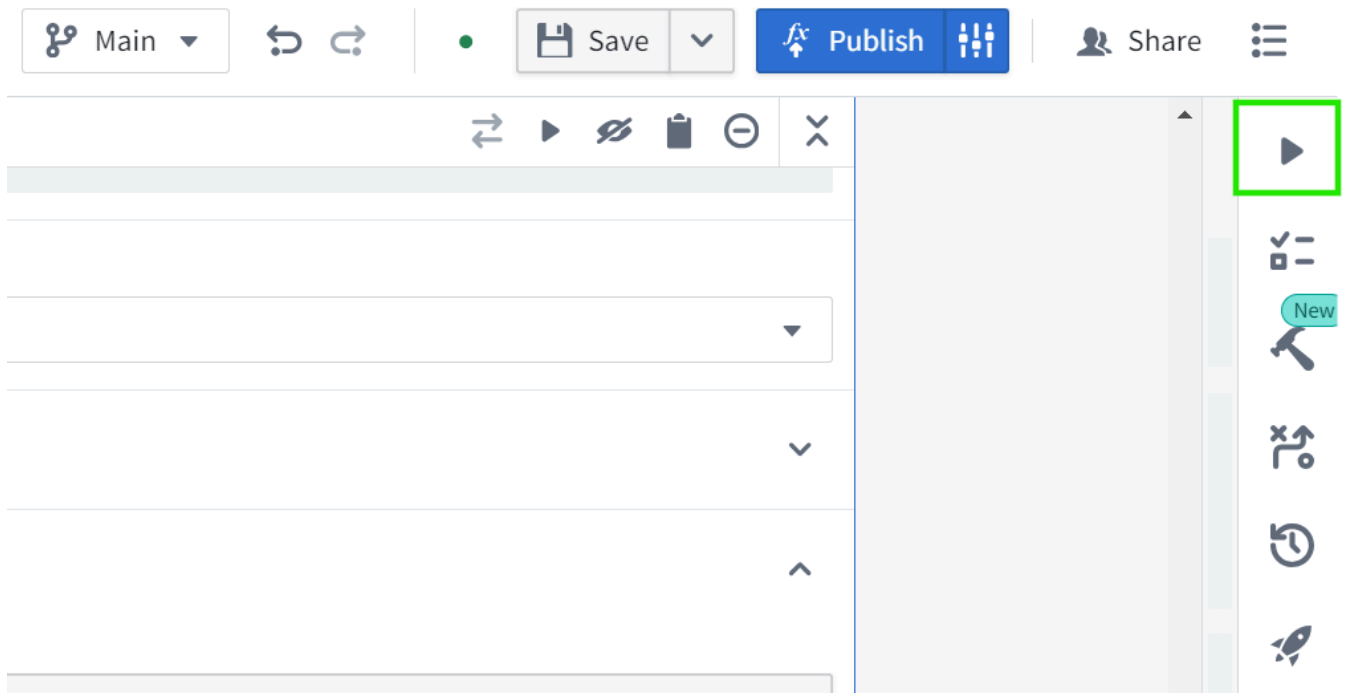
Select block

” Use LLM

Test Your Function

Lesson 40 / 51 · [原文链接](#)

1. Click on the *play* button in the top right corner



2. For the *userQuestion* input, enter a test question such as:

- What is the recommended treatment for Tuberculosis?
- How common is Tuberculosis these days?

3. Click the blue *Preview run* button at the bottom of the drawer

- Alternatively, you can click `Ctrl + Shift + D`

Preview run

Do a preview run of this logic to ensure its output looks as expected.

Ontology edits are not applied when running a preview in Logic. Publish your Logic function to apply edits where functions are used.

Function inputs



Clear

userQuestion

What is the recommended treatment for Tuberculosis?

Function output

(x) Use LLM



New



▶ Preview run (ctrl + shift + D)

+ Add as test case

Recent preview runs

Completed preview runs will appear here.

4. Verify that the *Result* returned makes sense
5. (Optional) Modify your LLM prompt to change the output format (eg: length, language, style...)

Debugger

Use the detailed execution log below to troubleshoot any issues in your Logic.

Execution started

allChunks

Create object set

In 0.42s

Searched allChunks

Semantic search

In 0.21s

Searched allChunks...

Print object set as formatte

In 1.81s

Use LLM

Use LLM

In 1.08s

Result

Successfully ran

The recommended treatment for pulmonary tuberculosis involves a regimen given for 6 months. This can include four twice-weekly regimens or a daily regimen. It is important to use effective drugs initially, and at least three effective drugs should be used in the continuation phase. Additionally, the prolonged use of pyrazinamide and the initial use of

Show more

Preview run

Do a preview run of this logic to ensure its output looks as expected.

Ontology edits are not applied when running a preview in Logic. Publish your Logic function to apply edits where functions are used.

Function inputs

userQuestion

What is the recommended treatment for Tuberculosis?

Function output

(x) Use LLM

Result

Successfully ran

The recommended treatment for pulmonary tuberculosis involves a regimen given for 6 months. This can include four twice-weekly regimens or a daily regimen. It is important to use effective drugs initially, and at least three effective drugs should be used in the continuation phase. Additionally, the prolonged use of pyrazinamide and the initial use of streptomycin (or an alternative injectable drug in

Show more

In 3.88s

▶ Preview run (ctrl + shift + D)

Testing is essential for developing reliable AIP applications. This becomes increasingly important when working on complex applications using the Chain of Thought execution mode. In this mode, your function has access to various tools, such as running simulations and querying and editing the Ontology. Then AIP decides which tools, or sequence of tools, to use in completing the task. The *Debugger* tab in Logic allows you to review the decision-making process of your function.



Debugger



Preview run

Do a preview run of this logic to ensure its output looks as expected.

Ontology edits are not applied when running a preview in Logic. Publish your Logic function to apply edits where functions are used.

Function inputs



Clear

userQuestion

What is the recommended treatment for Tuberculosis?

Function output

(x) Use LLM

Result



Successfully ran

The recommended treatment for pulmonary tuberculosis involves a regimen given for 6 months. This can include four twice-weekly regimens or a daily regimen. It is important to use effective drugs initially, and at least three effective drugs should be used in the continuation phase. Additionally, the prolonged use of pyrazinamide and the initial use of

▶ Preview run (ctrl + shift + D)



+ Add as test case



Recent preview runs

1 completed run



Run at 13:01



New





Succeeded

Publish Your Function

Lesson 41 / 51 · [原文链接](#)

1. Click *Save* in the top right corner
2. Click *Publish* in the top right corner

The screenshot displays the Logic Studio interface for a function named "Ontology Augmented Generation". The top right corner features a "Save" button (highlighted with a green box) and a "Publish" button (highlighted with a green box). The main workspace shows a logic block configuration for "Use LLM" with a "String" output type, "GPT-4o" model, and "Single completion" prompting strategy. The "Function inputs" section contains a "userQuestion" input with the text "What is the recommended treatment for Tuberculosis?". The "Function output" section shows a "Result" of "Successfully ran" with a preview of the LLM response: "The recommended treatment for pulmonary tuberculosis involves a regimen given for 6 months. This can include four twice-weekly regimens or a daily regimen. It is important to use effective drugs initially, and at least three effective drugs should be used in the continuation phase. Additionally, the prolonged use of pyrazinamide and the initial use of...". The bottom right panel shows a "Recent preview runs" section with one completed run at 13:01 that succeeded.

3. If requested, select your Ontology from the *Bound to the following ontologies* dropdown
4. Set the *Name* to "<username> Ontology Augmented Generation"
5. Set the *API name* to "*yourUsernameOntologyAugmentedGeneration*"
6. Click *Publish*

Publishing this Logic function allows it to be used by other applications such as Workshop or Automate, or called from code repositories.

Bound to the following ontologies

 Ontology

Ontology

Referenced by 1 Object · (<username>) Chunk

Edit function details

Name

(<username>) Ontology Augmented Generation

The name used when calling this function in other applications like Workshop or Automate.

API name

yourUsernameOntologyAugmentedGeneration ✓

The name used when calling this function in code. Must be lowerCamelCase.

Documentation

Providing a description of what your function does can help other users and LLMs understand how to use your function correctly.

Version configuration

Automatic

Manual

Logic will determine the appropriate new version tag based on the scope of changes.

 Publish

Summary

Lesson 42 / 51 · [原文链接](#)

You now have a basic, AIP powered function which can be used to answer your questions based on the Ontology, your organizations knowledge graph. All you have left to release this tool to your team is to construct a user interface.

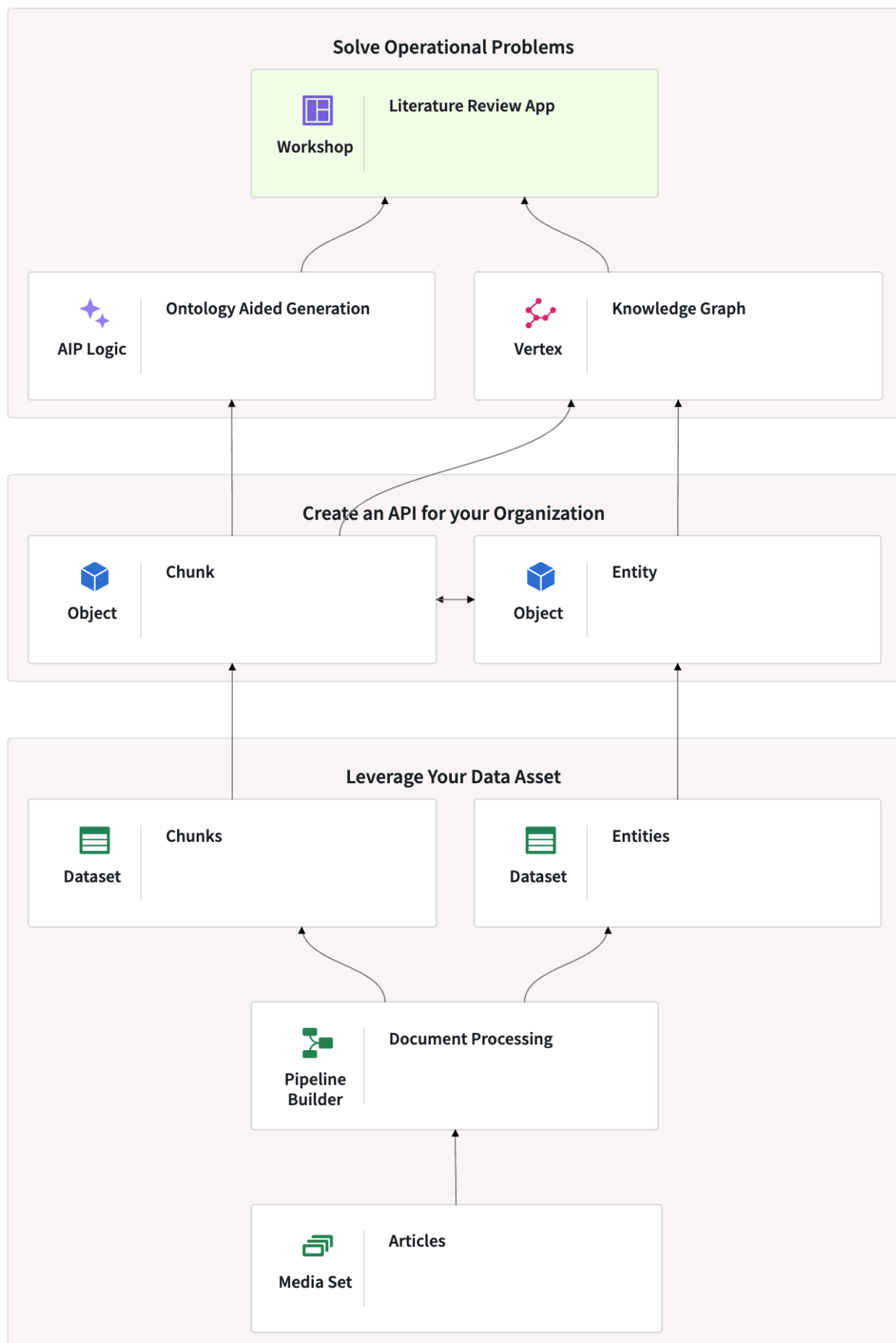
AIP Logic starts to truly shine as you start building on top of its Chain of Thought execution mode. This way, AIP becomes a true agent, working alongside you. It leverages its decision making ability to query data and pick and use tools made available to it via the Ontology. This allows it to tackle even tasks it was never specifically programmed to perform. At the same time, you didn't have to write a single line of boilerplate code!

If you're interested in learning more about this technology, you can review our [documentation](#). Additionally, Palantir's [YouTube channel](#) is continuously updated with demos of AIP across many applications and industries.

Introduction

Lesson 43 / 51 · [原文链接](#)

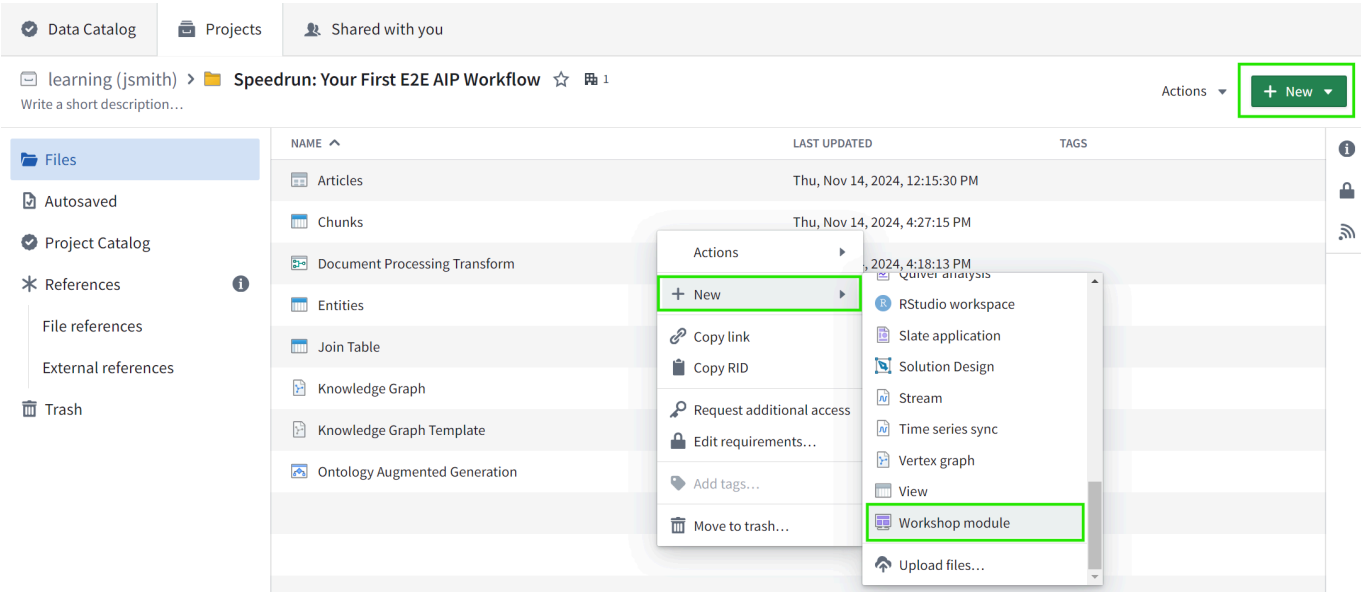
In this final section, you will create an interactive application which ties all your previous work together. This app will allow you to query your documents in natural language and receive trustworthy responses.



Create a Workshop App

Lesson 44 / 51 · [原文链接](#)

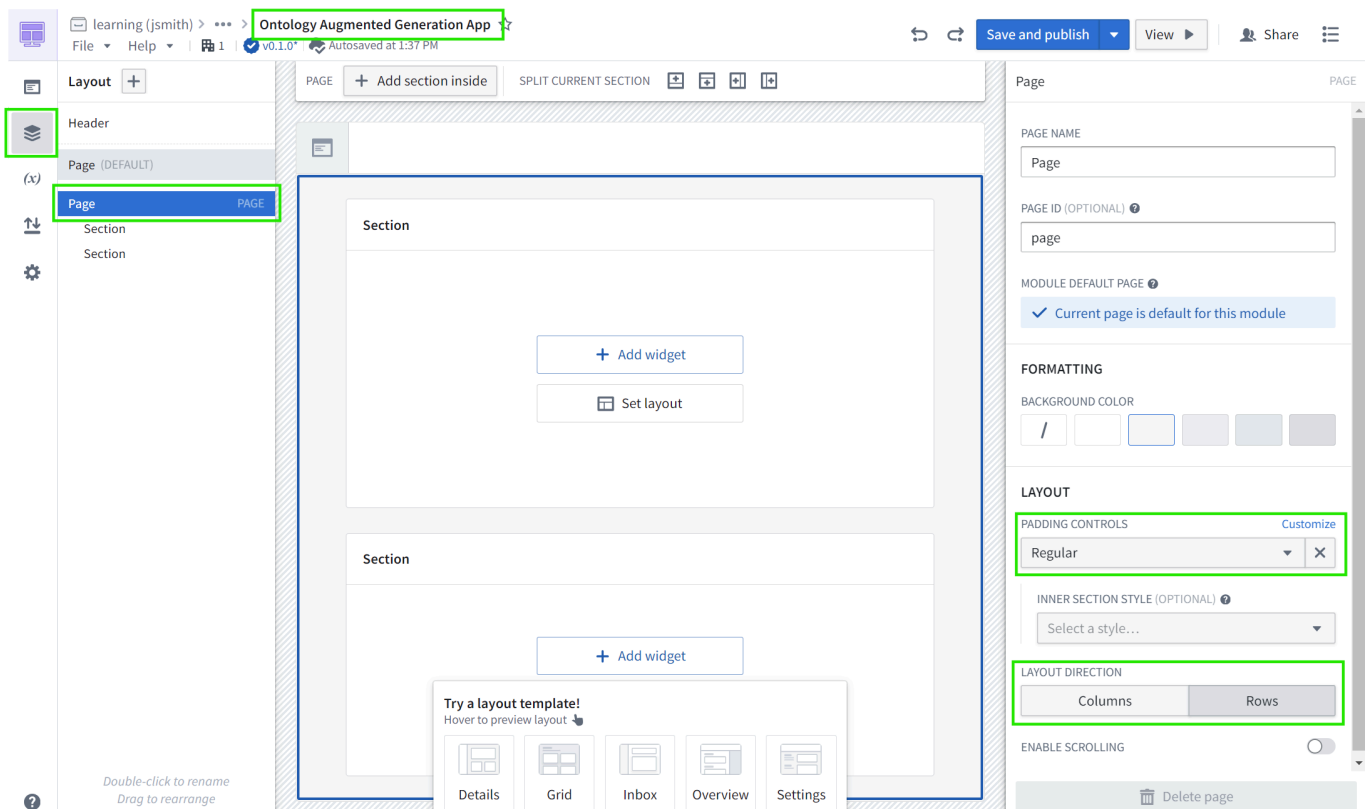
- 1. Navigate to your training project
- 2. Right-click in your Project and pick *New*
 - Alternatively, you can click *New* in the top right corner
- 3. Select *Workshop module* from the dropdown



Organize the Layout

Lesson 45 / 51 · [原文链接](#)

1. Name the app as "*Ontology Augmented Generation App*" on the top left corner
2. In the left side navigation bar, click on the *Layout* icon (the stacked squares)
3. Click on the *Page* layer
 - Not to be confused with the *Page (Default)* page identifier
4. In the right side configuration panel, under *Layout*, set padding to *Regular*
5. Change *Layout direction* to *Rows*

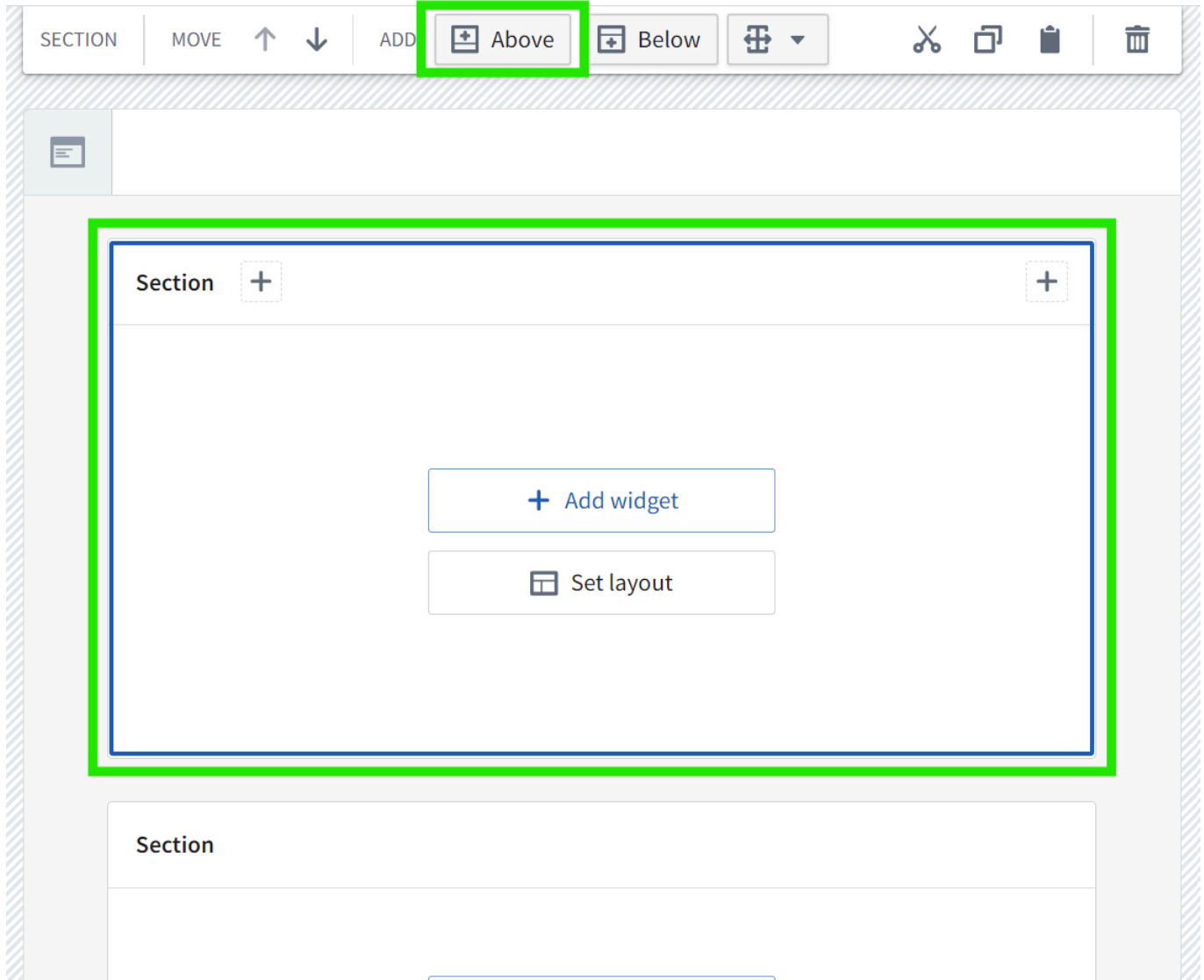


A Workshop app is organized as a tree of nested Sections and Widgets, which you can see on the left hand panel in the Layout tab. Each Section and Widget comes with configuration options to refine their functionality and appearance.

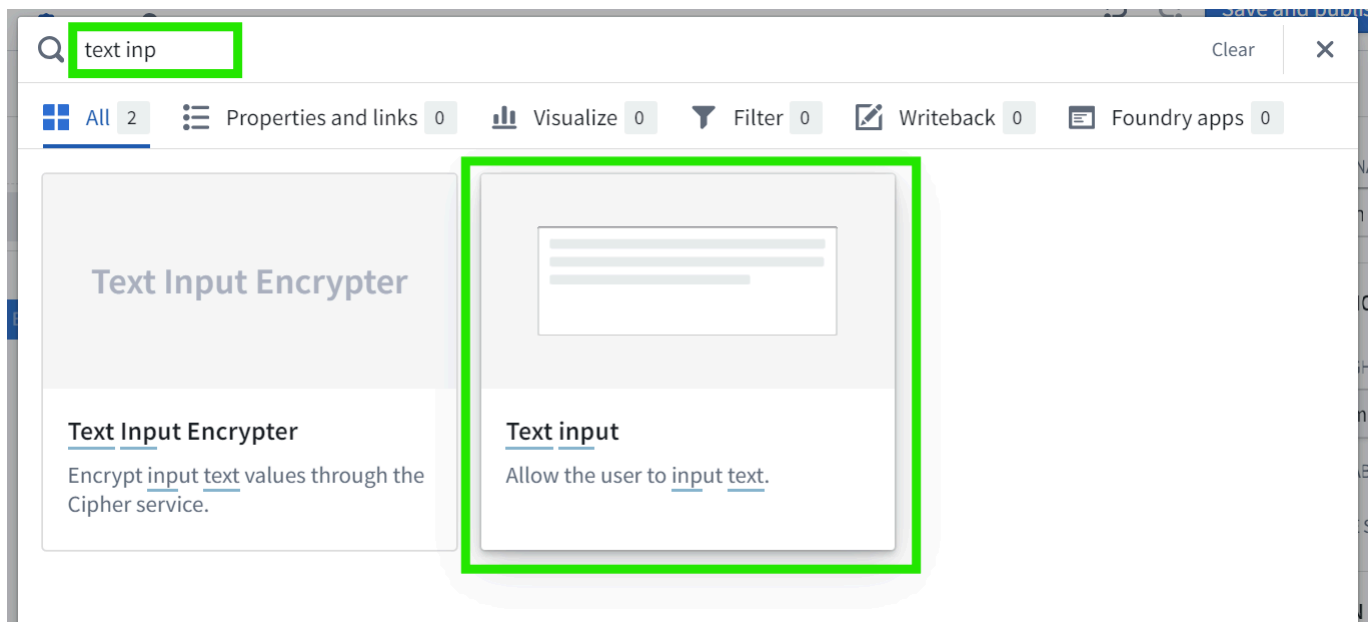
Get User Input

Lesson 46 / 51 · [原文链接](#)

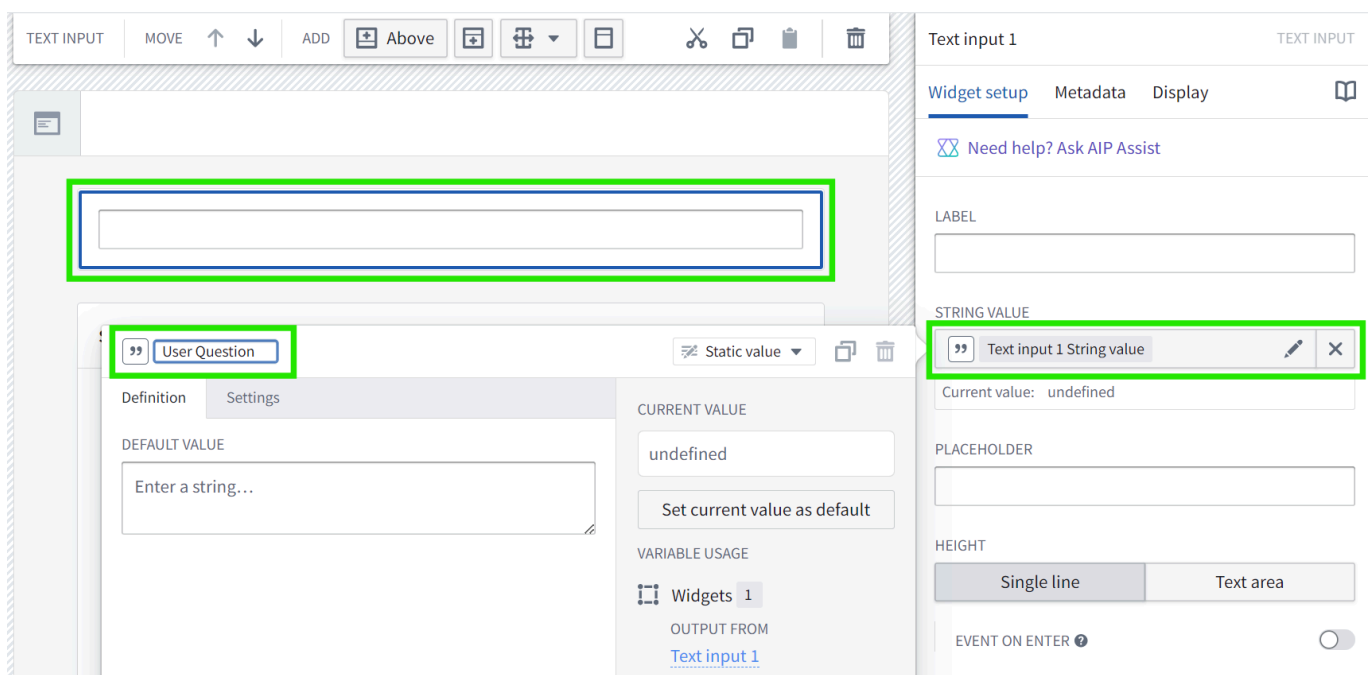
1. Click on the top *Section* to select it
2. In the top toolbar, click on the *Above* button to insert a new section above the one selected



1. Click on the *Add Widget* button in the newly created section
2. Search for the *Text input* widget
3. Click on the *Text input* widget to add it to the app



1. Click on the newly inserted text input widget
2. Click on the widget variable (under *string value*) in the configuration panel on the right
3. Rename the variable to "User Question"

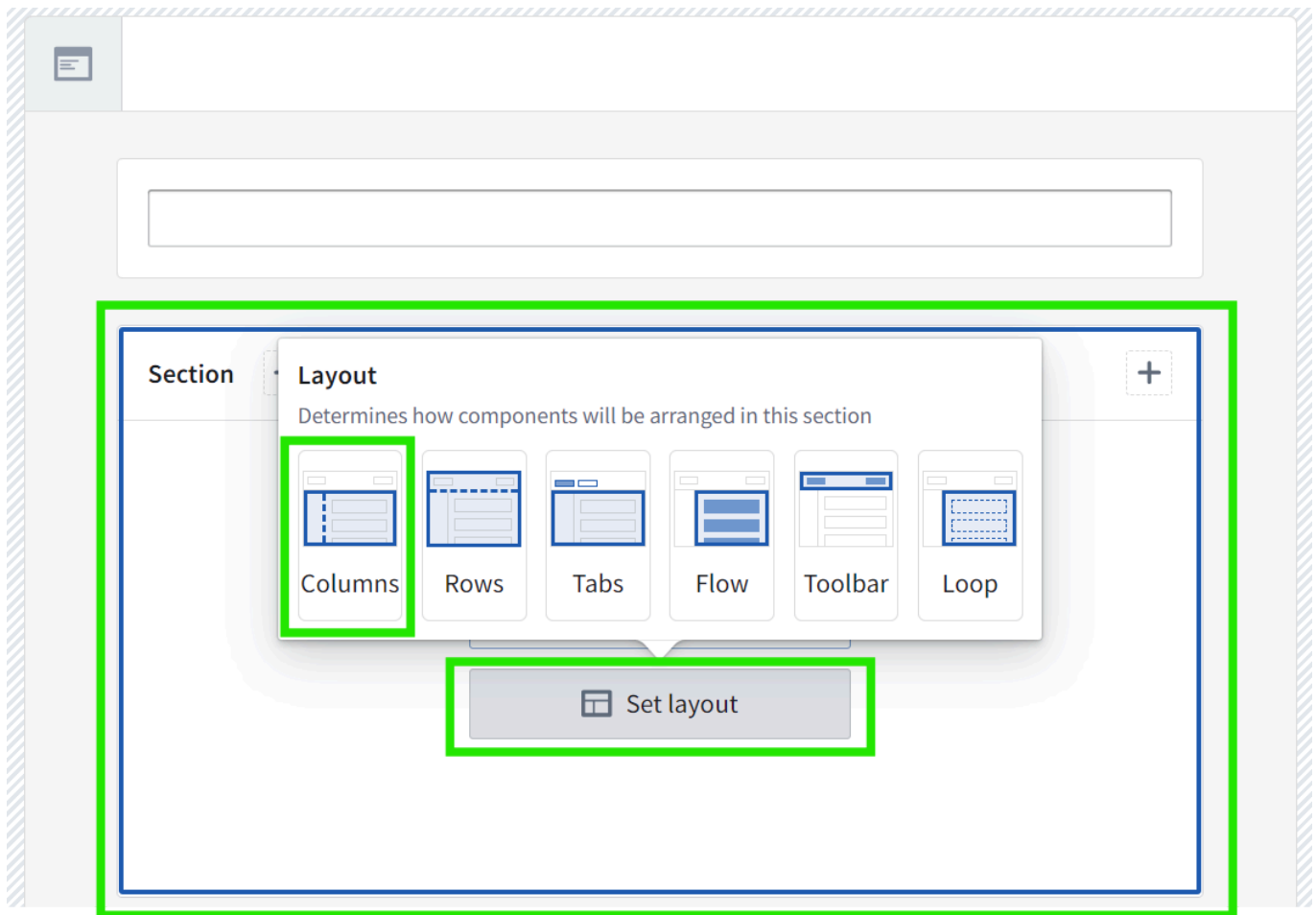


Each widget acts like a function, taking existing variables as input. Widgets also produce side-effects and populate output variables based on user interactions.

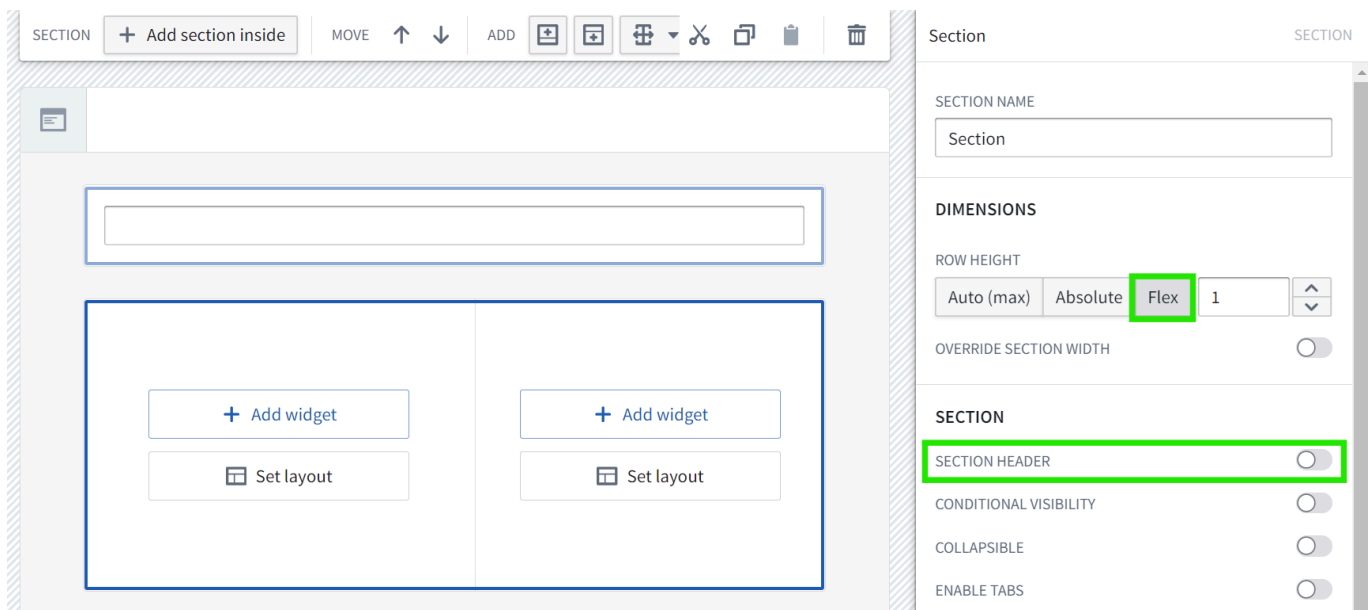
Display Objects

Lesson 47 / 51 · [原文链接](#)

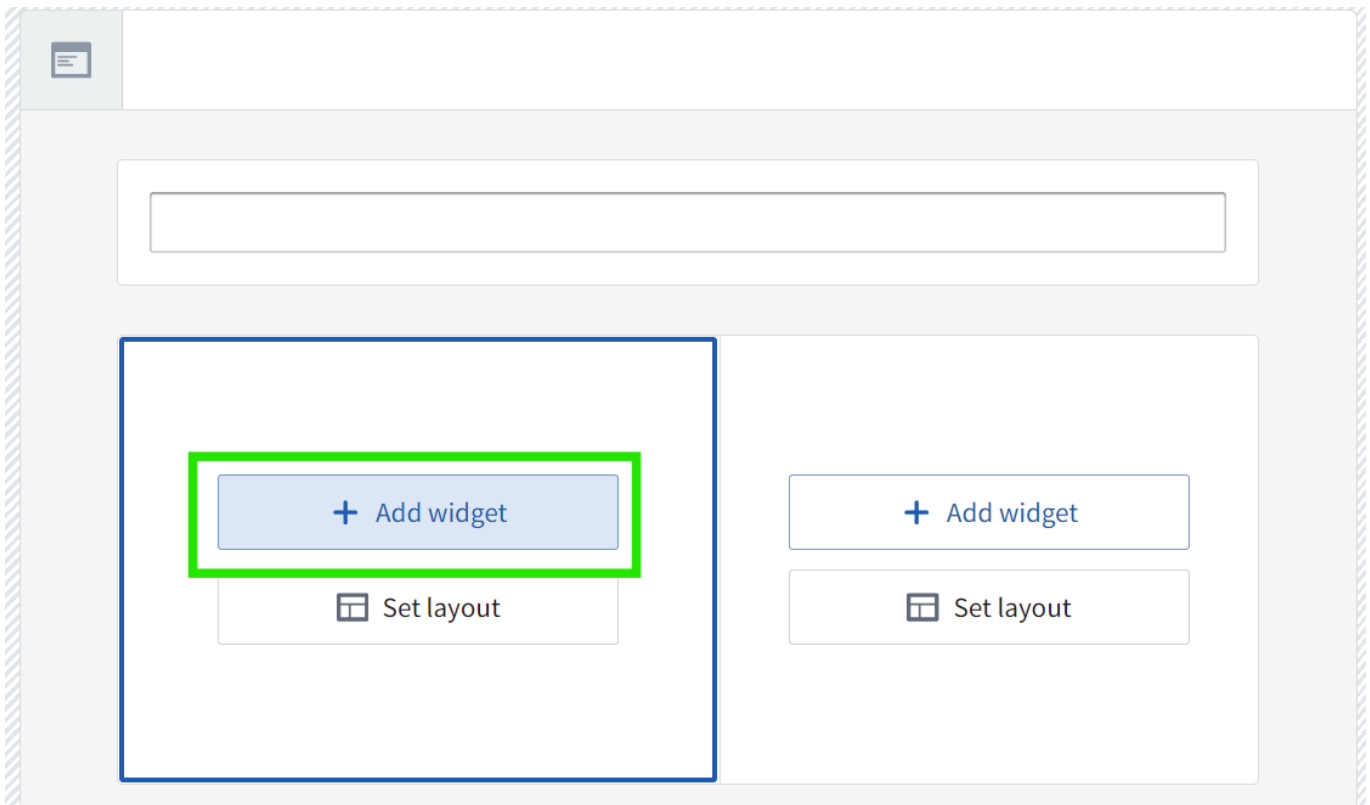
1. Select the *Section* below the text input widget
2. Click *Set layout* and select *Columns*



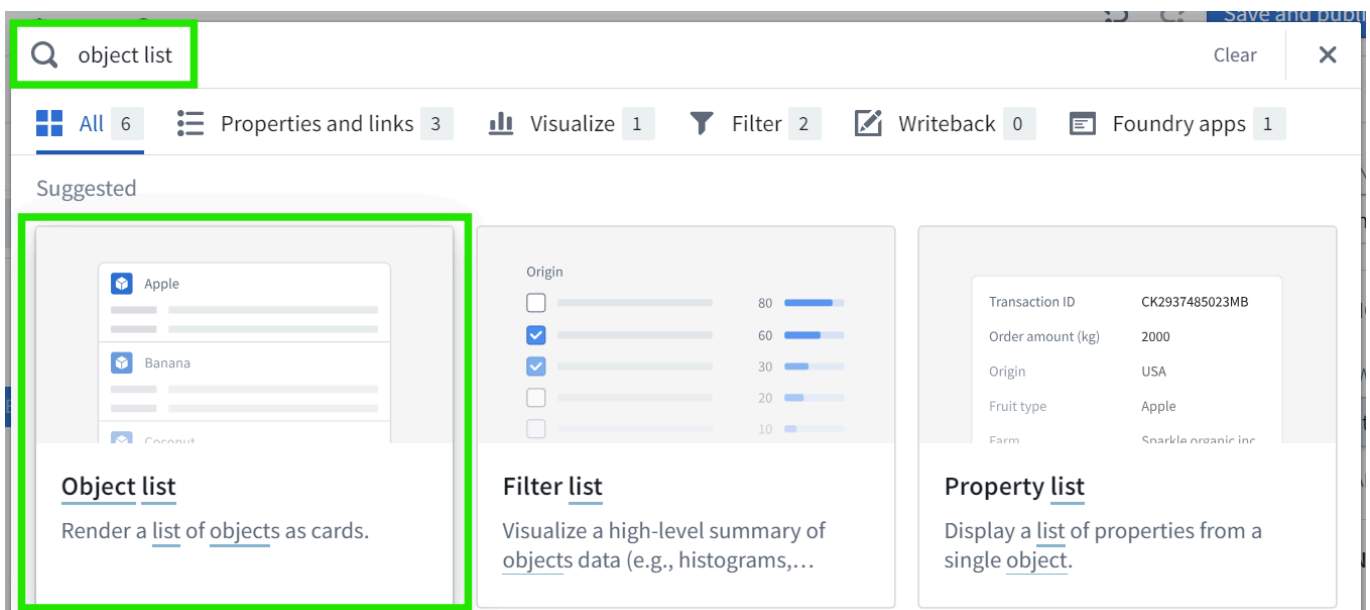
3. In the configuration panel on the right, toggle-off the *Section Header*
4. Set *Row height* to *Flex*



1. Click the new section on the left
2. Click *Add widget*

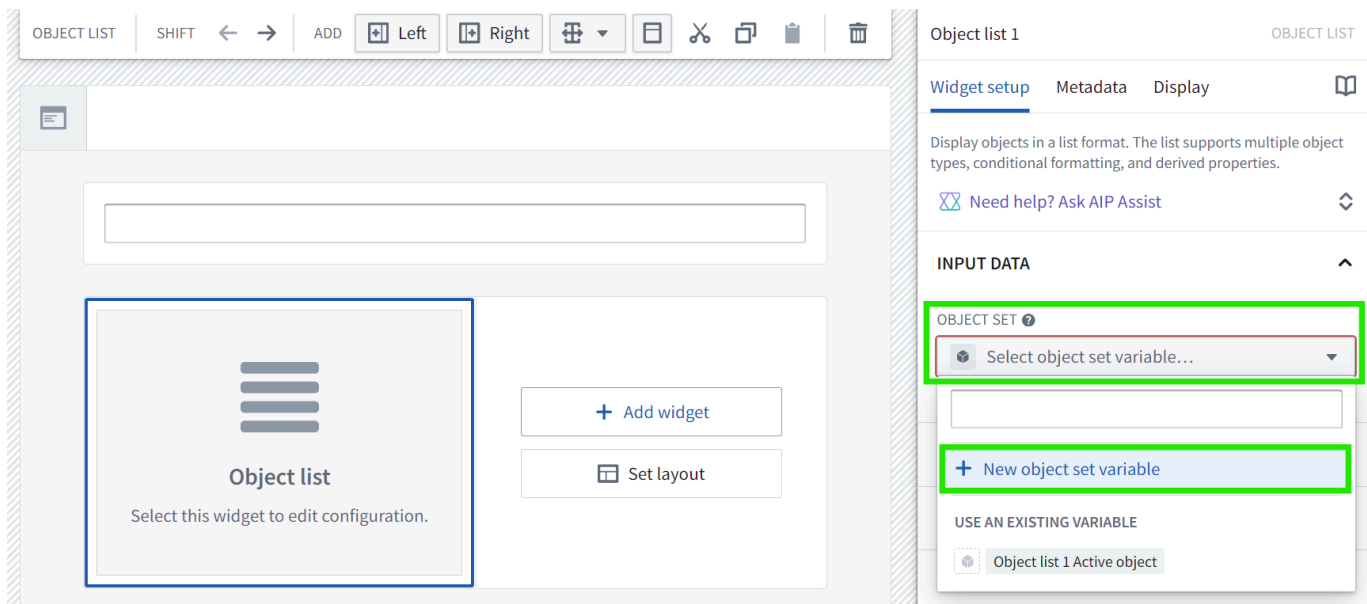


3. Search for and insert an *Object list* widget



4. Click on the *Object Set* input dropdown

5. Click *New object set variable*



1. Rename the new object set variable to "*Relevant Chunks*"
2. Select your Chunk object as the *Starting object set*

3. Click on *Filter... On a property*
4. Select the *Embedding* property

The screenshot displays a software interface with a dropdown menu for filtering objects. The dropdown menu is open, showing a search bar labeled "Filter..." and a list of properties. The "Embedding" property is highlighted with a green box. Below the dropdown, the "On a property" option is also highlighted with a green box. The background shows a list of objects with various properties and a "CURRENT VALUE" section.

Property	Value
Chunk Id	Experimental
123 Chunk Number	Experimental
Content	Experimental
[1.0] Embedding	Experimental
Media Item Rid	Experimental
123 Page Number	Experimental
Path	Experimental
Summary	Experimental

Below the dropdown menu, the "On a property" option is highlighted with a green box. The background shows a list of objects with various properties and a "CURRENT VALUE" section.

CURRENT VALUE

578 " (<username>) C...

" The article discusses t...

" ri.mio.main.media-ite...

" The text provides data...

" The article discusses a...

" The increase in U.S. T...

" ri.mio.main.media-ite...

" The text lists various d...

" The text references se...

" The study compares r...

" The article discusses t...

+ 568 more

VARIABLE USAGE

Widgets 1

USED IN

[Object list 1](#)

5. Set the User Question variable as the *Query*

6. Set the *K* value to "10"

Relevant Chunks

Object set definition

DefinitionSettings

Starting object set

<username> Chunk

EmbeddingK-nearest neighbors

Query

User Question

K value (1-100)

10

Filter...

On a property

(x) Using a variable

On a link

Traverse to

Get linked objects

CURRENT VALUE

578 <username> C...

The article discusses t...

ri.mio.main.media-ite...

The text provides data...

The article discusses a...

The increase in U.S. T...

ri.mio.main.media-ite...

The text lists various d...

The text references se...

The study compares r...

The article discusses t...

+ 568 more

VARIABLE USAGE

Widgets1

USED IN

Object list 1

The string variable passed to "Query" gets embedded with the same algorithm that you used for processing the text chunks. This embedding step represents it as a coordinate in such a way that text chunks with nearby embeddings are of a similar content. Using the Ontology, you can quickly retrieve the semantically most similar Object Types.

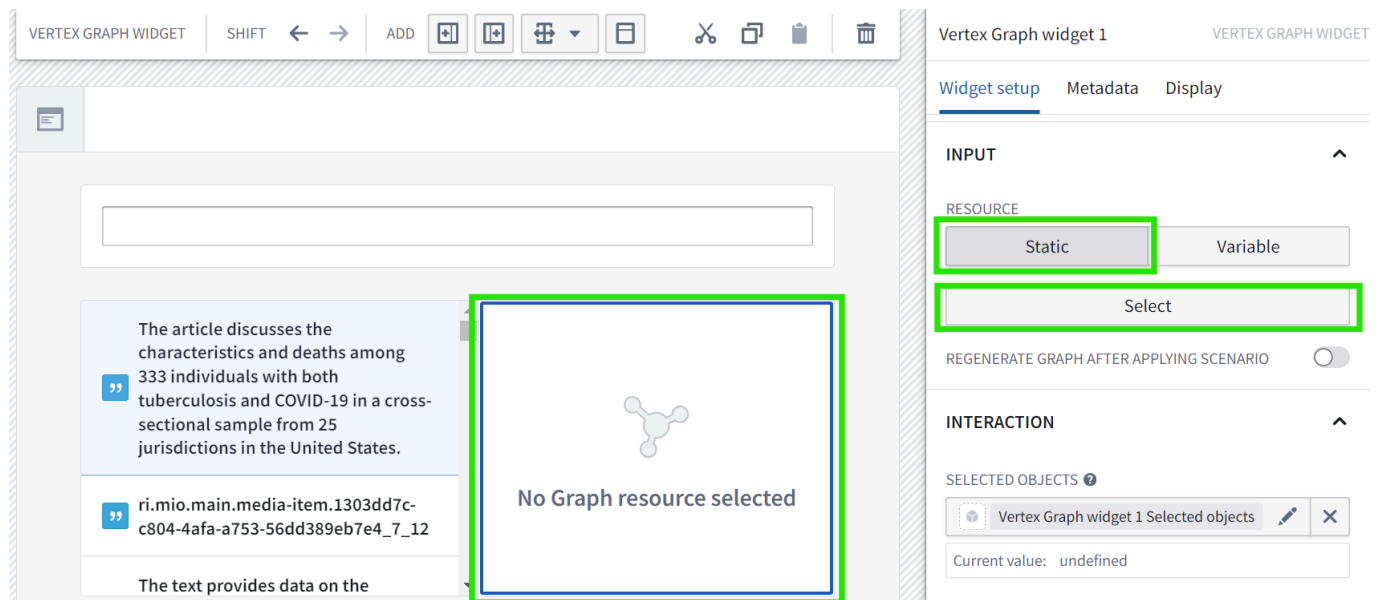
Display Object Relations

Lesson 48 / 51 · [原文链接](#)

1. Select the new *Section* on the right
2. Click *Add widget*
3. Search for and insert a *Vertex Graph widget*



4. Select the newly created Vertex Graph Widget
5. In the widget configuration drawer, under *Input*, select *Static* and press *Select*



6. Search for and select your *Knowledge Graph Template* as the backing resource for the graph

X

Cancel Select

7. Select the **Relevant Chunks** object set variable as the template input.

VERTEX GRAPH WIDGET

SHIFT ← → ADD Left Right Split section

The article discusses the characteristics and deaths among 333 individuals with both tuberculosis and COVID-19 in a cross-sectional sample from 25 jurisdictions in the United States.

ri.mio.main.media-item.1303dd7c-c804-4afa-a753-56dd389eb7e4_7_12

The text provides data on the effectiveness of Rifampin treatment over different durations and

Selection Search Histogram

Vertex Graph widget 1

Widget setup Metadata Display

INPUT

RESOURCE

Static Variable

Knowledge Graph Template /Gabriel Amaral-07f726/learning (jsmith)/S...

GRAPH OVERRIDE RESOURCE

” Select string variable...

Parameter inputs

USE LEGACY INPUT OBJECT SET

(<USERNAME>) CHUNKS *

Relevant Chunks

Current value: 578 (<username>) Chunks

This parameter expects multiple objects, so all objects of the right type in the object set will be used.

Display AIP Response

Lesson 49 / 51 · [原文链接](#)

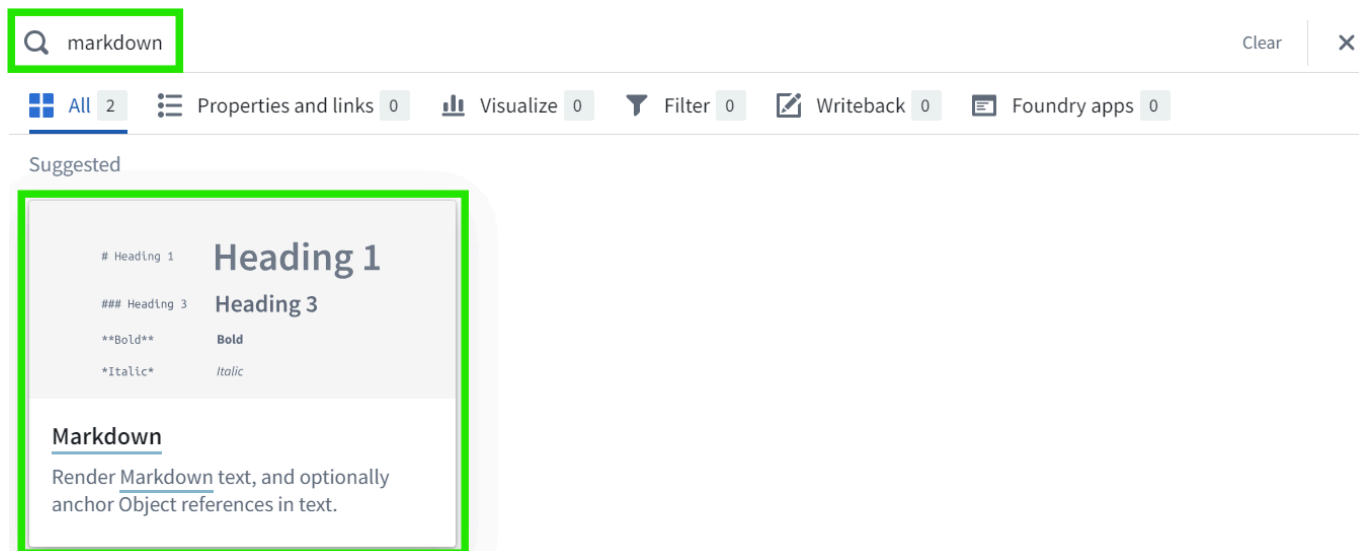
1. Select the last empty *Section* at the bottom of the app
2. Toggle-off the *Section Header* in the configuration panel on the right
3. Set *Row height* to *Auto (max)*

The screenshot displays a web application interface. On the right side, there is a configuration panel for a 'Section'. The panel includes the following settings:

- SECTION NAME:** A text input field containing the word 'Section'.
- DIMENSIONS:**
 - ROW HEIGHT:** A dropdown menu with 'Auto (max)' selected, and other options 'Absolute' and 'Flex'. A numeric input field shows '600' and a unit 'px'.
 - OVERWRITE SECTION WIDTH:** A toggle switch that is currently turned off.
- SECTION:**
 - SECTION HEADER:** A toggle switch that is currently turned off.
 - CONDITIONAL VISIBILITY:** A toggle switch that is currently turned off.
 - COLLAPSIBLE:** A toggle switch that is currently turned off.
 - ENABLE TABS:** A toggle switch that is currently turned off.
- FORMATTING:**
 - BACKGROUND COLOR:** A color picker with a dropdown menu showing a light blue color.
 - BORDER STYLE:** A dropdown menu with four options: 'None', 'Solid', 'Dashed', and 'Dotted'.
- LAYOUT:** A button labeled 'Set a layout to configure options'.

On the left side of the interface, there is a main content area. At the bottom of this area, there is a green-bordered box containing two buttons: '+ Add widget' and 'Set layout'.

1. Click *Add widget*
2. Search for and insert a *Markdown* widget

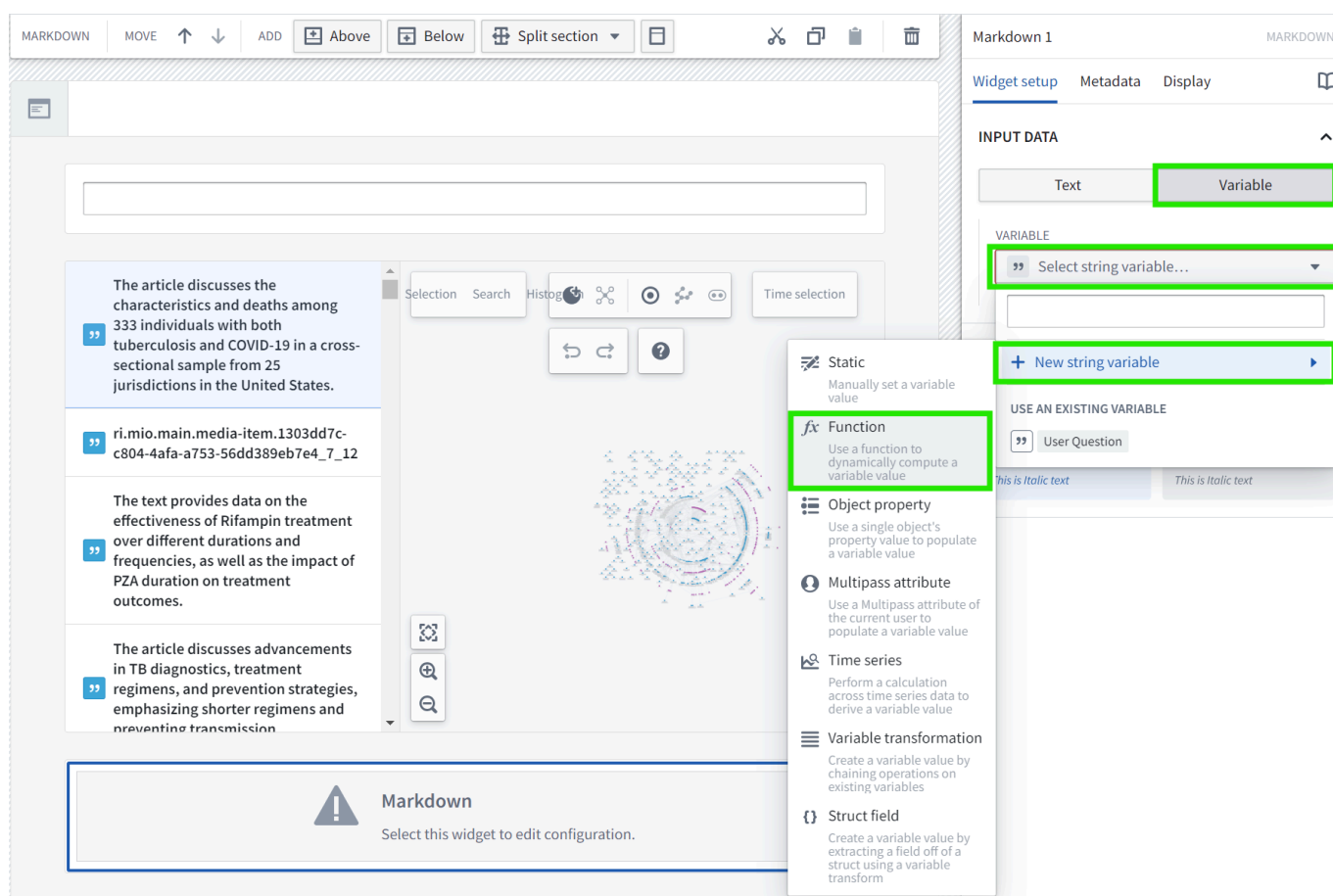


3. Change the *Input data* to be based on a *Variable*

4. Click on the input variable dropdown

5. Select *New string variable*

6. Select *Function*



1. Rename the new object set variable to "*AIP Response*"

2. Click on the *Function* dropdown and pick your Logic Function

- This will be your Ontology Augmented Generation function

3. Set the *User Question* variable as your *Function input*

” AIP Response

fx Function

Definition

Settings

FUNCTION

(<username>) Ontology Augmented Generation

1.0.0

Function inputs

USERQUESTION*

User Question

Use a static input

RECOMPUTE VARIABLE VALUE

Automatically

CURRENT VALUE

null

VARIABLE USAGE

Widgets 1

USED IN

[Markdown 1](#)

(x) Variables 1

USES

” User Question

You will see an error at the bottom of the screen, this is expected. Once you input a user question this will disappear

1. Click *Save and publish* in the top right corner
2. Click *View* in the top right corner
 - You've now connected all the components we've worked on throughout this course!

learning (jsmith) > *** > Ontology Augmented Generation App ☆

File Help 1 v0.1.0* Autosaved at 2:27 PM

Layout +

Header

Page (DEFAULT)

Page

Text input 1

Section

Object list 1

Vertex Graph widget 1

Markdown 1

TEXT INPUT MOVE ↑ ↓ ADD Above Below Split section

The article discusses the characteristics and deaths among 333 individuals with both tuberculosis and COVID-19 in a cross-sectional sample from 25 jurisdictions in the United States.

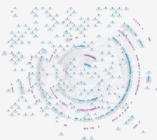
ri.mio.main.media-item.1303dd7c-c804-4afa-a753-56dd389eb7e4_7_12

The text provides data on the effectiveness of Rifampin treatment over different durations and frequencies, as well as the impact of PZA duration on treatment outcomes.

The article discusses advancements in TB diagnostics, treatment regimens, and prevention strategies, emphasizing shorter regimens and preventing transmission.

The increase in U.S. TB cases in 2023

Selection Search Histogram Time selection



Series

Show error

Save and publish View Share

Text input 1 TEXT INPUT

Widget setup Metadata Display

Need help? Ask AIP Assist

LABEL

STRING VALUE

User Question

Current value: undefined

PLACEHOLDER

HEIGHT

Single line Text area

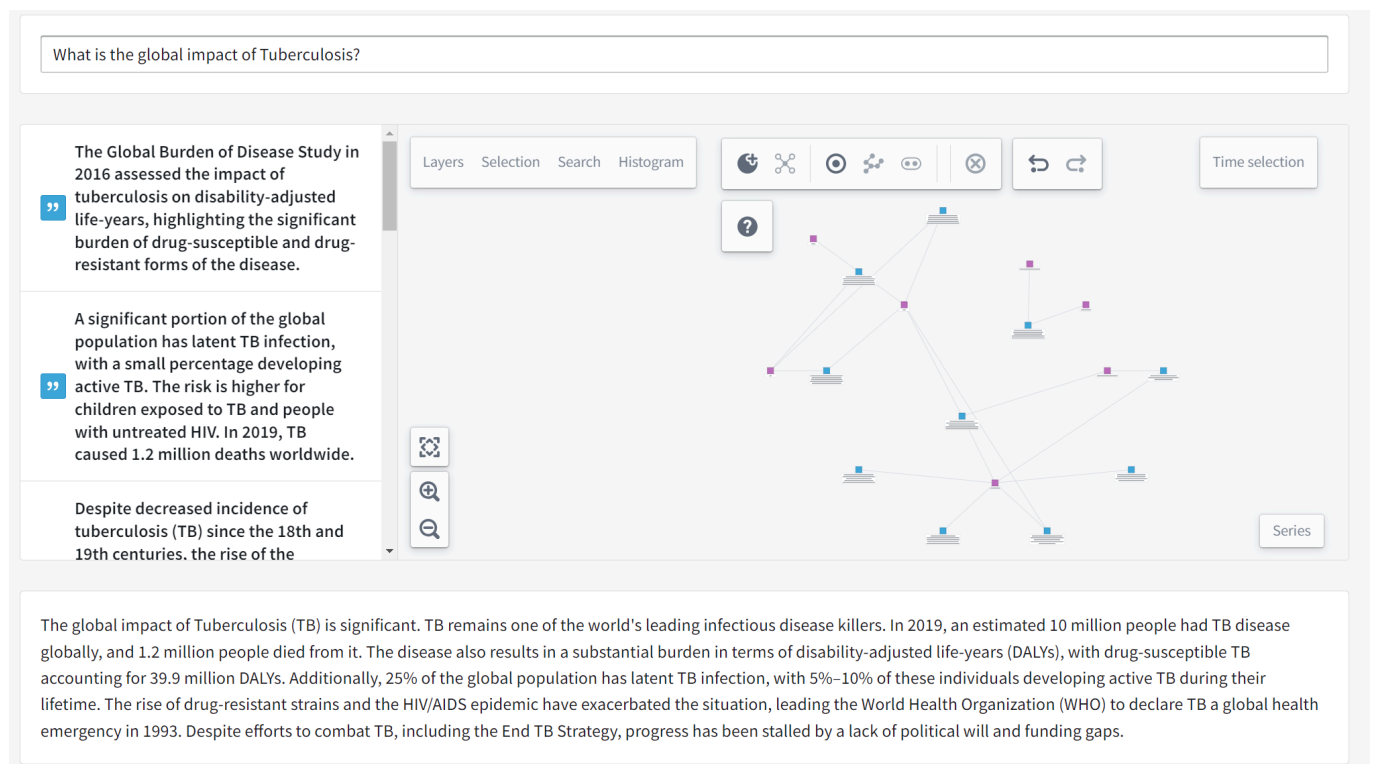
EVENT ON ENTER

Conclusion

Lesson 50 / 51 · [原文链接](#)

Well done! You have built an interactive app from scratch using Foundry and AIP. It allows you to ask a question about the topic focus of your team, Tuberculosis. Your AIP Logic function then finds all the relevant information from your knowledge base and compiles it into an answer rooted in trusted data. This subset of your Knowledge Graph is also displayed to you and other users in your new Workshop app to further inspect. Your task is complete - you no longer need to sort through endless pages to discover the answers to your questions and discover the most important concepts in the research literature. Try out the capabilities of your new tool using some of the questions below for inspiration.

- What are the biggest challenges in treating and preventing Tuberculosis?
- What is the global impact of Tuberculosis?
- Why is multidrug-resistant Tuberculosis a concern?



Note that this brief final section of the training material does no justice to the Workshop application in Foundry. Workshop allows you to tie your resources built with other Foundry applications together and expose them as a coherent workflow. On top of it, it allows you to curate and expose opportunities for users to write back their decisions to your data asset, truly making your work operational.

Looking to continue learning?

Explore the [AI Engineer Training Track](#) for more courses and resources related to this course.

As always, don't hesitate to reach out to your Palantir representatives if you have new workflow ideas and are ready to solve the problems of tomorrow.

If you have any feedback on this course, please leave a note in the optional feedback section of this course, or reach out to us directly at training-feedback@palantir.com so we can continue improving this course.

Loading...

Lesson 51 / 51 · [原文链接](#)

Loading...